The financing limits of local adaptation

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Abstract

What are the returns to investment in flood protection, and how does the financing structure shape its provision? Studying Louisiana's levee districts, I estimate a triple-differences regression using observations along the boundaries of districts that passed the levee tax. Property values inside the district rise about 32% on average, and increase by 3% per additional foot below the levee. The gains reflect demand for protection: occupancy increases more in high exposure areas. The tax burden is much smaller than these gains, but levee referenda often fail. A key financing friction is that uneven benefits and uniform taxes can cause the median voter to oppose aggregate improvements. I develop a spatial model with heterogeneous flood exposure and risk-reduction investments to quantify the welfare cost of this friction and evaluate the financing limits of local adaptation.

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1 Introduction

Adaptation to the changing climate is increasingly important, as communities across the world face unprecedented weather patterns. In particular, many coastal communities today are planning to or are already building infrastructure along the sea, seawalls, to protect themselves in the face of increasing flood risk. These investments will cost billions of dollars. For many other parts of the world, there is extensive built infrastructure and adaptation entails just upkeep and maintenance, that in practice can be quite cheap. However, estimates from a recent report by the United Nations environment program suggests that globally, developing countries need 12 to 14 times more funds for adaptation than existing flows.

Adaptation is inherently local. Infrastructure is built in particular places and protects particular properties. It is about local communities building resilience to remain in place, and minimize the losses they expect to face. Local communities can, in principle, self-organize to finance and maintain shared adaptation infrastructure (Ostrom, 1990). If they are unable to do so, it becomes important to understand where they fall short, and in the long run, how to assess the costs or the limits beyond which it is better for communities to migrate to safer land. In this paper, in particular, I focus on flood protection infrastructure - *levees*, and ask (1) what are the returns to levee investments, and (2) how does its financing shape its provision.

Levees or dykes are physical barriers that run parallel to water bodies that prevent settlements from flooding. They have been built across history, in societies all over the world to allow settlements near rivers on fertile soil, and to protect valuable land - places with immovable resources or geographic advantages like ports. With this built infrastructure came a dependence on their continued ability to prevent flooding. Like other infrastructure, levees require continual maintenance and investment in order to provide flood protection. Today, coastal land and land below levees are sinking, and so these levees require active 'lifting', along with everyday maintenance of mowing lawns and filling in any holes.

Financing shared flood protection infrastructure is challenging, as is the case with any local public good. There is a collective action problem; it is difficult to align who pays with who benefits. With floods, even a foot of elevation makes a big difference, so at a very granular level, there could be differences within neighborhoods, between neighboring properties, that share common levees. Levees protect the properties within their boundaries, but they may hurt neighboring properties via hydraulic spillovers: displacing flood waters into the unprotected region.

In this paper, I study Louisiana's levee districts - local government entities that are responsible for the operation and maintenance of levees. These districts primarily rely on

property taxes as a source of revenue. In order to raise additional funds over and above a threshold rate set by the constitution, they require voter approval. In recent years, many districts have taken the tax to vote in order to meet increasing costs, and as in any democratic process, this comes down to the deciding vote, or the median voter.

In the first part of the paper, I look at the districts that pass the tax to measure returns to levee investments. Specifically, I look at house prices to understand how the benefits of the better flood protection and the cost of the tax are capitalized, to quantify the returns. A key challenge in identifying the effects is that passing of the tax may not be exogenous. These passes are not random, and may be a leading indicator of growth in the district. I study house prices in narrow strips along the borders of the levee districts that passed the tax, relying on the local comparability of the properties outside the boundary. This now sets up a triple difference specification, comparing before and after the tax passage, inside and outside the boundaries of the levee districts, and finally within the treated side among properties more or less exposed to capture the protection benefits of the levee tax. The control group is the neighboring properties that otherwise trend similarly, absent the tax. I find that in my treated sample, property prices rise by about 32% following levee tax passage. The response varies by 3% with every foot of elevation or depth below the top of the levee: the deeper the property, the more the benefits from a better levee.

I then look at quantities along with other outcomes that characterize the response in the local economy following the tax passage. Levee investment reduced vacancy rates, as population and occupancy increase without any increase in the local housing stock. These outcomes are consistent with an increase in demand in the presence of inelastic supply, suggesting that the benefits are fully capitalized in the observed house prices.

Finally, I look at flood insurance premia and policy data from FEMA's National Flood Insurance Program, to find direct measures of the mechanism through which we expect to see the benefits accrue, through expected reduction in flood damages. The claims would allow us to see whether the realized damages were also lower after passage of the tax. There is a general reduction in observed claims post the tax passage, fewer floods affecting the treated regions, and the premia reduce significantly on average, and a similar gradient shows up per foot of depth reflecting the lower expected damages. Expanding the analysis beyond the boundary sample, I find that the flood insurance premia and claims patterns hold true across the protected region. So the benefits are widespread, across the district.

The returns are large, benefits greatly outweigh the costs in the passed districts and there is evidence of revitalization of the region broadly. However, some of these referenda fail. There are levee districts in Louisiana that request for additional funds to invest in the levees, and the median voter in these districts didn't see benefits outweigh costs. An in-built

constraint in the policy is that the tax is uniform, but benefits vary by exposure. By looking at the distribution of elevation across the districts, I find that the districts that failed were more likely to see their median voter at higher elevation than the average. This is a direct measure to capture the benefits of better levees, but in reality many other factors govern a voter's willingness to pay. This could be prior flood experiences, salience of the levee, beliefs about the effectiveness of the tax raised in preventing flood damages – which consists both of whether the wall will actually work, or the fact that the money may not reach the levee to begin with – and various other factors like a lack of awareness, a general distaste for taxes, financial constraints, propensity to vote in general. Precinct-level voting patterns suggest that many of these indeed play a role.

This evidence of various factors correlating with support for the levee tax suggest that there is significant heterogeneity across the districts. To understand how important these differences might be across districts, and quantify the welfare costs of the failed referenda, and in particular the median voter constraint, in part two of the paper I develop a spatial model that allows me to account for the heterogeneity and extrapolate the empirical estimates to measure exactly how much money was left on the table in the referenda that failed.

I set up a spatial model of house prices and local financing of flood protection, in the spirit of Rosen (1979)-Roback (1982). It closely resembles the state of affairs in the levee districts of Louisiana, and models a representative district. Households choose locations of different flood exposure within a levee district, and the cost of housing adjusts with the level of exposure. All households pay a common tax, set by the levee district through a referendum, that is used to fund the levees, which reduce the expected damages from floods. Households tradeoff the push of higher taxes, but the pull of better flood protection.

Aggregate land value, or property prices, is the object of welfare. This is because households are freely mobile, and their utility is pinned down by an exogenous outside option. Any gains from better flood protection is therefore accrued in the immobile factor, the house prices. The social planner maximizes welfare, and if the district were to set the tax at the optimal tax rate, the median voter may or may not agree, depending on their flood exposure.

To quantify the model, I combine various data sources and can observe and measure nearly all the core ingredients, except for an effectiveness parameter γ. This is the parameter that governs the conversion of dollars raised through levee taxes, into levee investments and therefore reduced expected damages, which captures many things: technology, ability of the levee district, and the conversion of dollars in practice into levees. To estimate this, I use the triple difference estimate from the data to match the corresponding coefficient derived from the model, taking the partial derivative of the difference in prices with respect to its flood exposure. This gives us a pooled effectiveness parameter.

Since the matched moment was the triple difference coefficient, or the estimate on the price gradient per foot of elevation, I simulate prices in the model to validate whether the average price response from the model matches the observed prices in the data. The model implied prices is lower by 5%, which suggests potentially that there are additional general equilibrium forces, the increase in economic activity in the regions revitalized by the better levees, the agglomeration benefits of people moving back in, that the model might miss. The estimates of price response may therefore be a lower bound on what could have been the case if we were to account for the agglomerative forces.

Taking this parameter to the failed referenda, and applying the model imposed structure on these districts, the model suggests that there is about $169 million dollars left on the table in these districts. However, this varies significantly by district, depending on their exposure profile. There are two districts where the aggregate price response predicted by the model is negative, denoting that the gains from better flood protection was very small relative to the amount spent by households in locations that may not have gained.

I next look at the the optimal tax rate for each district, and look at whether they would get majority support, to capture where the median voter's exposure constraint might bind. This constraint binds in two districts, suggesting that accounting for political economy frictions becomes an important consideration.

Finally, to examine how the median voter constraint might look in the face of increasing risks, I scale up the baseline expected damage parameter, constructed as a comparative static. Many of these levees are designed to protect a region from a 100 year flood. But if these regions start to experience these floods more frequently, for instance if these floods become 5 times more likely, or a 1 in 20 year flood, nearly all these districts will start to see the constraints bind. With increasing risk, more needs to be spent to make the risky properties safe, but the burden lies more on the safer houses to preserve their neighborhood, and it reaches a threshold, beyond which it may no longer make sense to continue to invest.

Ultimately, the goal of the model is to characterize the limits of investing in protection through local financing, given the constraints that regions currently face: how adaptation is financed matters as much as whether it pays.

Related Literature This paper contributes to literature on the capitalization of local amenities, investing under financing and governance frictions, and climate adaptation in spatial equilibrium. It makes three contributions. First, it provides estimates of the differential returns to local flood protection investments within the protected region and, through hydraulic spillovers, outside it. Second, it documents a financing friction central to investment in resilience in place: if the median voter does not see benefits outweigh costs, they will

not support on aggregate positive NPV projects. Third, it quantifies the welfare implications of the median voter constraint, and provides a framework to evaluate the existing financing structure and various other sources of heterogeneity that govern the efficacy of the public investments into flood protection, and how the constraints bind in the face of increasing flood risk.

First, I contribute to the literature on the returns to local public investments. Cellini, Ferreira, and Rothstein (2010) and Gupta, Van Nieuwerburgh, and Kontokosta (2022) use house prices to measure the value of local school and transportation investments. A growing set of papers measure how flood protection is capitalized into property values, including Fell and Kousky (2015), Kelly and Molina (2023), Benetton et al. (2022), and Bradt and Aldy (2026) also documents the spillovers from federal levee construction on nearby properties. I contribute to this literature by estimating differential returns to levee investments through house prices, at different levels of flood exposure who directly pay for the better levees through property taxes.[1][2]

This paper is closest in theme to Fan (2024) which shows that weaker housing markets lead to poor levee maintenance through the local property tax channel, and levees failures in turn reduce house prices and local tax revenues, creating a downward spiral. In this paper, I study the mechanism through which levee districts raise property taxes for levees. I focus directly on who pays and who benefits, and the frictions associated with the current financing arrangements which lead to lower spending on levees, which may then set off the mechanism described in Fan (2024).

Second, I contribute to a traditional corporate finance literature that assesses investment under financing and governance frictions (Fazzari, Hubbard, and Petersen (1987), Kaplan and Zingales (1997)). Specifically for adaptation, Rajan and Ramcharan (2023) show that access to finance shaped whether farmers invested in irrigation during the 1950s drought, and Lu and Nakhmurina (2023) show that financial constraints and short-term budget planning are linked to flood adaptation gaps in cities across the US. For public infrastructure, the relevant constraint is often whether the pivotal supporter of a budget or policy is willing to bear the costs.[3] In addition, traditional public finance literature suggests that the allocation

[1] A related literature in finance asks whether climate risk itself is capitalized. Flood risk capitalization in house prices may not be complete due misperception of the risk or beliefs of the buyers, as is documented in the Bernstein, Gustafson, and Lewis (2019), Baldauf, Garlappi, and Yannelis (2020), Bakkensen and Barrage (2022). Specifically land subsidence risk is in Hackett and Zamora (2026). In this setting, the salience is quite high, as captured for instance by the 'watermark' observed by households when their houses flooded during Katrina, which suggests these are places to expect full capitalization.

[2] In principle, capitalization makes local public goods self-financing: Arnott and Stiglitz (1979) show that if a region has optimal population, aggregate land rents equate to optimal expenditure on public goods (the Henry George theorem).

[3] Also related to Grossman and Hart (1986) - how control rights should relate to cash flow rights. Within

of public goods aligns with the preferences of the median voter Bergstrom and Goodman (1973), and is shaped by the agenda-setting entity Romer and Rosenthal (1979), and when voters choose to move to locations, is shaped by the location choice and who is the median voter in a particular region Epple and Sieg (1999). With the requirement of the median voter's approval on levee tax, these fundamental constraints show up in levee investment decisions, and this paper characterizes these constraints.

Third, I contribute to a growing literature on climate adaptation in spatial equilibrium, surveyed in Balboni and Shapiro (2025), focusing on how adaptation is financed. Bilal and Rossi-Hansberg (2021) treat location choice as a forward-looking investment, and quantitative spatial models show that migration, investment, and innovation are important margins of adaptation, to coastal flooding in Desmet et al. (2021) and to global warming in Cruz and Rossi-Hansberg (2024), while Bilal and Rossi-Hansberg (2023) show that anticipated climate damages reshape where people and capital locate. Yet exposure is persistent, Kocornik-Mina et al. (2020) find that low-elevation urban areas flood more often but concentrate more economic activity, and recover as rapidly as higher areas after floods, and public investment can reinforce this. Existing papers on flood protection investments in space document moral hazard (Hsiao (2025)), and path dependence in the face of increasing flood risk (Balboni (2024), Ouazad and Kahn (2025)). In assessing the choice to rebuild or relocate, Fu and Gregory (2019) show that rebuilding decisions in post-Katrina New Orleans generate spillovers across neighbors and that optimal rebuilding subsidies are biased against relocation, and Aron-Dine (2024) evaluates rebuilding and relocation policy after Hurricane Maria. In this literature, adaptation is chosen or funded by a planner or a higher level of government, or is an individual decision. This paper focuses on adaptation infrastructure, or public insurance, that is financed and chosen locally: freely mobile residents in spatial equilibrium vote on and pay a uniform tax for levee investments, while benefits vary across parcels within the district.

2 Institutional Background and Setting

The paper focuses on the levee districts of Louisiana, who (i) are responsible for the operation and maintenance of levees within political boundaries, (ii) are primarily locally financed, and (iii) rely on voter approval to raise funds above constitutionally set thresholds. This section provides a summary of the institutional framework governing levees and their and upkeep. Appendix **??** contains additional institutional detail.

the levee districts in Louisiana, control rights are allocated uniformly, as are costs, while benefits vary.

2.1 Levees, Louisiana, and the levee districts

Levees are physical barriers between water and land and have been built in the US since the 1700s. It is the primary structural flood defense in Louisiana, a state with extremely high flood risk. The responsibility of maintaining levees is largely local, and in Louisiana, lies primarily with levee districts. These intermediaries levy taxes and issue debt in order to fund levee maintenance, and predominantly rely on property taxes to maintain their aging levees.

Levees. Levees are an important flood protection infrastructure and are aging. Levees are physical barriers between water and land, and are a traditional form of flood protection infrastructure, built for a specific level of flood protection. Levees have been built across history, across the world to allow civilization to settle closer to water. They are also called dykes. The National Levee Database documents about 24000 miles of levees, built across history: the median age of a levee is 61 years.

Levees require active upkeep in order to keep the level of protection they were designed to provide. For example, a levee is designed to protect a region from a 100 year flood, which means it prevents flooding from a flood that has a 1 per cent chance of occuring each year. If the "1 per cent flood" becomes more common, keeping the protection at the 1 per cent flood level entails "lifting" the levee, or increasing the height of the levee to keep the same level of protection. Additionally, if the land sinks - land subsidence is a big issue in coastal regions across the world and in Louisiana - just to keep it at the same level of protection, levees need to be actively raised. Aside from height, mowing the lawns on the levees and filling up cracks are every day maintenance necessities to preserve the integrity of the structure.

Louisiana. Louisiana is highly prone to flooding by virtue of its geography, on the gulf coast and traversed by multiple rivers including the Mississippi. The first levee in the US in fact was built in New Orleans by the French in 1717 - to protect an important port from the might of the Mississippi River.[4] Over the years, more levees have been built and existing levees have been fortified and lifted, to withstand the increased force of water due to the constrained path of the river that the levees imposed and to keep up with the subsidence of the land itself. As of 2025, Louisiana has the highest density of levees in the US. It has about 3,300 miles of levees, 14% of the national total. Over the years, the levees have been increased in height. Where the construction year is recorded, the median age is 74 years.[5]

[4] History from the Army Corps here (accessed 2026-09-21)

[5] Construction year is sparsely recorded for Louisiana in the NLD: 68 of 2,348 segments (3%)

Levee districts. Levee districts are 'special purpose districts' - local entities with the specific purpose of managing the levees.[6] Louisiana was early to organize itself into these districts, with the first districts being established in the late 1800s. Levee districts are responsible for operating and maintaining the levees within their jurisdiction, and costs are borne by the district as a whole. Louisiana today has 33 levee districts and entities functioning as levee districts. Panel (a) of Figure 1 shows the levee districts across the state, and the levees they sponsor, manually mapped to the National Levee Database.

2.2 Financing the levees and the referendum constraint

Levee districts fund themselves via taxes, debt and state and federal grants.[7] Taxes account for the bulk of the funding, above 70%. Tax revenue is comprised of sales and property taxes (or *millage*), the latter accounting for over 60% of the total revenues dedicated to levee construction and upkeep.[8]

A critical aspect of tax financing is that above a constitutionally defined threshold,[9] the levee districts require voter approval to raise taxes. From 2010 through 2025, there were 29 referenda across 18 levee districts. The West Jefferson Levee District first called a referendum to increase the millage in November of 2015. The ballot measure read:

> Shall the West Jefferson Levee District (the "District") be authorized to levy and collect annually a special tax of 5.50 mills on all property subject to taxation in said District (the "Tax") (an estimated $5,500,000 reasonably expected at the time to be collected for an entire year) for a period of thirty years beginning with the year 2016, and ending with the year 2045; and (i) shall the revenues or proceeds of 5.50 mills of the Tax hereafter received from the authorization of the Tax be dedicated and used for the purpose of constructing and maintaining levees, levee drainage, flood protection, hurricane flood protection and for all other purposes incidental thereto, including payments required in connection with debt obligations incurred for any of the above described capital purposes, as well as a portion of the operation and maintenance of the Gulf Intracoastal Waterway - West Closure Complex (the "West Closure Complex"), which is part of Louisiana's comprehensive integrated coastal protection plan and from which the jurisdiction of the West Jefferson Levee District derives benefit?

[6] Such special purpose districts are growing in number, see Ruggieri (2025)'s Figure 1.

[7] Other sources of funding, which are the occasional bp oil spills settlement, the gulf of mexico oil revenue-sharing. See Barnes et al. (2021).

[8] Appendix Table A.1 documents the municipal bond issuances. There are 13 independent issuances across seven levee districts from 1989-2023. Table A.2 compares these bonds with other municipal securities issued in overlapping parishes. In addition, these districts issue private placement bonds, which are not recorded in the database and are difficult to obtain data on systematically.

[9] 5 *mills* or 0.5% property tax for districts created before 2006, 2.5 mills for Orleans, districts created after 2006 need voter approval for any levy (Louisiana Constitution).

There are are many reasons why residents may vote down a levee tax. For instance, as the text of this ballot shows, levee taxes are typically flat.[10] Voting therefore is contentious because the benefits of levee upkeep for flood protection are not evenly distributed across houses within a district: Some houses benefit more than others from better levee protection and the tax rate fails to capture those differences.[11] Of the 29 referenda from 2010 through 2025, 14 passed and 15 failed, which is what happened in the West Jefferson Levee District. Failure of the referenda have important consequences for levee upkeep. For instance, the fiscal year 2015 Annual Financial Report for the West Jefferson Levee District, filed in October 2016, flatly states the consequences associated with the failure to pass the November 2015 ballot:[12]

> In November of 2015, voters rejected WJLD's request for a 5.5 millage increase for 30 years. Without this additional revenue, WJLD has scaled back its proposed schedule of levee lifts. While this revised maintenance schedule will allow WJLD to provide satisfactory near term flood protection, present funding will be inadequate to cover required RR&R or maintenance lifts beyond fiscal 2016. Because of the failure of the 2015 proposed millage election for WJLD, the annual budgets will not be able to cover future levee lifts and structural repairs with projected resources beginning in 2016.

It is for this reason that West Jefferson Levee District voters were called to the ballot once again in April of 2018. This time, the residents of the levee district approved the special tax. From the Annual Financial Report for FY 2018:[13]

> SLFPA-W has been concerned with the long-term funding of the OMRR&R of these facilities. In November of 2015, West Jefferson parish voters rejected a 5.5 millage proposition over 30 years. To keep levee lifts building apace, Commissioners were obliged to seek alternative sources of funding for immediate construction and flood defense needs. As a result, in December 2016, WJLD bonded (borrowed) $7.5 million to fund the impending levee lifts prior to armoring projects. Based on the outcome of

[10] Louisiana's uniform levee millage (La. Const. art. VI, §39) is the prevailing design among large flood-exposed states: Texas levee improvement districts, though permitted to tax on a "benefit basis" (Tex. Water Code §§57.012, 57.260), levy uniform ad valorem taxes in practice (e.g., Fort Bend County LID No. 7, Official Statement, 2020), and Florida's water management districts levy uniform millage (Fla. Stat. §373.503), with courts upholding even "benefit" assessments apportioned by assessed value (*City of Boca Raton v. State*, 595 So.2d 25 (Fla. 1992)). California is the exception: Proposition 218 requires each parcel's assessment to be proportional to its special benefit, so agencies such as SAFCA price parcels by modeled avoided damage rather than by value (SAFCA, Final Engineer's Report, Natomas Basin Local Assessment District, 2011)

[11] See here for a lawsuit concerning how the levee boards were used to funnel valuable mineral rights and revenues in the Plaquemines Parish, who has subsequently never been able to pass the levee tax. Here is an article where they tried to convince the voters: "What we want the public to understand is this money can't be moved to other projects"

[12] See here for the original financial report (accessed 07-06-2025).

[13] See here for the original financial report (accessed 07-06-2025).

> the 2015 millage election, WJLD re-analyzed the near future funding needs and reduced the millage request from 5.5 mills over thirty years to 4.75 mills over ten years. The reduction extends some of the maintenance cycles and reduces other contingencies. In April of 2018, West Jefferson parish voters passed the requested 4.75 mills for a period of ten years. Beginning in 2019, WJLD will have sufficient funds to continue raising and armoring levees for the next ten years to keep pace with ongoing natural settlement and subsidence of the system. WJLD will also have sufficient funds to perform the long-term maintenance and repairs of its three pump stations and sector gate complexes."

In sum, taxes are an important source of levee upkeep financing. The Louisiana Constitution requires the district to get voter approval to increase the millage, and failure to pass a millage increase referendum constrains the ability of the district to finance levee construction and maintenance.

2.3 Fragmentation and Spillovers

This paper is concerned with measuring the returns of local adaptation strategies geared towards minimizing the effects of climate change. Because these strategies are designed and implemented by local authorities, who may lack coordinating institutions, there is room for externalities to get in the way of optimality. This externality is very much at work in the case of levee construction.

Flood protection works within the boundaries it is designed to protect. The decision to construct or lift a levee to enhance the protection of one district, increases the risk of flooding in another, adjacent, district: The fragmented jurisdiction of the levee districts means that investment decisions are made without accounting for the externality.[14]

This phenomenon of *hydraulic* spillovers is well documented (Bradt and Aldy (2026) and Wang (2021)), and applies to Louisiana's levee districts as well. The levee districts have historically defined political boundaries, and each of them maintain their own sections of levees along the same water bodies. Any policy evaluation of these local adaptation strategies thus needs to take into account these externalities.

[14]It is because of this fragmentation and a need of more regional coordination, after Hurricane Katrina the coastal regions created umbrella authorities - Southeast Louisiana Flood Protection Authority - East and West, and the Coastal Protection and Restoration Authority. These authorities facilitate cost sharing, allow for contingencies and enable coordination that was otherwise not required within the mandate of the individual levee districts. The financing, however, is still reliant on the local tax base, and each of these districts have their own renewal cycles and rates.

3 Empirical Strategy and Data

In this section, I describe the empirical strategy and explain how I use features of the setting described above to obtain causal estimates of the returns to levee investments. I compare properties close to the levee district boundaries, before versus after a tax passes on one side, and across high and low flood risk properties isolating the protection benefits net of taxes, as well as the hydraulic spillovers.

The goal of this exercise is to quantify the benefits associated with levee investments. Levees are a local public good, and provide differential benefits within defined boundaries. In the literature of evaluating the benefits of local public goods, the object of interest is housing prices (Cellini et al., 2010). This is because, if those who choose to purchase a home in the region see the benefits of better flood protection outweigh costs, we would expect to see a positive effect on house prices, and this encapsulates any additional benefits that are difficult to measure. This difficulty, well studied for school districts in the literature, applies to levees and levee districts as well, and house prices capture these various channels.[15] Some of these benefits are direct, the reduced probability and severity of damage in the event of a flood. Some, indirect, such as the positive agglomeration externalities associated with levee district re-population on account of that better protection. My identification strategy will isolate the first, direct, effect.

Data To measure returns to levee investments, my primary object of interest is house prices. I construct a novel geo-spatial dataset that combines three broad data sources. First, I use standard property level data from Corelogic. Second, I collect, from different sources, data on levee location, as well as the height, length, and area protected by the levees within each of the thirty three levee districts; in addition I collect data on the elevation of the properties. Finally, I obtain data on the various aforementioned referenda, dates, as well as several details about the proposed millage and expected revenue from the ballots.

As is standard in the literature, I obtain house prices and characteristics for single family residences and condos from Corelogic, along with the transaction date, and the exact latitude and longitude of the parcel.

The event of interest is the passage of a levee tax referendum. I obtain all referenda by the levee districts through a Public Records Request to the State Bond Commission, and collect the corresponding date, ballot details, voting outcomes and voter precinct information from the Secretary of State website. Table 1 lists all the referenda held from 2010–2025.

Properties within the political boundaries of the levee districts that passed the levee tax

[15] See Chay and Greenstone (2005).

are *treated*. I obtain these boundaries from voting precinct shape files publicly available on the redistricting state of Louisiana website - I match the corresponding referendum-year precincts to those that voted for the levee tax.

I measure flood exposure at the property level using elevation. Specifically, I use the depth below the median height of the levee system that protects it. Elevation data comes from satellite imagery from the USGS 3DEP, and the levee heights come from a public records request to the Coastal Protection and Restoration Authority. The exact location of the levees and the areas it protects comes from the National Levee Database, which I associate with the levee districts by sponsor name or segment name (and geography for validation). I then compute the distances to the levees at the parcel level. [16]

I also use each property's current flood-zone classification. *SFHA* denotes a *Special Flood Hazard Area*, FEMA's mapped 1%-annual-chance floodplain (A and V zones). I obtain these classifications from FEMA's National Flood Hazard Layer (NFHL), which provides the digital flood maps used by the National Flood Insurance Program (NFIP), and assign them to properties by matching parcel coordinates to flood-zone polygons.[17]

A detailed description of the data sources and variable construction is in **??**.

Empirical design The goal is to isolate groups with differential treatment effects from the levee tax passage, and define a control group that is unaffected by the levee tax which serves as the counterfactual price path for the treated houses.

The underlying assumption is that houses close to the boundary of the levee districts and on both sides of it belong to the same housing market, and stand to be in the path of the same hurricane, or flooded by the same water. Passage of the ballot is positive news regarding the exposure of houses in the treated district to future climate events. This positive news affects, amongst other things, housing prices in the treated district relative to the housing prices of the untreated group exposed to the same risk. This allows me to measure the returns to levee investment.[18] Properties fall into three groups. Treated properties are inside the

[16] FEMA identifies ground elevation relative to surrounding terrain and nearby flooding sources as a determinant of flood risk; see NFIP, "What's Your Flood Risk?". McKenzie and Levendis (2010) use property elevation to study flood-risk capitalization in New Orleans before and after Katrina. These references highlight elevation as an indicator of flood exposure, and show how Katrina priced it into properties in the New Orleans area.

[17] See FEMA's National Flood Hazard Layer. The indicator uses the effective maps in the downloaded snapshot, rather than the map in force at each transaction or referendum date. Current classifications may therefore reflect subsequent map revisions or levee accreditation.

[18] The empirical design combines the geographic regression discontinuity of Black (1999), but across multiple border-pairs and referenda into a stacked difference in differences specification. This setup is similar to Russo and Ostriker (2026) which uses a spatial RD at FEMA floodplain regulation boundaries to study the effects of flood zone regulation on construction and damages. The levee district boundary creates a sharp discontinuity in tax jurisdiction. The referenda provide time variation in treatment, and the events

district that passed the levee tax. Within the treated group, properties receive benefits of a better levee depending on their flood exposure - the deeper the property (the lower the elevation), the higher the risk and therefore the higher the benefits. Outside the boundary, neighbor properties close to the treated district's levees form the spillover group, the ones we expect to be affected by hydraulic spillovers.[19] Neighbor properties beyond this close proximity from that levee form the control group. Appendix Figure A.1 maps this design for the Orleans Levee District and its neighbors, with each transacted property in the boundary sample shaded by its depth below its own district's median levee crest.[20]

Identification strategy To obtain causal estimates, the endogeneity concern still remains, that the district's investment or even the passage of the tax in the district could be confounded with other factors in the local economy. The levee taxes pass where we might see better growth prospects or amenity improvements, passage may be a leading indicator of unobserved local shocks. I do not require that passage was a surprise. Instead, I compare properties in close proximity which serve as the counterfactual price path for the treated properties.

The underlying assumption is that houses close to the boundary of the levee districts and on both sides of it belong to the same housing market, and stand to be in the path of the same hurricane, or flooded by the same water. Passage of the ballot is positive news regarding the exposure of houses in the treated district to future climate events. This positive news affects, amongst other things, housing prices in the treated district relative to the housing prices of the untreated group exposed to the same risk. This allows me to measure the returns to levee investment.

Sample construction For each district that passed the levee tax, I collect its shared boundaries with neighboring levee district. Each district-district pair, a boundary-pair, becomes an individual event study which I use to identify the effects of passage of the tax.

I start with all the property transactions within these boundary pairs and apply the fol-

are stacked to align on relative time. This is also similar to stacked boundary design in Oh et al. (2021).

[19]This is similar to Bradt and Aldy (2023) in how they handle the spillover group. The definition there is where the protection of the levee ends, but given the margin here is a competition between neighboring levees, the distance is from the treated levees rather than the end of the leveed area.

[20]Distance to the levee is an alternative measure of flood exposure. However, because levees are built along the water's edge, distance to the levee largely proxies for distance to the water, which is itself a poor measure of exposure. In southern Louisiana, the highest ground lies on natural levee ridges adjacent to the river, and elevation declines toward the backswamps, so distance can be inversely related to exposure. While properties in the immediate vicinity of a breach face additional damage from the force of the water, floodwaters that overtop or breach the levee collect in low-lying areas, so a property's elevation largely determines the depth of inundation.

lowing filters: (i) keep only pairs where the referenda passes (treatment requires an actual tax increase), (ii) exclude districts with more than one passing referendum (no repeat treatment within the main analysis event roster), (iii) exclude districts with any subsequent referendum after the treatment date (within the main analysis event roster), (iv) exclude districts where there were any confounding events differentially affecting the treated side (v) restrict to sales from 2008 onward to exclude post-Katrina recovery dynamics, (vi) restrict to boundary pairs where the control is never or not yet treated (vii) restrict to properties inside National Levee Database "leveed areas", so every property is protected by an existing levee, (viii) apply the Calonico et al. (2014) MSE-optimal bandwidth of 2,354 m around each boundary, dropping properties that fall inside the bandwidth of more than one pair of the same referendum.

Across referenda, each surviving property is assigned to its earliest referendum on either side of the boundary, so each property only shows up in one referendum. Table 2 documents the funnel. The bandwidth sample consists of 38,947 transactions of 29,595 properties across 8 passed referenda in 8 districts and 16 boundary pairs. Summary statistics for the analysis sample are in Table 3, and by group are in Table 4.

4 The returns to levee investments

I use transacted house prices around the passage of the levee tax, at the district boundaries, to obtain the causal effect of levee investments. Each of the boundary pairs define one individual levee passage event, and so I stack these events to get pooled estimates across the 8 referenda. Treatment is defined as the passage of a levee tax referendum, and post is post passage of the levee tax in a particular referendum. Appendix Figure A.1 shows an example of the boundaries used and the empirical design, using Orleans levee district, who passed their levee tax in 2012. It has 4 shared boundaries with neighboring levee districts, the East Jefferson, West Jefferson, Algiers, and Lake Borgne Basin levee districts. The colors denote depth below the property's own district median levee crest, and the bandwidth is highlighted in dashed lines. The pair of Orleans - East Jefferson is 1 boundary pair.

The regression specification:

$$\begin{aligned}\log P_{i,p(i),t} = \beta_1 \left(\text{Treated}_i \times Post_t\right) + \beta_2 \left(\text{Spillover}_i \times Post_t\right) \\ + \cdots + X_i'\gamma + \alpha_{n(i)} + \alpha_{p(i),t} + \varepsilon_{i,p(i),t}.\end{aligned} \tag{1}$$

where i is a property, $p(i)$ is the referendum-specific boundary pair that property i belongs to, t is relative time in years, and $n(i)$ is the census tract that property i belongs to. $\alpha_{p(i),t}$ are pair $\times$ year fixed effects absorbing any referendum-pair-specific time trend, X_i captures

hedonic controls - living sqft, lot size, stories, age. These are zero-filled when missing, and a characteristic specific missing indicator is also added to the hedonic controls. The specification is saturated with the interacted arguments (Treated_i, Spillover_i). Standard errors are clustered at the pair level.

Results: house prices increase 32% on average on passing a levee tax Following passage of the levee tax, Table 5 suggests that passing a levee tax raises the average property price by 32% when we include controls for the hedonic characteristics. Figure 2 shows the dynamic treatment effects for treated properties and the neighboring spillover group. The ideal specification would be the property fixed effects, where any unobserved property characteristics are controlled for, but this requires a repeat transaction of the same property both before and after the time of treatment, which is very demanding of the sample: it reduces the sample by 30,891 transactions, and the estimate is less precise. Column 3 reports the property fixed effect specification, which has a lower magnitude of about 19%, but preserves the direction.

Results: properties in the spillover group lose 22% The spillover group are the properties within 600 meters of the treated levees. From Table 5, we see that this group sees huge negative spillover effects, of nearly and equal and opposite magnitude. The group however is very small in comparison - with only 195 pre period transactions. The effect weakly decreases in magnitude as we extend the sample to include more properties in the spillover group and out of the control group, but we lose comparability between the groups in the pre-period (there are significant pre-trends). Figure 4 captures this effect, the estimates are very unstable. This should be read as suggestive evidence, that the channel operates and is important to consider to ensure we address the spillover concerns and explicitly handle SUTVA.

Heterogeneity by flood exposure Benefits vary by the degree to which properties are exposed to flooding. Specifically, properties that are more flood prone will directly and disproportionately accrue the benefits of better levees, while properties on a hill, or those least exposed, only face an increase in taxes with no benefits.

To capture this, I run a triple difference, interacting the depth of the property from the top of the levee in the district. Specifically, the regression:

$$\begin{aligned}\log P_{i,b(i),t} = \beta_1 \left(\text{Treated}_i \times Post_t \times Depth_i\right) + \beta_2 \left(\text{Treated}_i \times Post_t\right) \\ + \beta_3 \left(\text{Spillover}_i \times Post_t\right) + \cdots + \gamma X_i + \alpha_{tract(i)} + \alpha_{b(i),t} + \varepsilon_{i,b(i),t}.\end{aligned} \tag{2}$$

In the DDD notation, $b(i)$ denotes the same boundary pair as $p(i)$ above, and $tract(i)$ denotes the same census tract as $n(i)$. $Depth_i$ is depth below the district median levee crest, centered at the treated-sample mean, so β_1 is the response per foot and β_2 is the treated-post level response at mean depth. The omitted terms include the remaining lower-order and spillover-depth interactions in the saturated DDD specification.

Results: properties gain 3% with every foot of depth Table 6 reports the results of the regression specification above and the dynamic treatment effect plot is in Figure 3. In the sample of passed referenda, at the main-sample treated mean depth of 18.10 feet below the top of the levee, the gain is about 38%, with a 3% - per - foot differential response.[21]

Interpreting the magnitude The magnitude of the price response is quite large. This can be broken down into different reasons, the composition of the underlying sample - its concentration in an urban economic center - New Orleans, its historical experience with failed levees and unprecedented devastation during Hurricane Katrina, and a restored belief in its ability to withstand future hurricanes. The limited number of property transactions in these narrow boundaries necessary for tight identification, prevents us from being able to tease out each channel on its own.

First, over 50% of the the sample is of the Orleans levee district's levee tax, passed in 2012. The Orleans levee district contains the city of New Orleans, an economic center. In August 2005, when Hurricane Katrina hit, the levee breaking in various parts of the Orleans district was what led to about 80% of the city being flooded. The most costly disaster in the US, and households suffered lasting consequences. i) the sample was most affected, so keeping the levees strong is a top priority, and having commitment for that instills faith in that they will not break when the next hurricane hits. This becomes a story of revitalization of the neighborhood once they see that the levees will be taken care of, and the house prices capture this. ii) this experience showed households how risky their properties were, and this is priced in most strongly in the areas that experienced flooding. When there was renewed commitment to better levees, the places most at risk recognize the risk reduction in these areas the strongest. In Orleans, the risk contrast is sharpest, and the sample is quite deep.[22] iii) The referendum happened to coincide with Hurricane Isaac, which also proved

[21] For comparison, Fan (2024) estimates that levee failures during 2000-2021 reduce neighborhood housing prices by 3-9%, over the six-year post-event window. Bradt and Aldy (2026) estimate that federal levee construction increases protected home values by approximately 3-4%. These estimates concern different interventions and measure overall price changes in the flooded/ protected regions respectively. The price response I find is significantly larger, and the heterogeneity is unique to this paper.

[22] See McKenzie and Levendis (2010) which shows how following Katrina the price reflected the elevation more strongly, going from 1.4% per foot to 4.6% per foot.

to be a test that the rebuilt levees after Katrina work, further amplifying the effect of better flood protection in the future. This renewed faith in the technology, and potentially reflects residents now updating their beliefs that the levees will hold up with the next hurricane.

Appendix Table A.5 reports transactions, distinct properties, and represented boundary pairs by referendum. Orleans accounts for 56.2% of rows in the hedonic specification. Appendix Figure A.8 reports leave-one-district-out estimates, including the joint exclusion of Orleans and West Jefferson, for the hedonic specification. Without Orleans, the continuous-depth gradient does not show the same differential response, it turns insignificant and weakly negative. Appendix Figure A.9 replaces depth with the current SFHA indicator as the sole exposure interaction: the hedonic Treated $\times$ Post $\times$ SFHA coefficient is positive when Orleans is excluded. Elevation is correlated with the SFHA indicator, but it is more granular. Across main-sample transactions with observed flood-zone status, the correlation between SFHA and depth below the district median levee crest is 0.291 ($N = 38,227$), and 0.243 when Orleans is excluded ($N = 16,331$). Evidence points toward the same gradient affecting prices, but elevation differences may not be as defined in these areas, or recognized by households as a clear measure of risk. [23]

Second, it could be the salience of the levees, which run along the political boundaries of the levee districts being used in the analysis, and so the intensity of treatment is could be concentrated at the boundaries. This is hard to disentangle - we need close proximity to neighboring districts to ensure identical housing market trends, and most of the boundaries have a water body in between - like a river or a canal, where either side has a levee maintained by each of the districts. We may expect then that treatment intensity is the highest closest to the levees, they are the first line of defense. Another explanation is the positive externalities of the levee itself in the neighborhood, where the levee is also a playground, a place of social activity, to run on, and people would rather it be intact that broken.

The levee is placed exactly along the water, by design, and removes the exposure from being close to a water body. However proximity to a levee makes the quality of the walls more salient. Appendix Table A.8 compares depth below the district median crest, distance to the treated district's levee, and separate indicators for current SFHA and levee-protected Zone X. The estimated crest-depth gradient remains positive and changes little when the other risk measures enter jointly. The distance coefficient is negative, consistent with larger perceived benefits closer to the levee, but is not statistically significant.

Appendix A.7 and Table A.7 report the complementary hedonic close-passed versus close-

[23] McKenzie and Levendis (2010) also mention how the watermark in the flooded houses during Hurricane Katrina made households recognize elevation as a measure of exposure. Anecdotally, the flood insurance premium is what makes households realize their risk, and the SFHA classification would capture that too.

failed comparison, similar in spirit to Cellini et al. (2010). Here 'close' refers to +/- 10pp around 50, which removes the Orleans' referendum from the sample. In this case, the distance to levee captures the price gradient, but weakly, and there is an overall positive price response across the levee district. In addition, Appendix A.4, documents a host of robustness checks, that correspond to checking the sensitivity to various assumptions made in the main specification, along with testing other potential confounders.

4.1 Quantifying social returns

To measure household returns to investing in flood protection, I combine the pooled average house-price effect with a simple back-of-the-envelope property-tax calculation.

The baseline is the mean sale price across all 7,629 treated pre-period transactions (6,861 distinct properties), \$206,750 (Table 4). No additional tax-type or property-level restriction is imposed. Applying the pooled average estimate, $\exp(0.2792) - 1 = 32.2\%$, gives a capitalized gain of \$66,582 (95% confidence interval: \$16,256–\$128,264).

For the property-tax illustration, I use 6 mills (0.6% of assessed value), the median incremental millage among the property-tax referenda. With a 10% residential assessment ratio and a \$7,500 exemption from assessed value, the annual tax on the representative home rises from \$79 at the baseline price to \$119 at its predicted post-vote price of \$273,332. With immediate reassessment and payments discounted at 5% over 30 years, this post-vote levy totals \$1,829 in present value. The capitalized gain is therefore approximately \$36 per dollar of property tax paid.

The separate property-level calculation below restricts to property-tax referenda and uses each property's latest pre-vote sale. Gains vary sharply across properties. The shallowest depth decile averages 8.5 ft below the district's levee crest and gains about \$10,600, compared with \$112,100 in the deepest decile, at 25.4 ft. The fitted price response becomes positive at 7.2 ft below the crest; 97% of sampled properties lie deeper than this threshold.

5 Quantities and mechanisms

The previous section characterizes the returns to levee investments through house prices, and finds it significantly increases on average, with the effects concentrated in the places that are most exposed to the flooding. This section looks at the quantity response - occupancy, housing supply, and population - in neighborhoods that align with the boundaries of the same boundary discontinuity specification above, and directly capture the mechanism, of reduction in expected flood damages and realized damages to the extent, measured using

the National Flood Insurance Program (NFIP) policy and claims.

I consolidate census-tract-level outcomes from the American Community Survey (ACS: population, housing units, and vacancies), the HUD-USPS quarterly address data (occupancy and active addresses), the National Flood Insurance Program (NFIP: flood-insurance take-up, premiums, and claims), and the LEHD Origin-Destination Employment Statistics (LODES: jobs and resident workers). I assign each tract's treatment status using the properties within it that belong to the house-price boundary sample in the previous section. To measure tract-level flood exposure, I subtract the mean ground elevation of these properties from the median levee crest of the tract's assigned district, flooring the result at zero and applying the same district-specific depth cap as in the house-price analysis. Appendix Table A.3 reports tract support by boundary pair, and Appendix Figure A.2 illustrates the tract geography and crest-depth measure. Appendix **??** provides the tract-assignment rules and quantity-variable construction; Appendix **??** details the insurance-variable construction, eligibility rules, and geographic matching.

Figure 5 compares house-price and occupancy gradients before and after tax passage. The price panel uses real sale prices after removing tract fixed effects; the occupancy panel shows raw occupied shares. These descriptive patterns motivate the regressions below; they do not identify a treatment effect. Appendix **??** documents the plot construction, and Appendix Figure A.23 compares nominal and real house-price gradients.

Then, for each tract level outcome Y, I estimate

$$\begin{aligned} Y_{c,b(c),t} = \beta_{\text{treated}} \left(\text{Treated}_c \times Post_t \times Depth_c\right) + \gamma_{\text{treated}} \left(\text{Treated}_c \times Post_t\right) \\ + \delta \left(Post_t \times Depth_c\right) + \alpha_c + \alpha_{b(c),t} + \varepsilon_{c,b(c),t}. \end{aligned} \tag{3}$$

where c is a census tract, $b(c)$ is its referendum-specific boundary pair, and t is calendar year. The specification includes tract and boundary-pair $\times$ year fixed effects and the lower-order depth terms.

The quantity outcomes measure occupancy (rate), occupied addresses, population, resident workers, jobs, and housing units. Insurance outcomes measure premiums in dollars per \$1,000 of building coverage, policy counts, and claims payments as a percentage of claimant coverage. The main insurance table uses individual policy terms for premiums and individual positive claim records for severity, each assigned its tract's depth and boundary pair; policy counts and quantity outcomes use tract-years. Appendix Table A.17 reports the all-tract insurance counterpart. Claims dynamics and raw trends use individual positive claims; the binscatters retain tract-year ratios. Appendix Tables A.16 and A.18 report summary statistics. Since census tracts are coarser geography and few in number, I do not separate

out the spillover group. Regressions are unweighted and standard errors are clustered by boundary pair. Quantity, premium, and policy-count dynamics use $t-1$ as the reference period. By design claims from flood events are not every year for every census tract, so I use post-period groups of years 0–2, 3–5, and 6 onward relative to the pooled pre-period. The annual windows are reported in the figure notes and Data Appendix **??**. Identification requires parallel trends in the treated–control depth gradient.

Results: Occupancy and population increases in areas with higher flood exposure Table 8 shows the average treatment effects and the continuous gradient based on tract exposure. Occupancy and population across the various measures seem to go up on average, as well as the number of jobs. For occupancy, the gradient effect also survives at 5% significance, suggesting that the high exposure areas where we saw the price response the strongest are exactly where people are moving in and filling up the stock. There is no significant increase in the housing stock on average, suggesting that people are occupying existing housing rather than building more post passage.

The dynamic coefficient plots in Figure 6 suggest that there are pre-trends. The triple-difference specification is demanding for these census tracts: there are very few observations, and most of the tracts are concentrated in the dense urban centers (Orleans and Jefferson). Moreover, the census variables are slower-moving and [ACS: are only available at the tract level as five-year estimates, which pool responses over overlapping windows and smooth year-to-year changes], so it would be harder to isolate the treatment effect without more granular data. Appendix Figures A.19 and A.20 show the raw trends and binscatters with fewer controls. The patterns are consistent with filling up empty, risky housing rather than building new stock, although the raw trends alone do not establish this interpretation.

Results: Insurance premia drop and flood claims are lower Table 9 shows the same specification for individual policy premia per coverage, claim severity, and tract-year policy counts from the National Flood Insurance Program. Each policy term and claim record is assigned its tract depth and boundary-pair classification.[24] Figure 7 shows annual premium and policy-count dynamics through six years after passage, and claims dynamics grouped into years 0–2, 3–5, and 6 onward. The premium gradient becomes more negative in years four through six. I run this first for the boundary sample to highlight the flood risk reduction channel through which we observe the price gradient shift.

I find that the pooled average premium falls by \$0.708 per \$1,000 of coverage. In the

[24]Tract level aggregate results are in Appendix Table A.17. Appendix **??** gives the construction of the policy-term and aggregate outcomes.

DDD, the effect at mean depth is $-0.750, while the gradient is $-0.138 per foot per $1000 dollars of coverage. This is consistent with lower insurance prices when FEMA recognizes improved levee protection and maintenance.[25].

Claims measure severity among claimants: payments as a percentage of the coverage recorded on each eligible positive claim. The two columns cover 2005–2024 and 2009–2024, the second matches the premium-data period. The individual-claim depth gradients are -2.2% (standard error 0.846) and -1.6% (1.814) percentage points per foot, respectively. Overall the region sees fewer claims in the period post referendum, but the effects are not statistically significant in the period post 2009. Only the longer-window gradient, which includes Katrina claims, is statistically significant. The regressions retain 17,139 claim–referendum observations, against 437 observations from 2009–2024. Appendix Figure A.22 shows the binscatters and Figure A.21 the raw trends. These outcomes describe severity among claimants, they do not capture the probability of making a claim or total avoided losses. Claims dynamics use the same individual-claim severity and matched tract-year support as the table, grouping post-period years into 0–2, 3–5, and 6 onward, relative to the pooled pre-period.

It is not obvious how policy take-up would respond. More occupied houses and cheaper premiums could increase demand for insurance, while improved protection could reduce it – with the FEMA accreditation insurance is no longer mandatory. Table 9 suggests that policy take-up does not change significantly. Its pooled average coefficient is positive but imprecise, consistent with Vinnakota and Ziff (2023), who find that take-up increases once a levee is accredited.

Benefits of better flood protection extends to the entire district I now expand the binscatters to all tracts in the treated levee districts. Figure 8 shows the same four measures, absorbing referendum fixed effects. Premiums and claims are lower in the post period, especially on lower ground. The claims, as seen, are heavily dominated by Katrina, and no flood has been as severe in the sample, so the 2009-2024 sample doesn't reflect the same reduction.

Together, these results are consistent with expected reductions in flood damages contributing to the house-price response in the previous section.

Does the tax pay for itself by reducing insurance premia? There are many reasons to believe that the premium we observe does not measure the true price of flood risk, and the portion of the price response explained by premia reduction will be heavily understated. Not

[25]See here for documentation on the community rating system. FEMA recognizes the better levees and reduces the flood insurance premium for a community, by giving them a community wide rating that denotes how much risk they would have mitigated.

everyone buys the policy, take up is generally low.

Transferring the premium DDD of Table 9, in the same crest depth, to the full treated pre-period sample used in the simple price calculation, using the mean building coverage of $190,408 gives a projected saving of $143 per year, 120% of the tax. The tax effectively pays for itself, if we assume everyone in the district buys property insurance, but this is not true. Only 3.3% of the price gain can be explained by the premia savings in present value. The gain per dollar of annual premium saving is $467, against $37–152 in Ge et al. (2022) and Bellon et al. (2024). Observed premiums are not actuarially fair, so the implicit benefit is larger than the premium saving. NFIP premiums were set more than 50% below expected losses before Risk Rating 2.0 Congressional Budget Office (2017), Gruber and Solomon (2026), and households remain under-insured against the losses that do occur Sastry et al. (2024); at a 50% subsidy the fair-priced saving is about $285 per year, still only 6.6% of the gain in present value.

6 The political economy of the levee tax

The previous section documents that passing a levee tax leads to positive NPV within the district, and negative externalities. Within the district the tax leads to better economic outcomes. However, some of these referenda fail at the ballot. This section looks at one of the drivers of referenda passage, the risk profile of the district, and highlight the frictions associated with the current political setup to organize levee funds: uniform tax rate with unequal benefits, and it comes down to whether the median voter sees benefits outweigh costs.

Failure to pass the levee tax has important consequences. In Section 2 we highlighted the West Jefferson Levee District's failed referendum, and their subsequent changes to their planned levee lifts and maintenance schedule. Another example is this, where the St. James Parish realizes that the spillovers from the neighbor's new levee system is going to impact them during the next storm. They took the tax increase to the ballot, and it was overwhelmingly rejected by the residents:[26]

> "The voice of the people was loud; they are not ready for more taxes... With St. John Parish's 16-foot levee being imminent and Ascension Parish working to protect their community of Sorrento, we are left with being a funnel for backwater. May God watch over and protect us as weather conditions continue to worsen."

To characterize the failed referenda I start with a conjecture that comes from the heterogeneous flood exposure within the region subject to the tax. If the median voter is not the *average* beneficiary, they may not vote for the levee. In exposure terms, this means the median voter might live on higher ground than the average.

To test this hypothesis, I obtain elevation data for the entire housing stock recorded in Corelogic's most recent property deeds data. I then compute the median and mean elevation for each district. To proxy for the median voter, different from the median house in the housing stock, I superimpose the voting precinct boundaries on these properties, and weight the property level exposure within each precinct by the number of people that turned out to vote for that referendum - which sections of the distribution of flood exposure were actually represented in the referendum. With this information, I compute a 'weighted median' flood exposure, the closest to the median voter in the referendum. Figure A.3 maps the voting precincts, crest depth, and referendum approval. Data Appendix **??** explains the parcel-stock measures and turnout weights.

[26] Original article is here

For most failed referenda, the median voter is on higher ground than the average house Figure 9 compares the elevation of the turnout-represented median voter with the mean elevation of the residential parcel stock. Negative values place that median below the average parcel, indicating greater exposure by this elevation proxy. The grey lines show the parcel-stock median minus its mean, separating the settlement pattern from the turnout reweighting. Chenier Plain is the one outlier among the districts that always failed, where about 3% of their properties are at very high elevation, pushing up their mean. Appendix Figure A.27 shows the underlying pooled distributions, where the 'peak' is to the left for the passed districts, somewhat bi-modal for the districts that had mixed results, and slightly toward the left for the failed districts.

Local median vs mean effects survive with other precinct level characteristics Appendix Table A.20 compares the precinct's median residential-parcel elevation with the district-election stock mean, directly linking the precinct regression to the stock benchmark. On its own, the precinct-by-precinct difference from the mean doesn't predict support for the levee tax. However, the within-precinct housing stock median - mean gap supports the claim. Table 10 suggests that if the median voter is at higher elevation than the average within a precinct, for every meter of difference, the support is 2-3% lower for the levee tax. Columns (1) and (2) include only the median-minus-mean elevation gap. Columns (3) and (4) include measure of salience (proximity to a levee, an indicator for a levee crossing the precinct), experience (historical NFIP paid losses relative to coverage), and political leaning (Democratic share within precinct) and other characteristics like income, registered-voter density (proxy for urban-ness), education, renter share and FEMA's National Risk Index - social vulnerability. Every column absorbs referendum fixed effects, alternating columns weight by ballot-day turnout. Construction of the tract-based precinct covariates and claims-history measure is described in the Data Appendix **??**.[27] The correlations with the median voter gap remain significant at the 5-10% level within the precinct, but other factors are of similar economic magnitude and significance. Note that the democratic voter share does not predict the voting patterns in these referenda. This analysis is suggestive evidence that other factors govern voting behavior beyond raw exposure.

If voters were rational, in principle the only predictor would be the median vs the mean elevation. It is an inbuilt inefficiency - the uniform tax with unequal benefits. However, there are reasons other reasons why. People don't like more taxes, do not recognize the benefits due to lack of awareness, do not trust that the money will reach the levees.[28]

[27] I omit the residential-parcel share within precincts in the Special Flood Hazard Area (SFHA): requiring at least 95% coverage by the available FEMA maps would exclude 198 of the 1,724 estimation observations

[28] Chen and Cohen (2024) even suggest a gap in the translation of the tax to revenues through differences

Back-of-the-envelope: the forgone returns To quantify the cost of the failed referenda, I use the house price results and quantification of Section 4.1 (0.321 log points at the treated mean depth of 18.06 ft, 0.0294 per foot of depth) and apply it to the parcel stock of the six failed proposals. Each parcel is valued at the pre-vote price of a house at its elevation in its own district, fitted on the five years of sales before the vote, and taxed at the ballot millage over the same horizon as Section 4.1. Passage would have raised values by $904m in Lake Borgne and $900m in Plaquemines, whose stock sits 16–18 ft below a surveyed crest, and by $59m in St. Tammany, but lowered them by $54m in Chenier Plain, $362m in Iberia and $59m in St. James: $1,389m in total against $309m of levy in present value. Iberia and St. James have no built system, but Chenier has a very small section of levees, with plans for a much larger scale levee system. The depth measure used for these regions is based on the planned height of the levee the vote would have provided a funding base for (see Table **??** in Data Appendix **??**). From the passed referenda, the property price response is 32.1% at 18.06 ft of depth and falls 2.94% per foot, so it reaches zero at $18.06 - 0.321/0.0294 = 7.1$ ft and is negative for any property above that. 54–78% of the stock in these three districts is at elevation higher than that threshold.

Taking stock: So far, we have established that investing in the levees clearly has huge returns. Despite this, there are districts that fail to pass the tax to facilitate levee investments. We collect evidence to understand why this might be the case, and find that the median voter may not be as exposed to flooding, and that they voted rationally, but that there could be many other reasons why. The failed districts have not experienced as many flooding episodes historically, it is a more recent phenomenon, and households have not seen any levees built in the newer districts that are trying to raise the tax.

Another view is that these levees, in the long run with increasing climate risk, are a friction to effective adaptation: where moving to higher, dryer land is the superior strategy Bilal and Rossi-Hansberg (2023). So maybe the voters were rational, beyond the political structure and referendum frictions. This is not something we can easily assess without thinking about how the voters would value the region: to understand what is the limit upto which households would be willing to pay to live in a region vs leave. The previous extrapolation suggests that some of the districts rationally reject the tax, but is it because it was too expensive for everyone, or is it because the tax scheme was inefficient? In these cases, the financing structure is all the more important to assess and it will be important and useful think through as the forces outside of just elevation/ risk exposure - given we know about moral hazard concerns (with the commitments - Hsiao (2025), Balboni (2024)) is there a way we can account for various reasons why people may have still voted rationally.

in property assessment cycles relative to the market.

This is where the model helps us think through some of these forces.

7 Model

The empirical estimates measure the returns to protection when the referenda passed: house prices appreciate in the regions most exposed to floods, and occupancy increases. This section develops a spatial model of a levee district in the spirit of Rosen (1979) and Roback (1982), with freely mobile households, endogenous property values, and tax-financed flood protection. This helps us think about location choice in the face of flood risk, along with the various factors that govern the efficacy of investing in flood protection. This also allows us to think through how costly the median voter constraint is with the current voting protocol, and how much households would be willing to pay in the face of increasing flood risk.

Motivation The empirical section establishes that passing the levee tax raises the value of protected properties, with a gradient along various measures of flood exposure - the higher the exposure, the higher the protection benefit. The overall gains within the district are positive: net capitalization per dollar of the modeled residential levy is about 36× (back-of-envelope in Section 4.1). We also know from the previous section that many referenda also fail, despite the large returns.

When we project the back of the envelope calculations to the failed districts, we rely on the representativeness of the passed referenda. The failure is evaluated at the median passed tax rate, and at the pooled average revenue generated. Beyond this projection from the revenue side assumption, it assumes similarity on various other dimensions that are key in understanding how to translate the benefit cost analysis: the baseline flood risk, past flood experiences, underlying technology used to construct the levees – joint with the Army Corps or with federal government support, or locally constructed, and the translation of dollars raised through the levee taxes to better levees.

These assumptions become important considerations beyond just the counterfactual pathway of the failed districts. Each of these factors become important when we think about the longer term investments into a location faced with increasing flood risk, as the baseline expected damages shifts upward and makes it less viable to live in the riskiest areas without flood protection - specifically, it starts to become too expensive for households to live behind a levee, if it needs to be much higher to protect them from floods.

Added to this is the constraint of the median voter, whose benefit-cost ratio dictates the passage of a levee tax, that needs to increase to meet the increasing costs of preserving the higher levee. In Section 5, we see correlations in voting patterns with characteristics

beyond just geography or flood exposure. This suggests that there are more factors govern how the median voter may perceive the benefits and costs of the levee tax, and warrants thinking deeply not just about the location choice of households in flood prone areas, but of the various underlying forces that shape a decision to invest in this public good.

The model asks three questions. (i) For each rejected proposal, what was the counterfactual price response had it passed? (ii) Does a majority of pre-policy voters gain at the proposed rate, and how much additional residential value could be reached at the planner's optimum or at the best rate a majority accepts? (iii) How do these comparisons change with higher expected damage, lower protection effectiveness, or a tax that is proportional to exposure?

7.1 Environment

This is a model of residential location choice with two types of agents: households and a levee district government. Households can either live in the levee district or obtain utility ($\bar{V}$) elsewhere. Living in the district provides a common benefit (h), capturing locational benefits (amenity value), but households face flood damage, housing costs, and property taxes.

Flood exposure within the region Properties in a levee district are indexed by elevation j, with distribution $F(j)$ across the district. J is the district's flood-protection threshold: the levee crest height. Properties below J are exposed, with exposure $(J-j)_+$ in feet. properties at or above J have no exposure. Each property delivers housing value h per period. The housing stock is fixed.[29]

Expected flood damages Expected flood damage declines linearly with elevation up to a threshold, and exponentially with the district's levee revenue:

$$D(j, R) = \Phi_0 \, e^{-\gamma R(\tau)} \, (J - j)_+. \tag{4}$$

Φ_0 is baseline expected annual damage per foot of exposure. It combines the probability of flooding and the damage when it happens. R is annual levee revenue per mile of levee, and γ converts revenue into protection. $R = 0$ is the inherited, pre-referendum levee system, not a district without levees: all revenue in the model is incremental to what the district already collects.

[29]The empirical section finds no adjustment of the housing stock around the passed referenda, but occupancy responds. Fixed supply is the data-consistent case and implies full capitalization into house prices.

The exponential form is motivated by the approximately exponential decline in the probability that extreme water levels exceed a given level of effective protection (van Dantzig (1956)). From an engineering point of view, for a levee to be higher, it also needs to be wider, so the costs are convex, which is parsimonious with the exponential form. Appendix Figure A.29 plots pre-referendum annual NFIP payouts per dollar of insured coverage against average tract elevation. Insured losses decline with elevation and are generally small at higher elevations. We summarize this pattern with the parsimonious exposure function $(J - j)_+$, treating its linearity and threshold as modeling assumptions.

γ is a reduced form dollars-to-flood protection parameter. It combines the effectiveness of flood-protection investments with the district's ability to translate revenue into maintenance, levee lifts, filling the cracks, mowing the lawns, and other risk-reducing inputs. We do not observe this parameter directly and calibrate it using the empirical estimate.

Households. Identical, risk-neutral households are freely mobile within and across districts, with outside option $\bar{V}$. A household at elevation j gets utility

$$u(j) = h - D(j, R(\tau)) - (1 + \tau)P(j),$$

housing value less expected damage less the annual cost of the house including the levee tax. τ is the levee tax per dollar of annual housing cost, so $(1 + \tau)P$ is what a household pays each year. A property is occupied if and only if its price is nonnegative; j^* is the elevation at which the price is zero.[30] There is no mobility/ sorting within the district, the margin of entry/exit is only to the outside option if prices go negative. Also, with free mobility, households do not incur any costs to leaving.

Government. The district levies a uniform tax τ on occupied properties and spends the revenue on L miles of levee. The budget balances – all revenues generated by the tax are spent on levee upkeep.

$$L\, R(\tau) = \tau \int_{j:\, P(j)>0} P(j)\, dF(j). \tag{5}$$

7.2 Equilibrium

Given a tax τ, an equilibrium is a price schedule $P(j)$, revenue per mile $R(\tau)$, and an occupancy cutoff $j^*(\tau)$ such that

[30] In the estimation, we convert observed house prices into the annual user cost of housing using $\rho = 0.05$, so the annual cost is $(\rho + q)P^{asset}$ where q is the effective annual tax rate per dollar of market value. The model's $\tau = q/\rho$ and $P = \rho P^{asset}$.

(i) *spatial indifference* on occupied properties: $h - D(j, R(\tau)) - (1+\tau)P(j) = \bar{V}$ for $j \geq j^*$;

(ii) *indifference at the occupancy margin*: $P(j^*) = 0$;

(iii) *budget balance*: equation (5), with D given by (4).

Condition (i) is the compensating-differentials condition: prices absorb all cross-sectional variation in flood risk and taxes, so identical households are indifferent across occupied locations. Condition (iii) makes the tax base endogenous: it consists of the property values that protection creates.

Prices and occupancy. Writing $A \equiv h - \bar{V}$ for location value net of the outside option,

$$P(j;\tau) = \max\left\{\frac{A - \Phi_0 e^{-\gamma R(\tau)}(J-j)_+}{1+\tau},\, 0\right\}, \qquad j^*(\tau) = J - \frac{A}{\Phi_0}e^{\gamma R(\tau)}. \tag{6}$$

Prices fall linearly with exposure and are divided by $(1+\tau)$: the tax is capitalized like a permanent claim on the property. The cutoff is where expected damage exhausts location value. More protection lowers the cutoff, $dj^*/dR = -\gamma(J - j^*) < 0$.

Revenues. Substituting prices into the budget gives one equation in R:

$$R = \frac{\tau}{L(1+\tau)} \int_{j \geq j^*(\tau)} \left[A - \Phi_0 e^{-\gamma R}(J-j)_+\right] dF(j). \tag{7}$$

More revenue lowers damage, which raises prices and may bring properties back into occupancy, which enlarges the tax base and supports more revenue. The feedback slope is the extra tax revenue generated by one more dollar of revenue per mile.

7.3 Welfare and the Optimal Tax

Free mobility pins every household at utility $\bar{V}$. Investing in flood protection therefore is capitalized into property value rather than changing the equilibrium utility of the marginal resident. We further assume here that there is no behavioral distortions due to the tax, so there is no additional loss due to the tax borne by the households other than the direct tax burden.

The levee district's objective is to maximize aggregate land value. This is because prices on this 'immobile' factor capitalize the benefits of reduced flood damage and expanded occupancy net of the taxes required to finance protection. The planner chooses the tax rate

to maximize aggregate land value:

$$\max_{\tau \geq 0} W(\tau) \equiv \rho \int P(j;\tau)\, dF(j), \qquad P(j;\tau) = \frac{\left[A - \Phi_0 e^{-\gamma R(\tau)}(J-j)_+\right]_+}{\rho + \tau}, \tag{8}$$

subject to the balanced-budget condition

$$LR(\tau) = \tau \int P(j;\tau)\, dF(j). \tag{9}$$

Using this condition, the objective can equivalently be written as

$$W(\tau) = \int_{j \geq j^*(\tau)} \left[A - \Phi_0 e^{-\gamma R(\tau)}(J-j)_+\right] dF(j) - LR(\tau).$$

The planner therefore balances the benefits of reduced damage and expanded occupancy against the resources devoted to protection.

$$\gamma \Phi_0 e^{-\gamma R(\tau^*)} \int_{j \geq j^*(\tau^*)} (J-j)_+\, dF(j) = L. \tag{10}$$

Voting. Incremental taxes must be approved at the ballot. Voters are households with positive value at zero incremental tax, $A - \Phi_0(J-j)_+ > 0$. This electorate is fixed while candidate taxes are compared. A proposal τ passes if at least half the electorate weakly prefers $P(j;\tau)$ to $P(j;0)$. Each voter compares their own capitalization gain, $\gamma D(j)$ per dollar of revenue, with their own tax discount, $P(j)$ per unit of τ: properties near the crest gain little and pay the same rate, so they will not vote for the tax.

The median voter's optimal need not be the socially optimal Three tax rates matter. τ^* maximizes aggregate land value, which comes from the social planner. τ_{med} is the median voter's individually preferred rate, which need not equal the rate that maximizes aggregate value subject to majority acceptance. The welfare cost of the referendum constraint is aggregate land value at τ^* minus aggregate land value at the best majority-acceptable rate. The planner counts every property, including those made viable by protection. That difference, plus the fact that the median voter is less exposed than the value-weighted average property, is the wedge.

7.4 Estimation

The objective of the model is to evaluate the existing system of taxes for levee investments using observable data from the levee districts, and develop counterfactuals. In this section,

I explain how I translate the triple-difference empirical estimate of 3% per foot of elevation to the model to back out the implied, pooled technology measure γ. This then allows us to run counterfactuals for the passed referenda. Additionally, if we translate just this γ to the failed referenda, we can relax the requirement of these districts being exactly the same as the passed ones in all observable dimensions, and simulate their counterfactual house price response.

To estimate γ, I reconstruct the empirical triple difference setup, but use a counterfactual price for each of the properties in the sample based on the model. Specifically, I reconstruct the model-specified damage reduction factor with every gamma, for every property with its own characteristics, and run the same regression until the model-specified coefficient on the price difference post levee tax passage is the same as the empirical estimate.

I impose that the ratio of baseline expected flood damages per foot of depth normalized by the amenity value of the region (Φ_{0n}/A_n) is a fixed, and estimate this from pre-vote prices: the gradient of log price in depth, after hedonic controls, tract effects, and pair-year effects. γ is then so that the model's predicted treated-post price change, projected through exactly the empirical DDD design (controls, spillover group, tract and pair-year effects), reproduces the pooled crest-depth coefficient. District damage scales are recovered as $\Phi_{0n} = \psi A_n$ from prices; the descriptive NFIP claims comparison is not an independent calibration target. The estimation sample is the 38,930 sales of the main specification, 20,010 of them treated; Orleans supplies 61% of treated sales, so transferring ψ and γ to other districts assumes comparable capitalization elsewhere.

Appendix Tables **??** and **??** report the inputs used for each eligible referendum, ordered by district median elevation. They distinguish the effective household tax rate from the broader tax-base multiplier, and the ballot's reported expected revenue from revenue generated by the model's post-policy tax base. They also report miles of levee, the residential elevation distribution, the share outside built or planned protection, and the price and mean ground elevation of the baseline properties used to recover district location values. The assumed discount rate is $\rho = 0.05$, and the normalized damage sensitivity $\Phi_{0n}/A_n = \psi$ is shared across districts.

7.5 Validation

To validate the model, I do two checks. First, I compare the price of the average property in the triple-difference estimation from the data, with the model generated average price. Estimation targets just one moment, the triple difference coefficient. At the mean treated depth of 18.06 feet, the model gives 0.267 against 0.321 in the data (SE 0.043) from the

triple-difference estimation, which suggests that the absolute level gap is about 5%. By design, the model doesn't account for any of the general equilibrium effects associated with the better expected flood protection. The gap could potentially correspond to improvements in local economic activity and amenities with the increased occupancy.

Second, I compare the prices of the sub sample of repeat-sale properties, with the model generated prices. This means that the model applies 'treatment' based on each properties individual elevation, after accounting for the fixed effects and the hedonics. Figure 10 compares observed and predicted prices for the same 1,661 repeat-sale properties. Mean predicted post-sale price is $396,130, versus $405,428 observed. The results show that the distribution moves similarly overall (the average is very close to the DDD estimate). There could be many idiosyncratic reasons why prices individually may not line up with the model predicted one, which are not modeled explicitly.

7.6 Counterfactuals

I first compare the actual ballot tax with the rate that maximizes residential value and the best rate a majority accepts. Second, I return to baseline risk and change who pays by making the tax proportional to exposure. Third, I then increase expected damage, holding protection effectiveness and the tax rule fixed. Finally, I ask what district-specific protection effectiveness would make rejection of the observed uniform-tax proposals consistent with the model. This last exercise computes proposal-specific voting thresholds for γ.

Gains at the actual ballot rate. For each proposal, I compare passage at the requested incremental ballot millage with no incremental levy. Tables 11 and 12 report the resulting changes in aggregate residential value and each district's baseline optimal millage. Across the six districts whose latest eligible proposals failed, the baseline model implies about $0.17 billion in forgone housing value (2024 dollars), with positive gains concentrated in Chenier Plain and Lake Borgne and negative gains in some other districts. The extrapolated-gain column repeats the crest-depth transfer of Section 6 (Table **??**); this compares two predictions using different valuation bases, rather than providing independent validation of the model.

Optimal taxes and the majority constraint. Figure 11 first shows only the baseline, uniform-tax case. The circle marks the rate that maximizes aggregate residential value, the square the best rate accepted by a majority of pre-policy residents, and the vertical line the actual ballot millage. These are distinct from the median voter's individually preferred rate. In Orleans, the planner's and majority-acceptable rates nearly coincide, at about 48 mills. In Chenier Plain, the planner chooses about 13 mills, while the best majority-acceptable

rate is zero: most parcels lie outside modeled built or planned protection and would pay the uniform tax without receiving a direct protection benefit. Across the districts whose latest proposals failed, the majority constraint removes about 26% of the planner's aggregate gain; it is nearly nonbinding among the passed districts. This compares gains from optimizing the rate with the cost of the voting constraint, separately from gains at the actual ballot rate.

Exposure-proportional taxation. Figure 12 returns to baseline expected damage and holds fitted γ fixed, isolating the tax reform from the increase in climate risk. The tax schedule becomes $q_i = qx_i/\bar{x}_n$, where $\bar{x}_n$ is mean exposure across the full parcel stock, including zeros. Properties with zero modeled exposure pay no incremental levy; including them in the mean preserves the average rate across parcels, rather than total revenue. The baseline pre-policy electorate and one-vote-per-parcel rule are retained. The best majority-acceptable and planner-optimal rates coincide in all fifteen districts, eliminating the measured majority-constraint loss. In Chenier Plain, both rates are approximately 16.1 mills, compared with a zero majority-acceptable rate under baseline uniform taxation; in Orleans, both are approximately 48.7 mills. Zero-exposure properties are indifferent under the reform, and the model counts indifference as approval. This illustrates how allocating tax costs in proportion to modeled protection benefits can relax a financing constraint; it does not establish the distributional effects or administrative feasibility of that tax rule. Table 13 compares baseline uniform taxation with the separate changes in risk and the tax schedule, holding effectiveness fixed.

Higher climate risk. Figure 13 next doubles baseline expected damage, holding the uniform tax rule and γ fixed. This is a comparative-static increase in risk rather than a dated climate forecast. The mean planner-optimal rate rises from 89 to 145 mills in passed districts and from 71 to 163 mills in failed districts. The annualized value lost to the majority constraint in the failed group rises from about $13 million to $48 million. More risk increases the value of protection, but does not eliminate opposition among properties receiving no direct benefit. The pre-policy electorate is defined at the higher scenario risk and then held fixed across candidate rates.

Protection effectiveness and failed referenda. The final counterfactual asks how low expected protection effectiveness, γ, would have to be to rationalize rejection of each district's latest eligible failed proposal. I return to baseline flood risk and the actual uniform ballot tax, holding the protection footprint, housing stock, fiscal multiplier, other calibrated parameters, and pre-policy parcel electorate fixed. At each candidate γ, I jointly solve

property prices and tax-funded protection and compare each voter's property value with its value under no incremental levy. The cutoff is the lowest effectiveness at which at least half of voters weakly gain from passage; effectiveness below this cutoff predicts rejection, with equality counting as approval. Lake Borgne, St. James, Plaquemines, and West Jefferson would reject at effectiveness below 22.3%, 48.7%, 53.6%, and 7.3% of the fitted pooled γ, respectively. The West Jefferson calculation uses its March 2025 failure, preceding its later passage. Iberia already rejects at the fitted effectiveness. In Chenier Plain and St. Tammany, even eliminating all modeled flood damage yields approval from only 19.3% and 4.1% of voters, respectively, so no finite γ delivers majority approval. For these two districts, the constraint is the mismatch between the taxing jurisdiction and the protection footprint: most properties pay the uniform levy without receiving a modeled direct protection benefit. These calculations give conditional bounds on expected effectiveness rather than estimates of district technologies or a fit to observed vote shares. They hold tax incidence and the parcel-to-voter mapping fixed and exclude indirect benefits through infrastructure, employment, or protection of neighboring properties.

Discussion. The model connects the estimated capitalization response to a financing mechanism: an incremental levy funds protection, protection raises property values, and those values help finance the levy. It provides a common framework for evaluating each proposal at its actual ballot rate and for comparing the rate that maximizes aggregate residential value with the best rate accepted by a majority of pre-policy residents. The counterfactuals show how the distribution of protection benefits and the allocation of tax costs can create a financing constraint even when aggregate residential gains are positive.

The model but abstracts from many considerations that become important as we assess policy in these regions: income-dependent consumption costs, risk aversion, moving costs, endogenous wages and amenities, and firms' location decisions. It also omits the hydraulic spillovers documented in the empirical analysis, borrowing and construction dynamics, and changes in insurance or federal support.

8 Conclusion

In this paper, I estimate the returns to levee investments through Louisiana's levee districts. I find that levee tax passage raises protected property values, with larger gains at greater flood exposure. The tax pays for itself, for the average household along the boundaries, they gain $36 in property value per dollar of tax. Occupancy rates increase in the high exposure regions, and the flood insurance premia is lower, reflecting the expected lower flood damages.

Despite these high returns, levee referenda fail. I find that there is significant heterogeneity within the districts in terms of their flood exposure. Through the model, I show that there is money left on the table for most of the districts that failed the referenda, and find that the political economy matters in two of the districts – the median voter doesn't see benefits outweigh costs. The median voter constraint bind across most districts if we increase the climate risk, highlighting the necessity of better policy as these communities face increasing risk going forward.

The model allows us to open up a conversation on optimal policy, and suggests many avenues future research: (i) characterizing the "who" – who are the households at lower elevation in these districts and what are their constraints in the face of increasing or proportional taxes, which are necessary considerations for policy design for these districts, (ii) characterizing the outside option, for communities across the world where moving inland is not easy to do.

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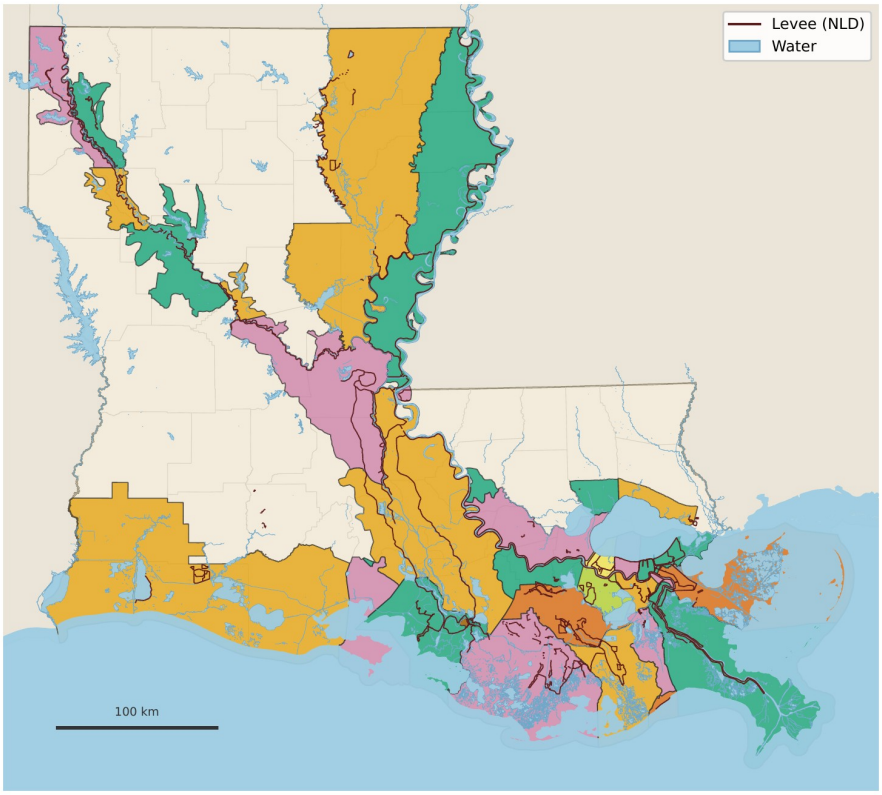
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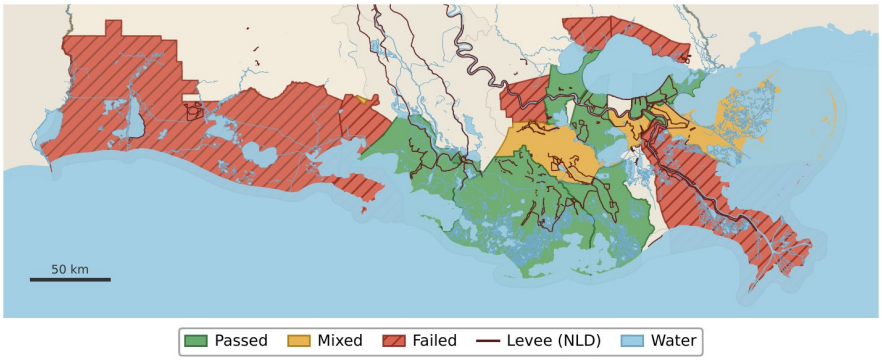
Figures & Tables

Figures

Figure 1: Louisiana's Levees and Levee Districts



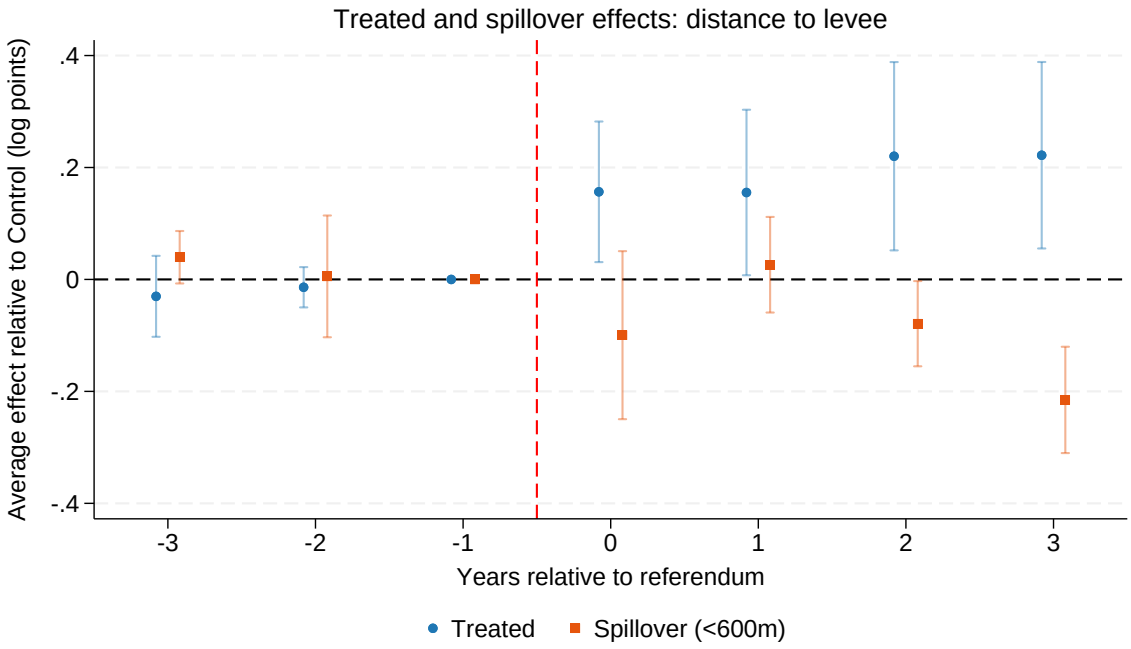
(a) Levee districts and levee (NLD) lines, statewide



(b) Districts that held a levee tax referendum, by outcome

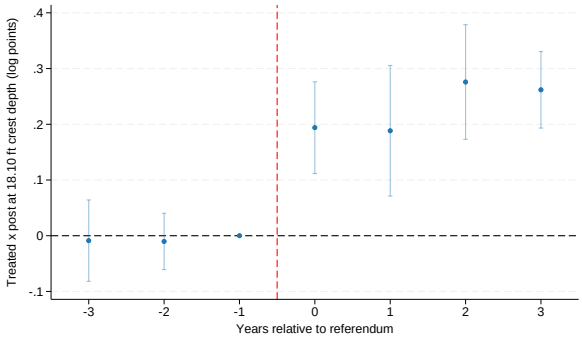
Notes: Panel (a) a digitized map of Louisiana's 33 levee districts, along with their corresponding levees from the National Levee Database, spanning the state. Panel (b) focuses on the districts that held at least one levee tax referendum, shaded by their referendum record over **2010–2025**: every measure passed (green), every measure failed (red), or a mix of passing and failing measures (amber).

Figure 2: Event Study: Average House-Price Effects of Levee Tax Passage

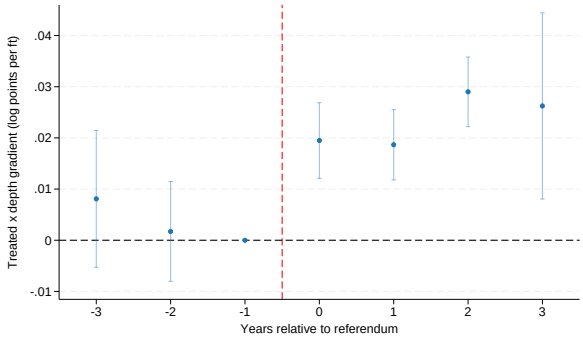


Notes: Dynamic coefficient plot for the average treatment effect from the main regression specification (equation 1). Treated (blue circles) and neighboring spillover (orange squares) dynamics relative to neighboring controls. Spillover properties are within 600 m of the treated district's levee, remaining neighboring properties are controls. Bars show boundary-pair-clustered 95% confidence intervals.

Figure 3: Event Study: House price effects, by depth below levee crest



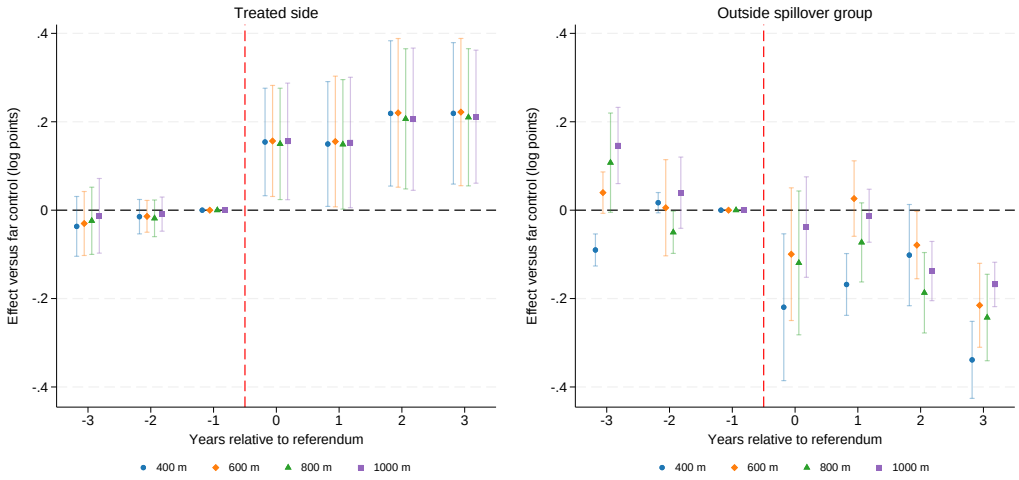
(a) Treated × Post level



(b) Treated × Post × Depth gradient

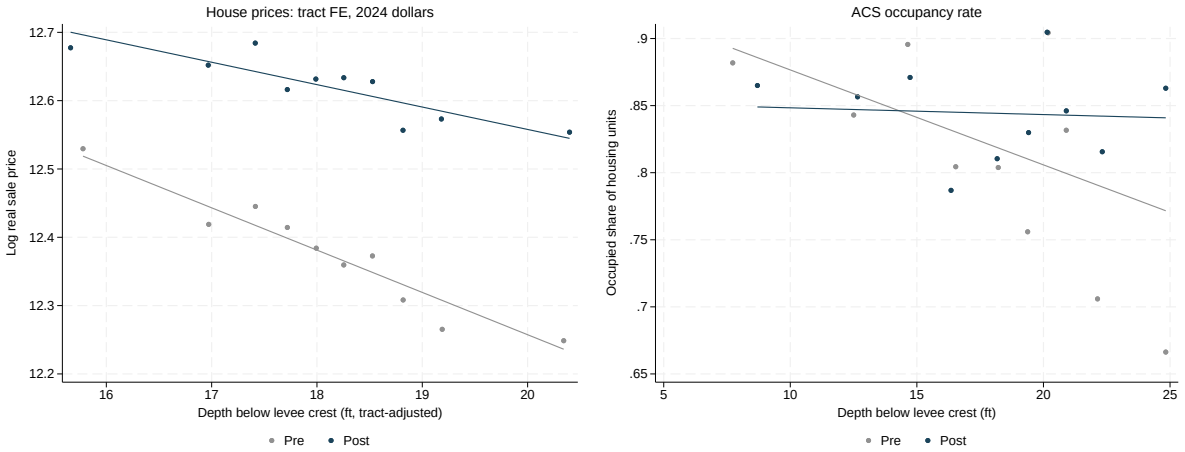
Notes: Dynamic coefficient plot for the triple difference specification (equation 2). The specification includes hedonic controls and tract and pair-by-year fixed effects and the main hedonic controls. Year $t-1$ is omitted. The left panel reports the treated level at the full-sample treated mean crest depth (18.10 ft), the right panel reports the treated depth gradient in log points per foot. Estimation uses event years -10 through $+10$; the plot displays -3 through $+3$. Bars are boundary-pair-clustered 95% confidence intervals.

Figure 4: Spillovers: Price Effects by Distance to the Treated Levee



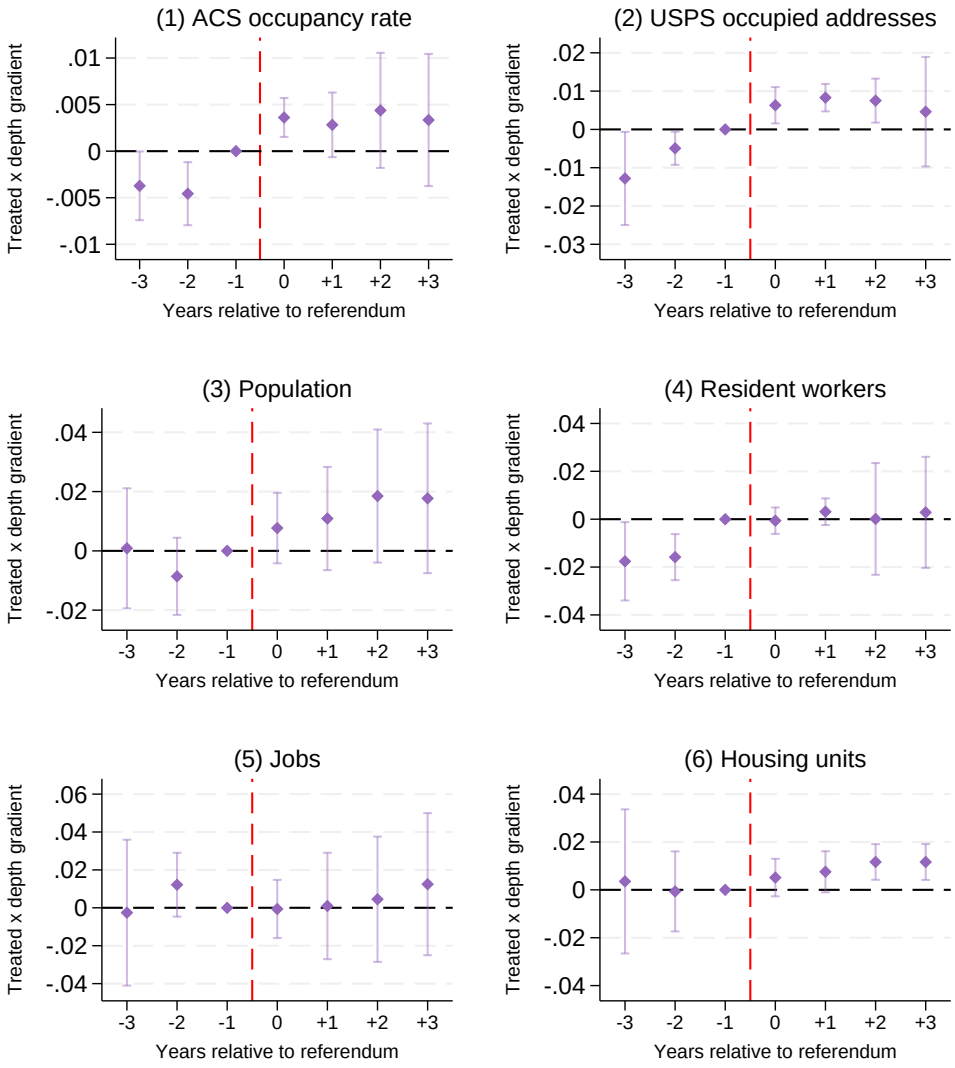
Notes: Dynamic coefficient plot for the average treatment effect from the main regression specification (equation 1), for treated properties (left) and neighboring spillover properties (right), relative to neighboring controls beyond the indicated cumulative cutoff from the treated levee (400, 600, 800, or 1,000 m). Bars are boundary-pair-clustered 95% confidence intervals. The spillover group is a small sample of properties and fairly unstable across specifications, but the average treatment effect is stable.

Figure 5: Pre- and post-referendum price and occupancy gradients



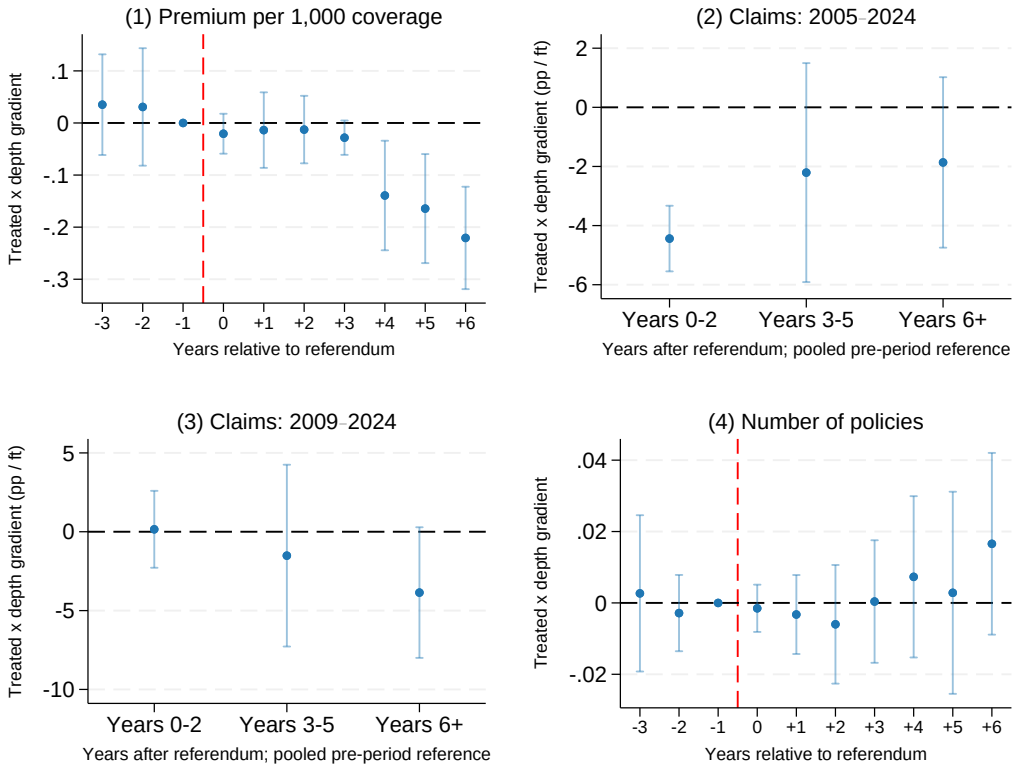
Notes: Treated-side observations and ten equal-count bins, with separate pre/post fitted lines; grey denotes pre and blue post. The left panel uses transaction-level log sale prices deflated to 2024 dollars with annual CPI-U. Census tract fixed effects are removed from both log prices and property-level crest depth, and overall means are restored for display. The right panel shows the ACS occupied share of housing units, one minus the vacancy rate, against tract-level crest depth, without fixed effects or centering. Both panels use the own-side district median crest and the main depth cap. No year effects or hedonic controls are included. Transactions and tract-years are equally weighted within their respective panels. This is a descriptive comparison.

Figure 6: Dynamic quantity responses across flood exposure



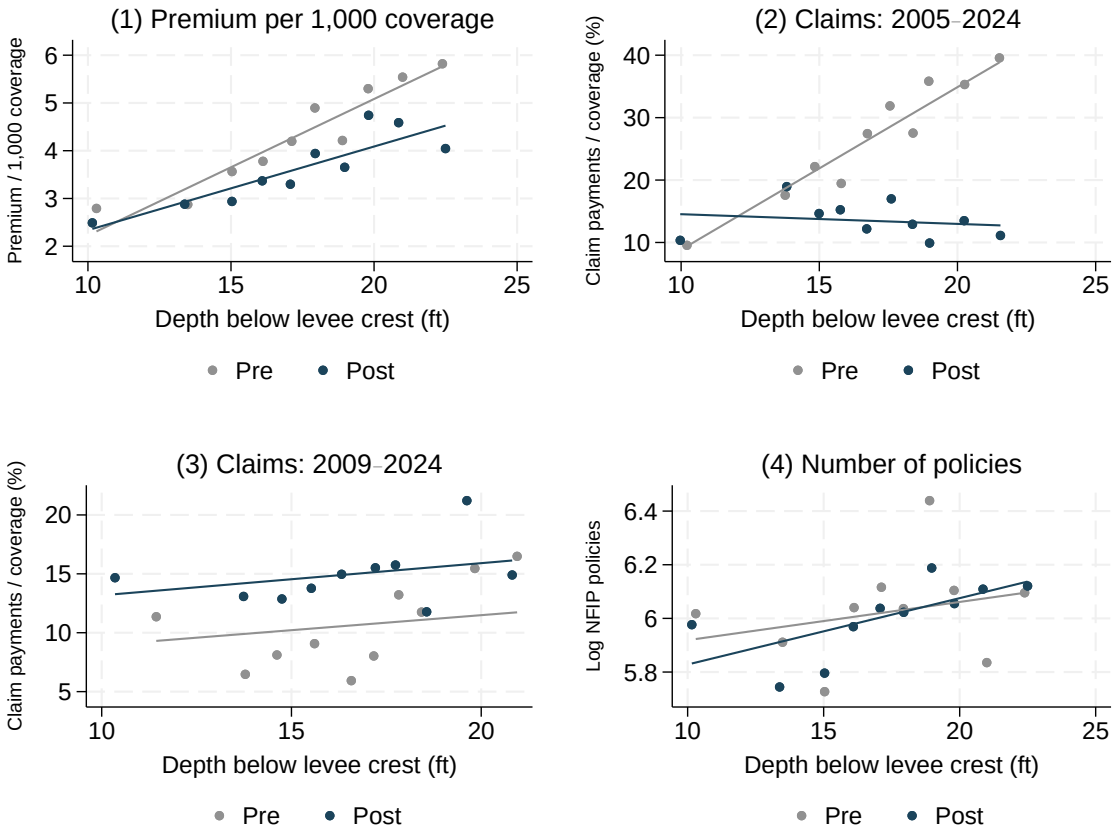
Notes: Dynamic coefficient plots from the tract level triple difference regressions from equation 3. Tract and pair-by-year fixed effects. pair-clustered 95% confidence intervals. Depth is measured in feet below the district median levee crest. ACS occupancy is one minus vacant housing units divided by total housing units. USPS occupied addresses are annual means of quarterly residential addresses net of no-stat and vacant addresses, in logs. Population and housing units are ACS five-year counts in logs. Resident workers and jobs are LODES counts transformed using the inverse hyperbolic sine.

Figure 7: Dynamic insurance responses across flood exposure



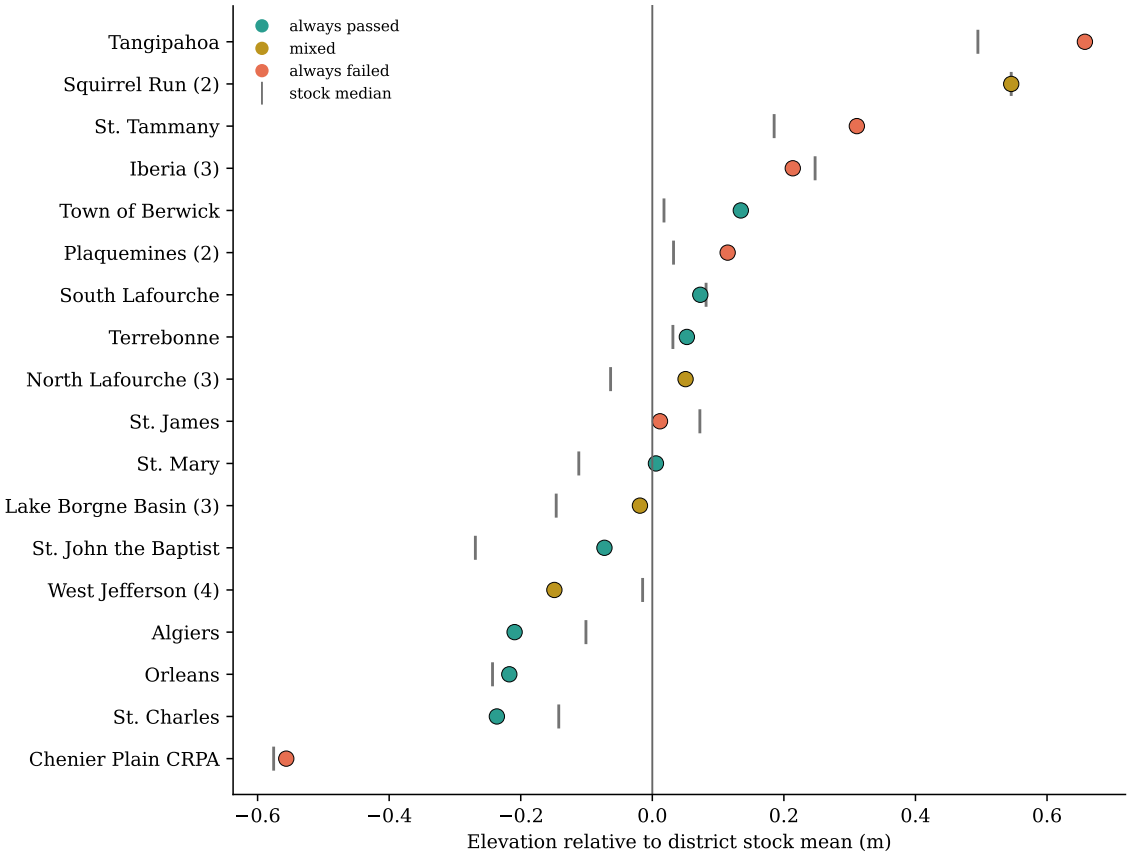
Notes: Dynamic coefficient plots for the National Flood Insurance Program policy level premia and claims data, to characterize the mechanism of reduction in expected (and realized) flood damages. Regressions are at the policy level for the premia and claims, individual positive claim records for severity, and at the tract level for policy counts. Pair-clustered 95% confidence intervals. Premium and policy-count panels use annual event indicators over -10 through $+10$, displaying -3 through $+6$, relative to $t-1$. Claims use the same individual payment-to-coverage ratios and tract–referendum–year support as Table 9, separately for 2005–2024 and 2009–2024. Claims periods are years 0–2, 3–5, and 6 onward, relative to all pre-period claims pooled. Claim eligibility and the common one-to-one tract assignment are described in Table 9 and Data Appendix **??**. Years without eligible claims are not assigned zero losses. Depth is measured in feet below the own-side district median crest. Grouping improves support but does not provide a test of annual pre-trends.

Figure 8: Insurance gradients across the treated districts



Notes: Treated-side observations, unweighted, with ten equal-count bins and separate pre/post fitted lines. Grey denotes pre-referendum and blue post-referendum. Both axes are residualized on the indicated fixed effects with their pooled means added back; the exposure measure is depth below the own-side district median levee crest; no year effects are removed. These plots are descriptive. Referendum fixed effects are absorbed. The universe includes all tracts in each treated district. All four panels are tract-year measures; premium is total premium divided by total building coverage, times 1,000. Claims retain the tract-year severity ratio of Appendix Table A.17; the policy-count definition matches the main table.

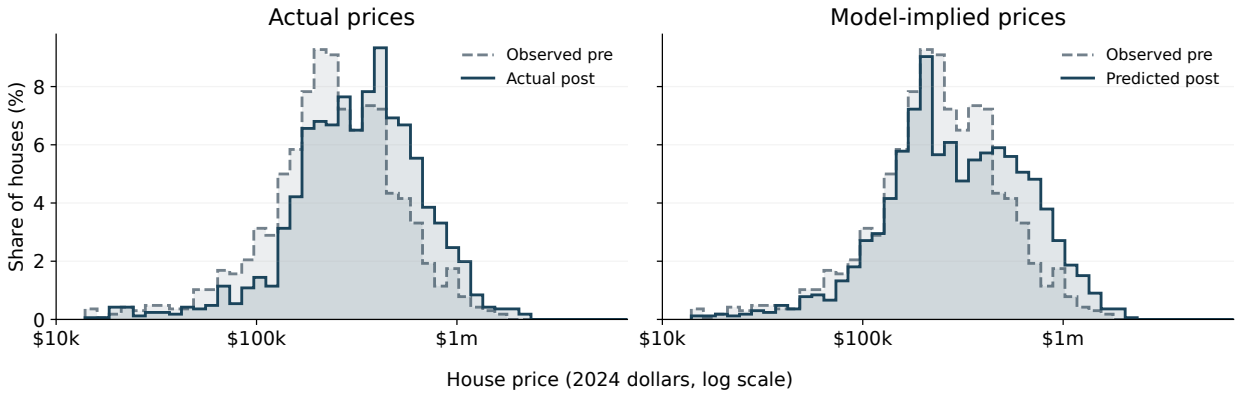
Figure 9: The median voter's elevation relative to the district average



Notes: Dots are turnout-represented median elevation minus mean residential-parcel elevation, in meters. Turnout is allocated equally across the precinct's residential parcels with observed elevation. Grey ticks show the unweighted parcel median minus the parcel mean. A negative gap indicates lower elevation than the average parcel. Each parcel receives equal stock weight regardless of dwelling count or value. Repeated referenda are averaged within district, using the same elections for both marks; parentheses give the number of referenda where greater than one. Colors classify districts using official outcomes, including early votes, across the 2010–2025 sample. Parcel allocation uses matched geographic precincts and ballot-day turnout, so it does not locate early voters. All districts, including Chenier Plain, are retained. This is a descriptive proxy for voter exposure.

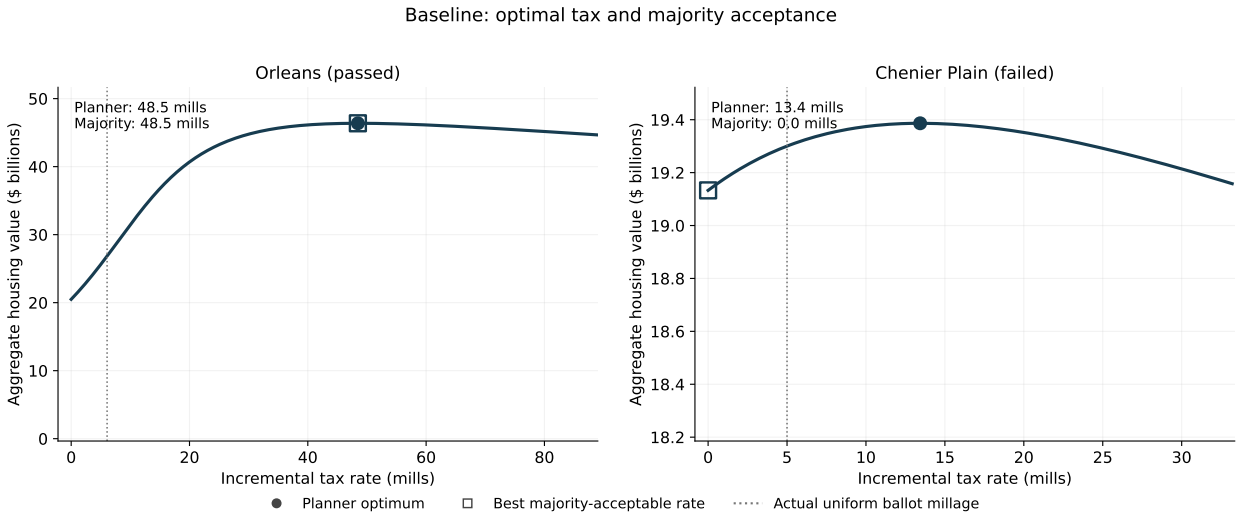
Chenier Plain: The high-elevation tail raises the mean to 4.59 m; the represented median is 4.04 m. Appendix A.17, Figure A.28, shows the full distribution and its upper tail.

Figure 10: Observed and predicted prices for repeat-sale properties



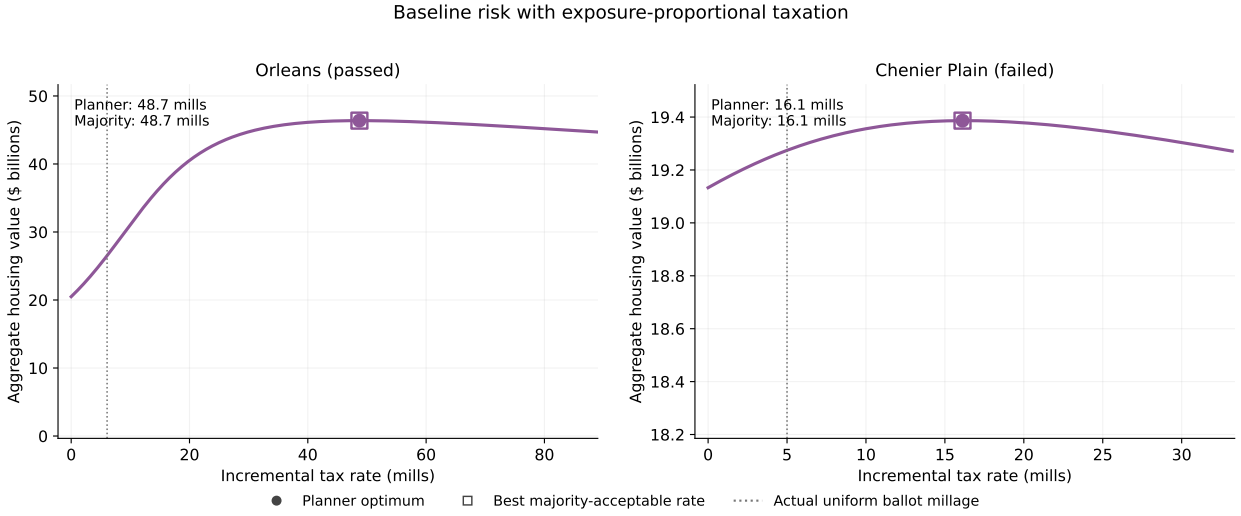
Notes: Each distribution contains the same 1,661 treated houses. Predicted post-sale price equals observed pre-sale price times the fitted non-policy price factor (adjustments for hedonics, tract fe and pairxyear trends) times the model policy factor. Prices are in 2024 dollars, both panels use common bins and logarithmic axes. The model implied prices seem to underestimate the price response on the lower end, and overshoot for the few properties at the top end of the distribution. On average, it predict a slightly lower price response, potentially missing out other benefits associated with the better levees (agglomeration, boosted local economic activity, neighborhood revitalization).

Figure 11: Optimal taxes and majority acceptance



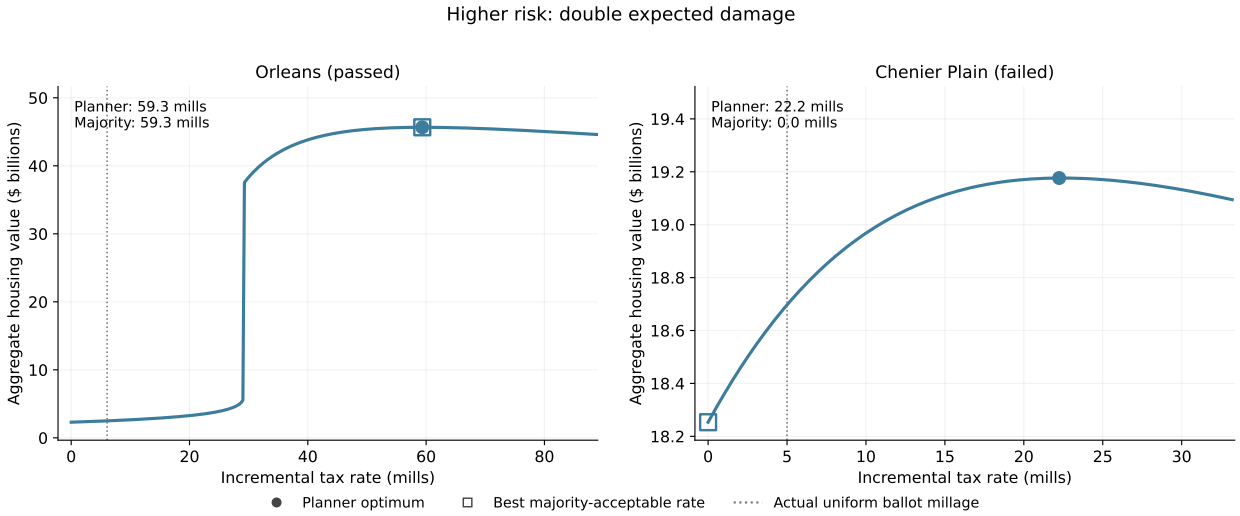
Notes: Aggregate residential value in one-time dollars. Circles denote the planner's optimum, squares the best majority-acceptable rate, and dashed vertical lines the actual ballot millage. The electorate is fixed at pre-policy occupants within each scenario. Only the baseline uniform-tax case is shown. Prices and funding are solved jointly from zero incremental funding. The following figures separately change risk and the tax schedule relative to baseline, holding fitted protection effectiveness fixed.

Figure 12: Exposure-proportional taxation at baseline flood risk



Notes: Expected damage is at baseline and γ remains at its fitted value. Only the tax schedule changes: $q_i = qx_i/\bar{x}_n$, with the full-stock mean including zeros. The baseline pre-policy electorate and one-vote-per-parcel rule match Figure 11. Zero-exposure properties pay no incremental tax and indifference counts as approval. The horizontal axis converts the average full-stock effective rate to equivalent mills using the same district taxable share; the dotted line is the original uniform ballot millage, not an observed proportional-tax proposal. Markers and axes follow the preceding figures. All housing values are one-time 2024 dollars.

Figure 13: Higher expected flood damage under uniform taxation



Notes: Baseline expected damage is doubled; fitted protection effectiveness and the uniform tax schedule are unchanged. Circles denote planner-optimal rates, squares the best majority-acceptable rates, and dotted lines actual uniform ballot millage. The electorate consists of parcels with positive no-levy value at this higher risk and stays fixed across candidate rates. Axes match Figures 11 and 12. All housing values are one-time 2024 dollars.

Tables

Table 1: Levee Tax Referenda in Louisiana, 2010–2025

Date	Levee District	Tax	Rate	Yes (%)	Result
2010-05-01	St. Mary Levee District	Property	0.5%	66.7	Passed
2011-10-22	Lake Borgne Basin Levee District	Property	0.3%	61.2	Passed
2012-11-06	Orleans Levee District	Property	0.607%	61.3	Passed
2012-12-08	North Lafourche Conservation Levee and Drainage District	Sales	1%	45.8	Failed
2012-12-08	Terrebonne Levee and Conservation District	Sales	0.5%	71.5	Passed
2013-04-06	Iberia Parish Levee, Hurricane and Conservation District	Property	0.5%	49.8	Failed
2014-12-06	Lake Borgne Basin Levee District	Property	0.75%	38.9	Failed
2015-05-02	Lake Borgne Basin Levee District	Property	0.75%	41.6	Failed
2015-05-02	North Lafourche Conservation Levee and Drainage District	Sales	0.25%	57.0	Passed
2015-05-02	St. Charles Parish Government	Property	0.4%	74.3	Passed
2015-11-21	Algiers Levee District	Property	0.635%	54.1	Passed
2015-11-21	West Jefferson Levee District	Property	0.55%	41.6	Failed
2017-03-25	South Lafourche Levee District	Property	0.5%	90.3	Passed
2017-04-29	St. John the Baptist Parish Government	Property	0.7%	62.8	Passed
2017-10-14	North Lafourche Conservation Levee and Drainage District	Property	0.6%	64.8	Passed
2017-11-18	Iberia Parish Levee, Hurricane and Conservation District	Property	0.5%	27.0	Failed
2018-04-28	West Jefferson Levee District	Property	0.475%	59.6	Passed
2019-05-04	St. James Parish Government	Property	0.7%	18.8	Failed
2020-08-15	Town of Berwick	Property	0.6%	82.3	Passed
2020-12-05	Plaquemines Parish Government	Property	0.5%	43.7	Failed
2021-11-13	Chenier Plain Coastal Restoration and Protection Authority	Property[a]	0.5%	40.2	Failed
2021-11-13	Iberia Parish Levee, Hurricane and Conservation District	Property[a]	0.5%	30.1	Failed
2021-11-13	Plaquemines Parish Government	Property	0.5%	43.7	Failed
2021-11-13	Squirrel Run Levee and Drainage District	Property[a]	0.5%	37.4	Failed
2021-11-13	St. Tammany Levee District	Property[a]	0.5%	39.0	Failed
2021-11-13	Tangipahoa Levee District	Property[a]	0.5%	25.5	Failed
2021-12-11	Squirrel Run Levee and Drainage District	Property	3%	69.3	Passed
2025-03-29	West Jefferson Levee District	Property	0.475%	37.2	Failed
2025-11-15	West Jefferson Levee District	Property	0.475%	80.8	Passed

Notes: This table lists all levee district tax ballot measures from 2010–2025 from the Louisiana Secretary of State website. *Yes* is the share of yes votes within the district. Of the 29 measures across 18 districts, 14 passed and 15 failed. [a]The five 2021 measures are a single constitutional amendment (a property tax) voted on across the state: the Yes share is calculated based on the district-specific precinct votes.

Table 2: Sample Construction

Restriction	Transactions	Properties	Pairs
All boundary transactions	1,345,609	179,280	85
Passed referenda	816,316	164,639	44
Single-pass districts	756,687	163,565	34
No subsequent referenda	674,253	160,883	31
Excl. Algiers	563,795	159,508	27
Post-2008	436,933	122,107	27
Control not yet treated	242,015	97,907	18
In leveed area	233,755	92,436	18
Single-parcel sales	226,592	89,409	18
2354m CCT bandwidth, unique pair	38,947	29,595	16
Singleton FE cells dropped (estimation sample)	38,930	29,579	15

Notes: This table reports the sample construction using CoreLogic single-family and condominium transactions of properties in the boundary-pair universe, one row per filter in Section 3. *Passed referenda* keeps pairs whose treated district passed the levee tax: 13 passed referenda across 12 districts in the boundary universe (North Lafourche passed twice). *Single-pass districts* drops districts with more than one passed referendum within the main analysis event roster. *No subsequent referenda* drops districts that held a later referendum, pass or fail, within the main analysis event roster. *Excl. Algiers* removes federal project completion confounding in its pre period, *Post-2008* excludes earlier sales because of sparsity and Hurricane Katrina recovery. *Control not yet treated* drops pairs whose control district had passed its own tax on or before the treated vote and censors later-treated controls at their own pass year. *In leveed area* keeps properties protected by a levee according to the National Levee Database. *Single-parcel sales* drops multi-parcel deeds. The final row applies the 2,354 m Calonico et al. (2014) MSE-optimal bandwidth and drops any parcel lying within the bandwidth of more than one of its own district's boundary pairs, yielding 8 passed referenda in 8 districts and 16 boundary pairs. The last row drops sales that are alone in an absorbed tract or pair-year cell, which the fixed-effect estimator cannot use: 17 sales, including all five sales of one Terrebonne pair, so 15 pairs survive as clusters. This is the main static hedonic estimation sample and the sample for the pooled summary statistics; restricted and repeat-sales specifications report their own sample sizes.

Table 3: Summary Statistics

	Mean	SD	p25	Median	p75	N
Sale Price (thousands)	271.62	254.55	128.80	205.00	335.00	38930
Elevation (ft NAVD88)	1.32	5.20	-3.63	1.29	5.01	38930
District Median Levee Crest (ft)	18.49	2.53	15.86	19.70	19.70	38930
Depth Below District Crest (ft)	17.17	5.49	13.54	17.42	21.04	38930
Living Sqft	3291.61	1922.36	1856.00	2830.00	4228.00	14098
Lot Size (sqft)	8339.45	14988.03	3600.00	5779.00	7425.50	38920
Stories	1.25	0.43	1.00	1.00	1.50	13030
Property Age (years)	67.33	39.50	42.00	66.00	94.00	13409
Dist. to Boundary (km)	1.32	0.63	0.82	1.36	1.86	38930
Dist. to Levee (km)	1.29	2.24	0.48	0.91	1.46	38930
Riverine Flood Freq. (events/yr)	1.67	0.44	1.75	1.75	1.96	38930
Coastal Flood Freq. (events/yr)	0.39	1.05	0.01	0.01	0.01	38930
Combined Flood Freq. (events/yr)	2.06	1.06	1.76	1.76	1.97	38930
SFHA	0.17	0.38	0.00	0.00	0.00	38930
Flood Zone (SFHA $\cup$ Levee X)	0.76	0.43	1.00	1.00	1.00	38930

Notes: Summary statistics for the estimation sample (Table 2), pooled across treated, spillover, and control properties. *Sale Price* is the CoreLogic transaction price. *Elevation* is the parcel's ground elevation from USGS 3DEP 1/3 arc-second (≈10m resolution) data, in feet NAVD88. *District Median Levee Crest* is the median crest elevation (CPRA survey, NAVD88 feet) of the levee systems enclosing the property's own levee district, taken over the district's protected parcel stock; where a passed referendum funded a system not yet built, the authorized design crest is used (see Table **??** in Data Appendix **??**). *Depth Below District Crest* is the positive part of that crest minus the property's elevation, capped at the district's 99th percentile of positive exposure. It is the continuous exposure measure used in the crest-depth specification and is zero for properties above their district's crest. Structural variables (living area through property age) are from CoreLogic assessments and reported conditional on non-missing; living area and lot size are winsorized at the 1st/99th percentiles of the full statewide CoreLogic sample, and lot size is heavily right-skewed. *Dist. to Boundary* is the distance to the pair's reference boundary and *Dist. to Levee* the straight-line distance to the nearest NLD levee centerline of any district. *Riverine* and *Coastal Flood Freq.* are FEMA National Risk Index tract annualized event frequencies (events/yr); *Combined* is their sum. *SFHA* is the current FEMA Special Flood Hazard Area indicator. *Flood Zone* equals one for properties in an SFHA or in levee-protected Zone X (would be SFHA without the accredited levee).

Table 4: Summary Statistics by group

	Treated		Neighbor		
	All	≥1 SD deeper	Spillover	Control	Deep−All
Sale Price ($K)	206.75	185.04	192.74	191.34	-21.71
Elevation (ft NAVD88)	0.37	-5.51	-2.12	3.94	-5.87
District Median Levee Crest (ft)	17.88	19.70	20.14	18.08	1.82
Depth Below District Crest (ft)	17.51	25.20	22.20	14.15	7.69
Living Sqft	3803.75	4179.35	2639.71	2633.40	375.60
Lot Size (sqft)	8083.44	6669.48	6812.29	9608.73	-1413.97
Stories	1.25	1.31	–	1.13	0.06
Property Age	72.59	49.99	57.00	48.53	-22.61
Dist. to Boundary (km)	1.30	1.24	0.38	1.42	-0.07
Dist. to Nearest Levee (km)	1.30	0.65	0.27	1.65	-0.65
Dist. to Treated Levee (km)	1.91	0.71	0.39	2.39	-1.20
Flood frequency (NRI, events/yr)					
Riverine	1.65	1.96	1.70	1.63	0.31
Coastal	0.39	0.24	1.35	0.44	-0.15
Combined	2.03	2.20	3.05	2.07	0.17
SFHA (share)	0.14	0.27	0.50	0.16	0.13
N (pre-period)	7,629	831	195	7,197	

Notes: Pre-period (event year < 0) means for the groups in the crest-depth specification. *Treated, All* is every sale inside the district that passed the tax: the regression level is evaluated at the full-sample treated mean depth, rather than the pre-period mean reported here. *Treated, ≥1 SD deeper* keeps the treated sales whose depth below the district crest is at least one treated-side standard deviation above that mean, the exposure at which the depth gradient is read. *Spillover* is neighbor sales within 600 m of the treated district's levee and *Control* the remaining neighbor sales. *Deep−All* is the difference between the second and first columns. *Dist. to Nearest Levee* is the distance to any NLD levee centerline; *Dist. to Treated Levee* to the levee of the district that passed the tax. *Flood frequency* rows are FEMA National Risk Index tract annualized event frequencies (events/yr) for riverine and coastal flooding and their sum. *SFHA* is the share of properties that FEMA's current National Flood Hazard Layer places in a Special Flood Hazard Area (A/V zones).

Table 5: Average Treatment Effect of Levee Tax Passage on House Prices

	(1) No hedonics	(2) Hedonics	(3) Property FE
Treated × Post	0.235**	0.279**	0.174*
	(0.079)	(0.095)	(0.082)
Spillover × Post	-0.218***	-0.249***	-0.113**
	(0.040)	(0.048)	(0.049)
Census tract FE	Yes	Yes	No
Boundary pair × year FE	Yes	Yes	Yes
Property FE	No	No	Yes
Boundary pairs	15	15	12
Observations	38,930	38,930	8,039
Within R^2	0.009	0.114	0.019
R^2	0.530	0.580	0.860

Notes: Difference-in-differences estimates for log transaction price (equation 1). Columns (1) and (2) absorb tract and boundary-pair-by-year fixed effects. Column (1) has no hedonics, column (2) adds hedonics: living area, lot size, stories, and age, with missing characteristics zero-filled and accompanied by missing indicators. Column (3) is the repeat sales sample, absorbs property fixed effects and pairxyear fixed effects. Spillover properties are neighboring sales within 600 m of the treated district's levee; remaining neighboring sales form the control group. Standard errors clustered by boundary pair are in parentheses. The preferred specification is column (2). Figure 2 reports treated and spillover dynamics using the structural hedonic controls used in the slides; unlike column (2), it omits nearest-levee distance as a control. Appendix Figure A.6 compares the no-hedonics, hedonic, and property-fixed-effects event studies. Appendix Table A.13 reports wild-cluster bootstrap p-values using boundary-pair and district clustering. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 6: Differential House-Price Effects by Depth Below the District Levee Crest

	(1) No hedonics	(2) Hedonics	(3) Property FE
Treated × Post (at 18.10 ft crest depth)	0.267***	0.321***	0.169***
	(0.037)	(0.044)	(0.036)
Treated × Post × Depth (per ft)	0.024***	0.029***	0.020**
	(0.007)	(0.007)	(0.007)
Spillover × Post (at 18.10 ft crest depth)	0.106	0.106**	0.132***
	(0.072)	(0.045)	(0.035)
Spillover × Post × Depth (per ft)	-0.056***	-0.059***	-0.042***
	(0.011)	(0.009)	(0.003)
Post × Depth (per ft)	-0.005	-0.007	0.000
	(0.005)	(0.005)	(0.003)
Census tract FE	Yes	Yes	No
Boundary pair × year FE	Yes	Yes	Yes
Property FE	No	No	Yes
Boundary pairs	15	15	12
Observations	38,930	38,930	8,039
Within R^2	0.020	0.129	0.032
R^2	0.535	0.587	0.861

Notes: Continuous-depth triple-difference estimates for log transaction price (equation 2), including all group, post, and depth lower-order terms and interactions. Depth is the measure of flood exposure, the difference between the property's own district median levee crest and ground elevation, centered at the full-sample treated mean, 18.10 ft. Columns (1) and (2) absorb tract and boundary-pair-by-year fixed effects. Column (1) has no hedonics, column (2) adds hedonics: living area, lot size, stories, and age, with missing characteristics zero-filled and accompanied by missing indicators. Column (3) is the repeat sales sample, absorbs property fixed effects and pairxyear fixed effects. Group-by-post levels are evaluated at average depth, gradients are log points per additional foot. Figure 3 shows the dynamic treatment effect plots for the hedonic specification, column (2). Appendix Figure A.7 compares the dynamic level and depth gradient across all three specifications. The same-sample comparison and controls ladder are Appendix Tables A.9 and A.10. Standard errors clustered by boundary pair are in parentheses. Appendix Table A.14 reports wild-cluster bootstrap p-values using boundary-pair and district clustering. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 7: Increasing the bandwidth of the spillover group

	(1) 400 m	(2) 600 m	(3) 800 m	(4) 1,000 m
Treated × Post	0.282**	0.279**	0.267***	0.256***
	(0.095)	(0.095)	(0.089)	(0.085)
Spillover × Post	-0.367***	-0.249***	-0.276***	-0.276***
	(0.051)	(0.048)	(0.054)	(0.043)
Census tract FE	Yes	Yes	Yes	Yes
Boundary pair × year FE	Yes	Yes	Yes	Yes
Treated sales	20,010	20,010	20,010	20,010
Spillover sales	642	1,169	1,926	2,671
Control sales	18,278	17,751	16,994	16,249
Observations	38,930	38,930	38,930	38,930
Boundary pairs	15	15	15	15

Notes: Difference-in-differences estimates for log transaction price (equation 1), redefining spillover properties as neighboring sales within 400, 600, 800, or 1,000 m of the treated district's levee. Remaining neighboring sales, including missing treated-levee distances, form the control group, treated properties are unchanged. All columns include tract and pair-by-year effects, and hedonics. The 600 m column reproduces the main pooled hedonic specification. Standard errors clustered by boundary pair are in parentheses. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 8: Tract-level quantity responses to levee tax passage

	(1) ACS occupancy	(2) Log USPS occupied	(3) Log population	(4) IHS workers	(5) IHS jobs	(6) Log housing units
Treated × Post (at mean depth)	0.035***	0.119**	0.111*	0.218***	0.275***	0.064
	(0.008)	(0.040)	(0.055)	(0.058)	(0.066)	(0.036)
Treated × Post × Depth	0.006**	0.020**	0.018	0.031*	0.016	0.008
	(0.003)	(0.008)	(0.014)	(0.014)	(0.013)	(0.008)
Post × Depth	0.001	-0.001	-0.002	-0.005	-0.009	-0.004
	(0.002)	(0.003)	(0.005)	(0.003)	(0.010)	(0.004)
Average treatment effect (separate DD)	0.025	0.094	0.086	0.187*	0.263***	0.056
	(0.015)	(0.052)	(0.056)	(0.087)	(0.070)	(0.034)
Observation unit	Tract-year	Tract-year	Tract-year	Tract-year	Tract-year	Tract-year
Census tract FE	Yes	Yes	Yes	Yes	Yes	Yes
Boundary pair × year FE	Yes	Yes	Yes	Yes	Yes	Yes
Depth centering (ft)	17.49	17.49	17.49	17.49	17.49	17.49
Boundary pairs	10	10	10	10	10	10
Observations	1,983	2,405	1,983	2,723	2,711	1,983
Within R^2	0.072	0.138	0.070	0.128	0.027	0.038
R^2	0.831	0.968	0.962	0.933	0.937	0.973

Notes: Tract level triple difference estimates following (equation 3) with tract and pair-by-year effects. Depth is centered at 17.49 feet, the treated tract-year mean. Standard errors clustered by boundary pair are in parentheses. The average treatment effect is from a separate pooled DD on the same column-specific sample, with no depth interactions. ACS occupancy is (1- vacant housing units/total housing units). USPS occupied addresses are annual means of quarterly residential addresses net of no-stat and vacant addresses, in logs. Population and housing units are ACS five-year counts in logs. Resident workers and jobs are LODES counts transformed using the inverse hyperbolic sine. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 9: Insurance responses to levee tax passage

	(1) Premium per 1,000 coverage	(2) Claims / coverage (%) 2005–2024	(3) Claims / coverage (%) 2009–2024	(4) Log policies
Treated × Post (at mean depth)	-0.750**	-12.892***	-4.629	0.093*
	(0.245)	(2.886)	(7.420)	(0.044)
Treated × Post × Depth	-0.138**	-2.200**	-1.576	0.007
	(0.044)	(0.846)	(1.814)	(0.011)
Post × Depth	0.055	-2.209**	-1.255	0.009*
	(0.034)	(0.800)	(1.108)	(0.005)
Average treatment effect (separate DD)	-0.708**	-0.538	7.487	0.075
	(0.296)	(13.206)	(9.103)	(0.056)
Observation unit	Policy term	Claim record	Claim record	Tract-year
Census tract FE	Yes	Yes	Yes	Yes
Boundary pair × year FE	Yes	Yes	Yes	Yes
Depth centering (ft)	17.52	17.52	17.52	17.52
Boundary pairs	10	10	7	10
Observations	1,275,553	17,139	437	2,031
Within R^2	0.002	0.003	0.032	0.015
R^2	0.118	0.589	0.338	0.925

Notes: Depth is centered at 17.52 feet, the treated tract-year mean. The average treatment effect is from a separate pooled DD on the same column-specific sample, with no depth interactions. Standard errors clustered by boundary pair. The main premium outcome is an individual policy premium divided by its building coverage, multiplied by 1,000. The claims columns use individual eligible positive claim records, with outcome 100 times building-plus-contents payments divided by the building-plus-contents coverage recorded on that same claim (cause codes 1, 2, 3, and B). The policy-count column remains tract-year. The claims columns use 2005–2024 and 2009–2024, respectively, with the latter matching the calendar-year coverage of the premium data. This outcome measures severity among claimants, not losses across all insured properties. Policy take-up is the log count of policy terms by effective-date year, not the fraction of households insured. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 10: Precinct characteristics and levee-tax approval

	Risk only		+ Characteristics	
	(1)	(2)	(3)	(4)
	No wt.	Turnout wt.	No wt.	Turnout wt.
Risk				
Median minus mean elevation (per m)	-3.05* (1.50)	-2.20** (0.93)	-2.61* (1.47)	-1.88* (1.04)
Salience				
Distance to levee			0.80 (1.46)	0.13 (1.62)
Levee in precinct			2.08 (1.24)	-0.13 (0.67)
Experience				
NFIP claims history			2.14 (1.44)	0.88 (1.11)
Other characteristics				
Median household income			0.55 (1.35)	-1.16 (1.21)
Voter density			1.57* (0.84)	0.80 (0.73)
BA-or-higher share			1.32 (1.10)	1.96 (1.23)
Renter share			2.22** (0.93)	1.08 (0.66)
NRI social vulnerability			-0.93 (0.56)	-0.53 (0.49)
Political				
Democratic voter share			0.67 (1.89)	-0.81 (2.68)
Constant	52.08*** (0.17)	51.93*** (0.12)	51.50*** (0.40)	50.83*** (1.17)
Referendum (district × date) FE	Yes	Yes	Yes	Yes
Other characteristics	–	–	Yes	Yes
Turnout weighted	–	Yes	–	Yes
Clustering		Precinct + ballot item		
Stable precincts	1,092	1,092	1,092	1,092
Ballot items	24	24	24	24
Observations	1,724	1,724	1,724	1,724
Within R-squared	0.001	0.001	0.047	0.068
Overall R-squared	0.667	0.726	0.682	0.744

Notes: Outcome: yes votes as a percentage of yes + no votes on a levee-tax proposition. Observations are precinct–proposition pairs from 2010–2025 with at least ten votes in the precinct, all listed characteristics observed, and positive recorded ballot-day turnout. All four columns use the same eligible sample and absorb district-by-date (referendum) fixed effects. The first two columns include only the within-precinct parcel-stock median-minus-mean elevation gap; the last two add all characteristics shown. The gap is measured in meters, so its coefficient is the percentage-point change in approval associated with a one-meter increase in the gap between median vs mean elevation. The levee-in-precinct indicator is binary. Other regressors are standardized: income is log median household income, claims history is log(1 + annualized paid losses per $1,000 coverage), and voter density is log(1 + registered voters per square kilometer), before standardization. Turnout weights are total recorded ballot-day voters. Standard errors are clustered by stable precinct and ballot item. The district-level figure uses a different, turnout-imputed median. Area interpolation, nearest-year fills and the 2023 NRI vintage are described in Data Appendix **??**. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table 11: Passed proposals: district housing-value gains

District	Ballot mills	No levy ($m)	Passage ($m)	Gain ($m)	Optimal mills
Algiers	6.35	3,041.8	3,289.8	248.0	93.91
North Lafourche	6.00	6,622.1	6,852.0	229.9	135.17
Orleans	6.07	20,489.7	26,843.7	6,353.9	48.46
South Lafourche	5.00	955.2	1,019.4	64.2	83.31
St. Charles	4.00	3,320.8	3,658.0	337.2	40.31
St. John the Baptist	7.00	2,229.4	2,227.4	-1.9	1.49
St. Mary	5.00	1,545.8	1,555.1	9.4	192.39
Town of Berwick	6.00	504.7	519.1	14.4	116.00
West Jefferson	4.75	6,827.3	7,293.0	465.7	89.27
Total	—	45,536.8	53,257.5	7,720.8	—

Notes: Each district's latest eligible proposal is evaluated at its incremental ballot millage. No levy and passage report aggregate residential value; gain is passage minus no levy, including the modeled property-tax cost. All values are one-time 2024 dollars. Optimal millage uses the baseline calibration.

Table 12: Failed proposals: district housing-value gains

District	Ballot mills	No levy ($m)	Passage ($m)	Gain ($m)	Extrapolated gain ($m)	Optimal mills
Chenier Plain	5.00	19,132.9	19,300.3	167.4	-53.8	13.43
Iberia	5.00	1,640.2	1,636.3	-3.9	-361.5	0.00
Lake Borgne	7.50	1,479.7	1,515.2	35.5	903.6	234.34
Plaquemines	5.00	1,571.7	1,577.9	6.3	900.1	135.75
St. James	7.00	672.5	678.6	6.1	-59.2	43.13
St. Tammany	5.00	17,085.0	17,042.6	-42.5	59.4	0.00
Total	—	41,582.0	41,750.8	168.8	1,388.6	—

Notes: Each district's latest eligible proposal is evaluated at its incremental ballot millage. Forgone value is the one-time change in aggregate residential value from passage relative to no incremental levy; negative values indicate that passage would lower value. No levy and passage report aggregate residential value. Optimal millage maximizes aggregate residential value on the policy grid. Extrapolated gains apply the current empirical crest-depth response to failed-district pre-vote price profiles; they use a different valuation base from the model and are a prediction comparison, not validation.

Table 13: Optimal millage and the cost of the referendum constraint

Group	Scenario	Optimal mills	Loss ($m/year)	Loss / gain (%)
Passed	Baseline	88.9	0.010	0.001
Passed	Double expected damage	145.2	0.660	0.020
Passed	Baseline risk + proportional tax	92.2	0.000	0.000
Failed	Baseline	71.1	12.679	26.378
Failed	Double expected damage	163.1	48.156	29.699
Failed	Baseline risk + proportional tax	75.6	0.000	0.000

Notes: Groups use each district's latest eligible proposal. Optimal millage is the unweighted mean across districts. Loss is the sum of annualized residential value at the planner optimum minus value at the best majority-acceptable rate; loss/gain divides this sum by the group's total planner gain over no levy. The scenarios are baseline uniform taxation, doubled expected damage with uniform taxation, and baseline expected damage with exposure-proportional taxation. Fitted protection effectiveness is unchanged throughout. The electorate is fixed at pre-policy positive-value parcels within each risk scenario.

A Appendix

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A.1 Municipal bond issuance

Table A.1 lists the observed municipal bond issuances of the sample's levee districts, including parish governments serving as levee districts. Over the full available Refinitiv history, August 1967–February 2025, the sample contains 13 issuer–date groups across seven jurisdictions, issued between 1989 and 2023. Table A.2 compares these bonds with other municipal bonds in the overlapping parishes. The selected bonds account for 191 of 15,858 recorded securities (1.2%), documenting occasional use of this funding channel.

Table A.1: Historical municipal bond issuance by levee districts

Issuer	Issue date	CUSIPs	Recorded purpose	Par ($m)	Maturities
A. Stand-alone levee districts					
Lake Borgne Basin	1989-02-01	17	Not recorded	—	1992–2009
Lake Borgne Basin	1993-08-01	18	Public improvements	7.580	1995–2009
North Lafourche	2017-03-08	17	Flood control / drainage	13.620	2018–2037
Orleans	1995-12-01	8	Mixed purposes	110.770[†]	2001–2015
Orleans	2005-09-08	9	Public improvements	36.300	2006–2014
St. Mary	2018-12-06	20	Flood control / drainage	13.000	2020–2039
Terrebonne	2013-07-30	18	Public improvements	91.190	2014–2038
Terrebonne	2020-09-09	19	Flood control / drainage	103.595	2021–2041
Terrebonne	2023-08-22	10	Public improvements	14.550	2026–2039
B. Parish governments serving as levee districts					
Plaquemines Parish	2004-04-28	10	Water / sewer	10.000	2005–2014
Plaquemines Parish	2010-10-12	19	Flood control / drainage	18.000	2012–2030
St. Charles Parish	2003-12-23	11	Water / sewer	21.875	2004–2014
St. Charles Parish	2017-09-07	15	Flood control / drainage	15.000	2018–2037

Notes: Source: Refinitiv. Each row groups distinct maturity-level CUSIPs by issuer and issue date; one row may contain multiple financing series. Panel A includes stand-alone levee districts. Panel B includes parish governments that also serve as levee districts: their recorded purpose must be flood control/storm drainage or water/sewer, and Plaquemines bonds dated April 28, 2010 are excluded. Purpose labels reproduce source categories in abbreviated form; they do not establish that all proceeds financed levees. Orleans 1995 combines public-improvement and seaport/marine-terminal categories. Par amounts sum recorded face values in nominal millions of dollars. — denotes missing data: all 17 Lake Borgne 1989 securities lack face amounts. [†]Only four of the eight Orleans 1995 securities have recorded face amounts; the displayed sum is partial. Maturities give the earliest and latest maturity years. Matured and inactive securities are retained. Refunding securities are included, so par does not measure net new funding. This inventory covers observed municipal securities, not all district debt or loans.

Table A.2: Levee-district bonds and other municipal bonds in overlapping parishes

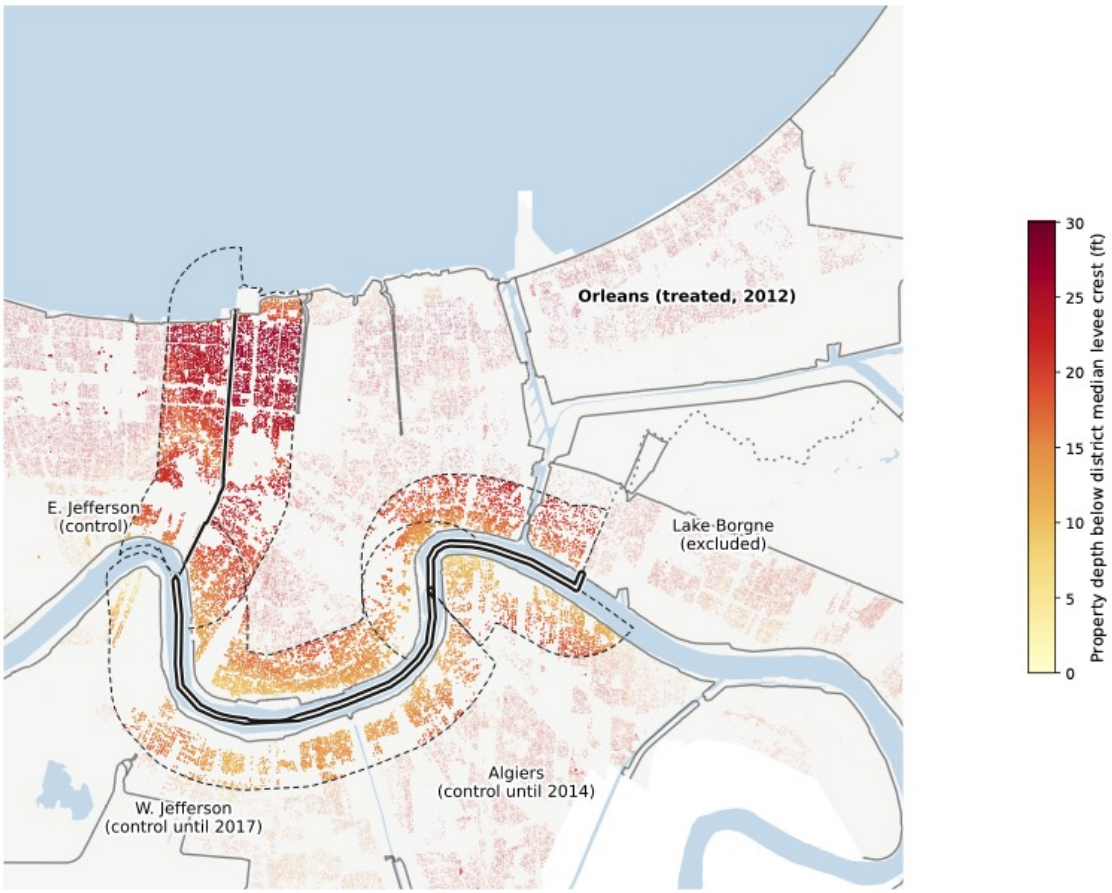
	Levee districts	Other issuers
Issuer–date groups	13	1,204
Maturity-level securities (CUSIPs)	191	15,667
Recorded par amount ($ million)	455.480	33,834.394
Securities with recorded par	170	14,684
Mean coupon (%)	4.42	4.47
Securities with recorded coupon	191	15,667
Mean term at issuance (years)	8.88	9.71

Notes: Source: Refinitiv, August 1967–February 2025. Both columns use the union of 16 parishes overlapping the 18 jurisdictions in the 2010–2025 referendum roster. A parish is included if it contains more than 5% of a district's area, using the existing district–parish overlay; St. James and St. John the Baptist use the parish-name fallback. The parishes are Calcasieu, Cameron, Iberia, Jefferson, Lafourche, Orleans, Plaquemines, St. Bernard, St. Charles, St. James, St. John the Baptist, St. Mary, St. Tammany, Tangipahoa, Terrebonne, and Vermilion. The levee-district column includes both panels of Table A.1. Other issuers include parish governments' remaining bonds, municipalities, school boards, utilities, and other municipal issuers. Multi-parish securities are counted once by CUSIP. Issuer–date groups use the reported issuer name and issue date and may combine multiple series. Par totals sum observed face amounts only; coupon and term means weight securities equally. Term is the existing bond panel's integer-year maturity measure (days from issuance to maturity divided by 365, rounded down). One placeholder issue date of January 1, 1900 is excluded; no 1990 cutoff or referendum event window is imposed. Historical coverage may be incomplete.

A.2 Analysis geographies and crest depth

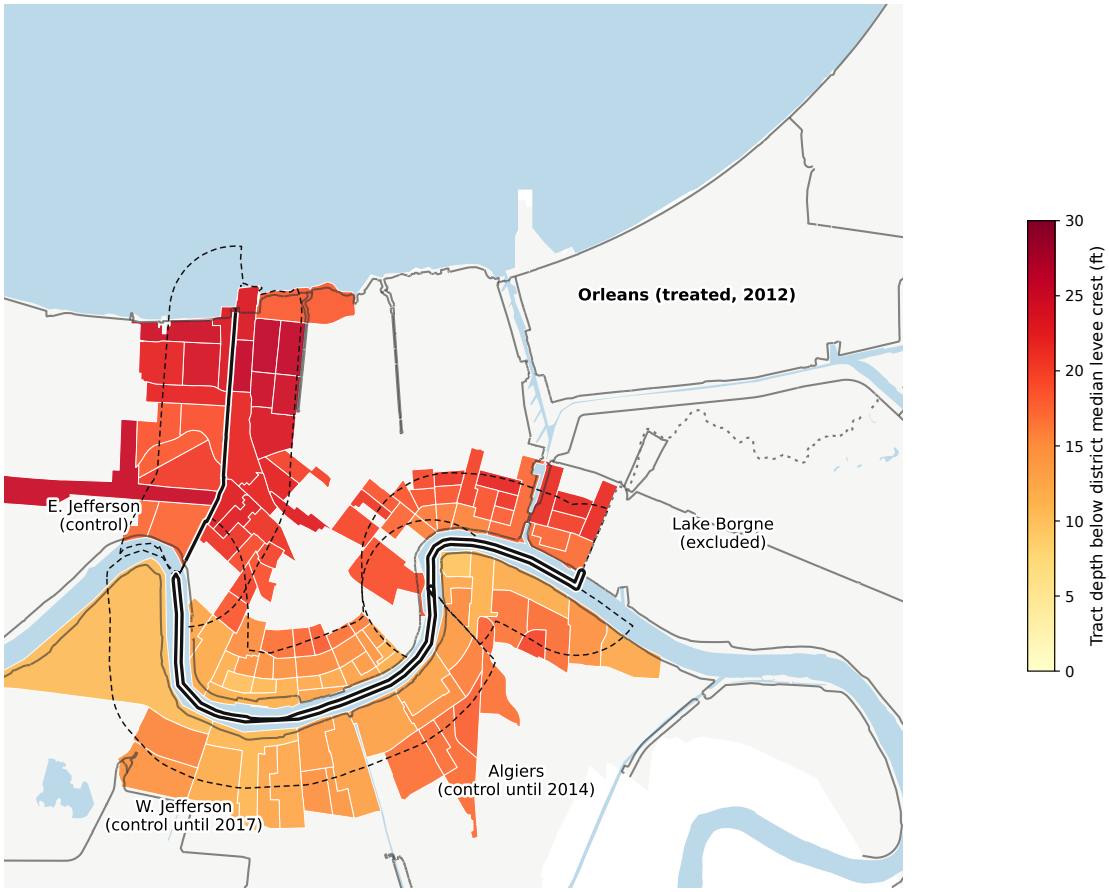
These maps illustrate the property, census-tract, and voting-precinct geographies for Orleans. Exposure panels share a common scale for depth below the own-district median levee crest, in feet. The approval panel reports the yes-vote share.

Figure A.1: Orleans levee district and its neighbors, properties, boundaries and crest depth



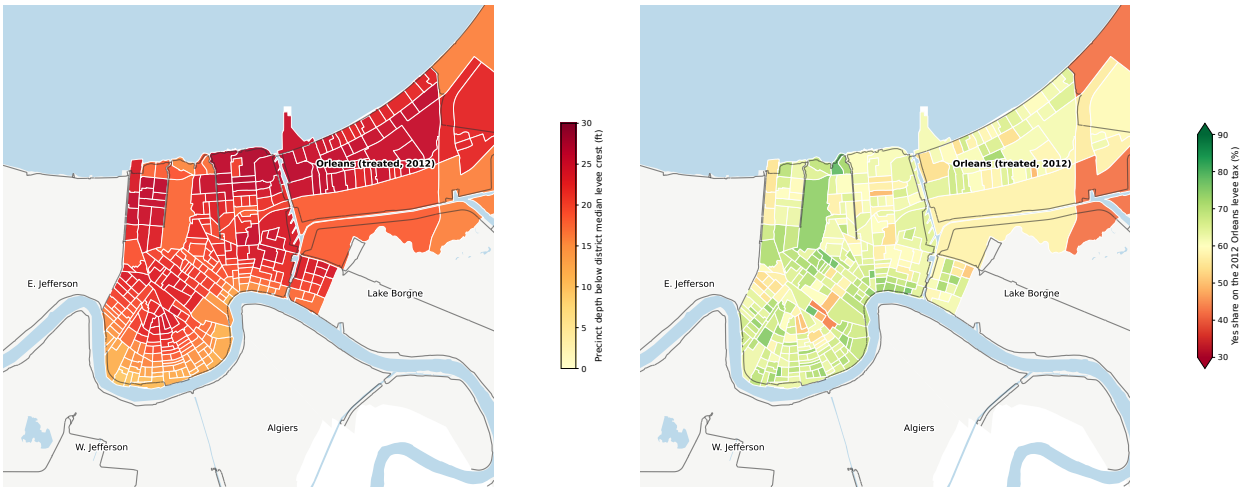
Notes: Geographic illustration of Orleans (2012 passage) and neighboring districts. Each dot represents a distinct transacted residential property in the prepared-data universe, shaded by depth below its own district's median levee crest (feet). Depth is the positive part of crest minus ground elevation; zero denotes a property at or above the crest. Properties within 2,354 m of surviving boundaries are opaque; others are faded. Levees are dark grey. Lake Borgne is excluded as a comparison because it passed its tax before Orleans. The illustration uses post-2008, leveed-area and single-parcel filters but does not reproduce every overlap, censoring, or estimator-specific exclusion in the regression sample.

Figure A.2: Orleans levee district and its neighbors, census tracts, boundaries and crest depth



Notes: Census tracts of the boundary tract panel, shaded by depth below the median levee crest of the tract's assigned district (feet). As in the tract regressions, depth is the positive part of the own-side district crest minus the mean elevation of qualifying transacted properties. Colors show uncentered depth. A tract enters on the side of its qualifying properties and its depth is interacted continuously with treatment, as for house prices.

Figure A.3: Unit of observation, referendum approval: voting precincts of the 2012 Orleans levee-tax referendum

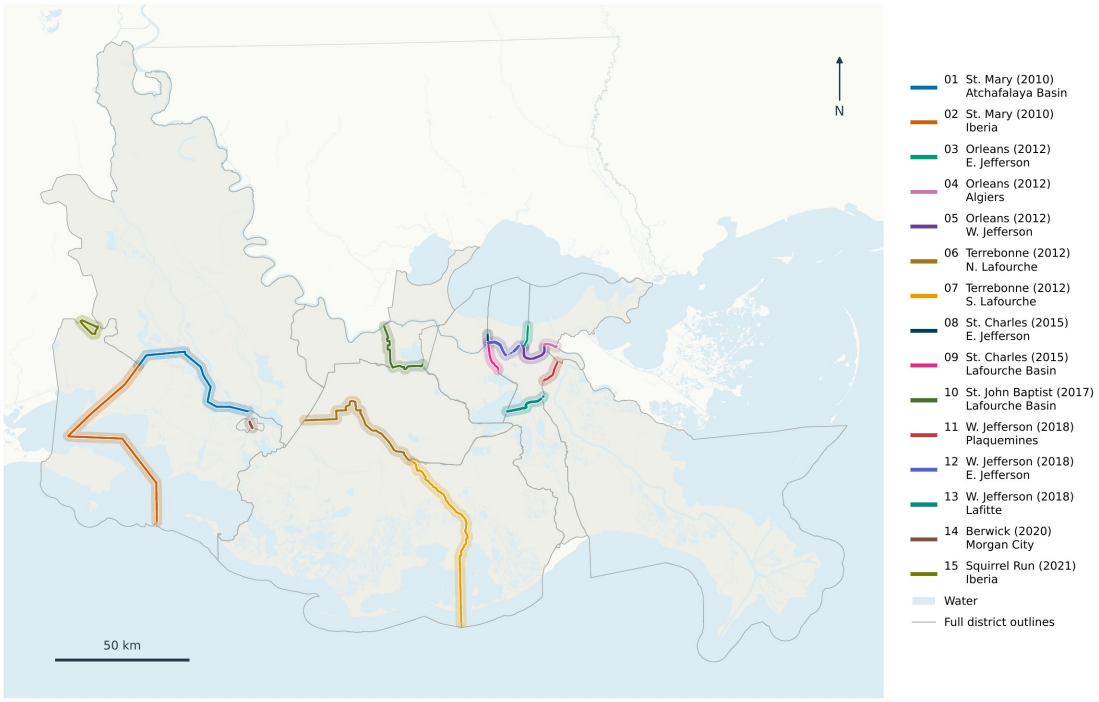


Notes: Voting precincts of the Orleans Levee District in the 2012 referendum: left, shaded by depth computed from the mean elevation of their residential parcels relative to the Orleans district median levee crest (feet), right, by the share voting yes. This geographic depth illustration does not change the median-minus-mean elevation regressor used in the voting analysis. The precinct approval regressions use precinct characteristics with referendum fixed effects.

A.3 Boundary Geography and Sample Support

Figure A.4 shows the comparison boundaries used in the current house-price analysis in their regional setting. Full levee-district outlines show the extent of the jurisdictions on either side. The numbered color key links each boundary to Table A.3, which compares reference-line length with the estimation bandwidth and reports transaction, property, and census-tract support.

Figure A.4: Boundaries used in the empirical design



Notes: Each color identifies a boundary pair in the current static hedonic estimation sample. Gray outlines show the full levee districts involved in these comparisons; blue areas show water. Colored land frontages are simplified for display by joining nearby gaps along district outlines and omitting tiny detached fragments; water-pair reference lines are retained. Matching shading extends 2,354 projected meters on each side of the displayed frontages, the optimal one-sided bandwidth. Colors overlap where comparison frontages or their bandwidths coincide. District outlines provide geographic context and may differ from the referendum-specific precinct geography. These continuous corridors illustrate the bandwidth rather than the exact estimation footprint: water exclusions, leveed-area restrictions, and overlap rules still apply to the sample. The section shared by the St. John–Lafourche Basin and St. Charles–Lafourche Basin comparisons is omitted from both colored frontages and both table lengths; sample counts are unchanged. No properties are plotted. The map uses NAD83 / UTM zone 15N.

Table A.3: Boundary Length, Bandwidth, and Census-Tract Support

						Transactions (unique properties)		Census tracts	
ID	Treated district (year)	Neighbor	Type	Length (m)	Width (m)	Treated	Control	Treated	Control
01	St. Mary (2010)	Atchafalaya Basin	Land	22,447	2,354	0 (0)	2 (2)	0	1
02	St. Mary (2010)	Iberia	Land	79,648	2,354	0 (0)	23 (22)	0	1
03	Orleans (2012)	E. Jefferson	Land	9,242	2,354	6,648 (4,517)	6,539 (4,587)	30	13
04	Orleans (2012)	Algiers	Water	15,738	2,354	1,230 (1,069)	745 (674)	25	11
05	Orleans (2012)	W. Jefferson	Water	32,347	2,354	4,380 (3,513)	2,354 (2,038)	40	21
06	Terrebonne (2012)	N. Lafourche	Land	67,073	2,354	727 (648)	611 (574)	11	10
07	Terrebonne (2012)	S. Lafourche	Land	74,118	2,354	148 (132)	6 (5)	3	1
08	St. Charles (2015)	E. Jefferson	Land	2,973	2,354	4 (4)	466 (369)	1	4
09	St. Charles (2015)	Lafourche Basin	Land	12,933	2,354	81 (70)	0 (0)	1	0
10	St. John Baptist (2017)	Lafourche Basin	Land	23,570	2,354	4 (4)	117 (95)	1	2
11	W. Jefferson (2018)	Plaquemines	Land	11,661	2,354	4,697 (3,677)	400 (301)	13	3
12	W. Jefferson (2018)	E. Jefferson	Water	34,397	2,354	1,672 (1,366)	6,372 (4,583)	4	19
13	W. Jefferson (2018)	Lafitte	Water	36,422	2,354	0 (0)	145 (115)	0	1
14	Berwick (2020)	Morgan City	Water	6,689	2,354	419 (302)	772 (628)	1	3
15	Squirrel Run (2021)	Iberia	Land	16,123	2,354	0 (0)	368 (284)	0	2

Notes: IDs match the color key in Figure A.4. Transactions and unique properties (in parentheses) are reported separately for the treated and control sides; the control side includes observations in the modeled spillover group. Counts use the current static continuous-crest-depth hedonic estimation sample after singleton fixed-effect removal, with the same sample restrictions, overlap rule, tract effects, and pair-by-year effects as the main price model. Length is the reference-line length after removing the 17,127-meter section shared by the St. John–Lafourche Basin and St. Charles–Lafourche Basin comparisons from both pairs; width is the one-sided boundary bandwidth (the full two-sided strip is twice this width). Both are measured in the analysis coordinate system, EPSG:3857. These projected meters exceed ground distance at this latitude. Census-tract columns count distinct 2020 tracts with sampled transactions on each side; they describe price-sample coverage, not the outcome-specific samples of the tract regressions. Tracts can appear on both sides or in multiple pairs. Properties appearing in multiple pairs within a referendum are excluded; across referenda, each property is retained only in its earliest surviving referendum, so every property maps to one boundary pair. A represented pair need not have treated-side support. The current continuous-depth model replaces the archived high-/low-risk split, so support is reported by boundary side.

A.4 Robustness

Sensitivity to controls and bandwidth choices: Appendix Figure A.15 shows the overall price appreciation of the treated tracts/zips that overlap with the boundary properties used in the analysis from the FHFA House Price Index, and from Zillow, and also highlights how splitting by risk at this level is coarse. Appendix Table A.11 and Figure A.17 show estimates when we vary the bandwidth from 1-5km, alongside the main bandwidth of 2,354 m. The hedonic depth gradient declines from 0.036 log points per foot at 1 km to about 0.026 at 5 km, while the level effect is not monotonic across bandwidths. This is consistent with the salience of the levee operating along with elevation itself. However, widening the bandwidth brings in more properties and boundary pairs, so this also changes the composition of the sample, but we lose comparability across sides - there are significant pre-trends. The static level effects are evaluated at each sample's mean treated depth. Appendix Figure A.18 holds the depth at which the level is evaluated fixed at the main-sample mean and shows how the dynamics change across these bandwidths. Appendix Figures A.13 and A.14 repeat the DDD and average-effect event study while retaining within-referendum overlapping properties with distance-based weights that favor the closer boundary.

Appendix Table A.10 shows the control ladder, starting with no fixed effects, adding tract effects, year effects, replacing year effects with pair-by-year effects, adding hedonics, and finally parish-by-year effects. The depth gradient is positive throughout, but becomes more precisely estimated with the pair-by-year effects. Adding hedonics takes it from 0.0241 to 0.0295 log points per foot. Adding parish-by-year effects brings it to 0.020 (standard error 0.005), with two additional singleton observations dropped. This is not driven by losing observations when I add the housing characteristics: the sample falls from 38,947 to 38,930 before adding hedonics, and the same pattern holds when all columns use exactly the same observations. The estimates are stable across all specifications.

The estimated depth gradient remains positive across the main specifications and similar in magnitude when alternative risk measures are included jointly. Appendix Tables A.13 report wild-cluster bootstrap p-values using boundary-pair and district clustering. Inference is sensitive to the clustering level, reflecting that the small sample may not be composed uniformly of similar regions, as we already have seen with the leave-one-out analysis.

Compositional changes: Is it compositional shifts or true capitalization of identical properties that causes differences in magnitudes across the three specifications? Ideally the property fixed effects specification is the number we use, but given the loss in sample it is harder to draw inferences from this. Appendix Table A.9 compares no hedonics, hedonics, and property fixed effects on exactly the same 8,039 transaction rows from repeat-sold properties, displaying the individual group coefficients. The magnitudes suggest that there is some unobserved heterogeneity, but the no hedonics spec is very similar to the property FE estimate.

Other risk specifications The capitalization patterns are robust to various configurations and specifications, highlighting the heterogeneity in benefits of the uniform levee tax within the districts. The horse race in Appendix Table A.8 uses the same observations across the separate and joint exposure models. The map indicators separately identify current SFHA and Zone X explicitly designated as levee-protected, current flood maps may reflect accreditation, so this is a sensitivity comparison.

There is almost no zero risk group in the treated sample The analysis require properties to be very close to the levees by design because many of the boundaries also have a levee running right through it. The river banks, where the levees are, are naturally on higher ground than the rest of the region, and so in the majority of these places, there is almost no property above the top of the levee - the new 'sea level' that needs to be overtopped. This makes it difficult to isolate the 'tax only' effect. I try to split into finer risk bins, but this results in samples being sliced too thin, and sub groups lose comparability with the control sample.

Are these just different school districts that have unobserved quality improvements? If a school district invested in better schools at the same time as the levee tax, we would capture that in house price response as well. What helps distinguish the flood-protection interpretation is that I look at how capitalization varies with observable flood-risk measures, especially depth below the levee crest, rather than only the average price increase. To explain this gradient, school-related gains would also need to be systematically larger in the deeper properties around the vote, and this is where the granularity of the measure helps rule out this confounder.

More generally, contemporaneous events on the treated side could also affect outcomes. Appendix Table A.12 lists the levee taxes and other ballot items reported in precincts eligible to vote on them, including school-finance measures and taxes for other local services. If these events produce changes that affect treated properties differently from nearby comparison properties, their capitalization could be attributed to the levee tax. The table documents potential competing changes at the election date, assessing their contribution requires evidence on passage, implementation, and exposure of the estimation properties. Again in this scenario, property level elevation granularity allows us to rule out this confounder.

Are the outcomes priced in already if the votes are not close? House prices are the present discounted value of future benefits, and therefore would reflect the better flood protection in expectation following passage of the tax. Given this is a tax for maintenance, it is not far out in the future when they would expect to see a change. This also means that they may price in the passage, if it is not a surprise. The referenda are not close, so we cannot immediately rule this out. Appendix Figure A.12 shows that there is no spike in sales in the months preceding the vote, so no changes in behavior in anticipation of the levee tax. The price response in this scenario, if households predict the outcome and price it in advance, would be biased downward.

Spillover effects due to people moving to the safer areas? Cannot distinguish this from flood risk capitalization. Would require more granular mobility data for this, or a general equilibrium model to disentangle the two.

A.5 Additional Price Specification Robustness

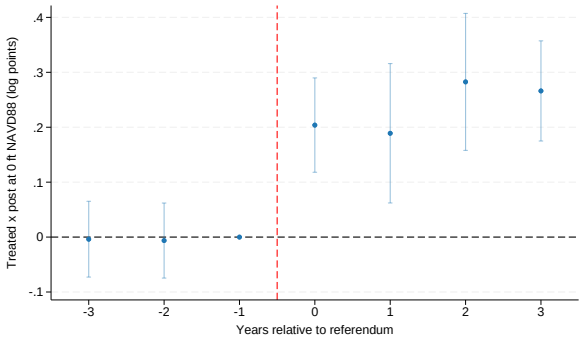
Dynamic specification comparisons. Figure A.5 reports the alternative absolute-depth dynamics, measuring exposure as negative elevation (relative to sea level). Figures A.6 and A.7 compare no hedonics, hedonics, and property fixed effects. The first two specifications have similar dynamic paths; property fixed effects use the smaller pre/post repeat-sales sample and show different spillover and depth-gradient patterns.

Table A.4: House-Price Responses: Absolute Depth

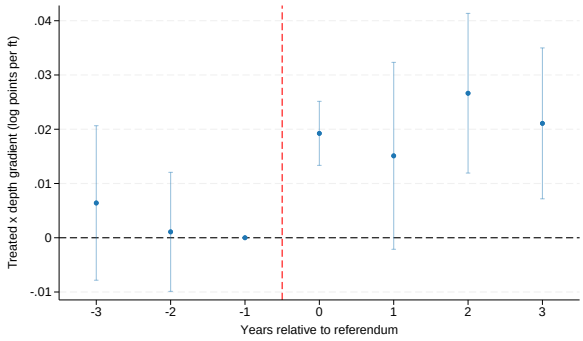
	(1) No hedonics	(2) Hedonics	(3) Property FE
Treated × Post (at 0 ft NAVD88)	0.269***	0.322***	0.180***
	(0.061)	(0.072)	(0.053)
Treated × Post × Depth (per ft)	0.021*	0.026**	0.019
	(0.011)	(0.012)	(0.011)
Spillover × Post (at 0 ft NAVD88)	-0.001	-0.023	0.049
	(0.063)	(0.062)	(0.048)
Spillover × Post × Depth (per ft)	-0.066***	-0.063***	-0.042***
	(0.008)	(0.006)	(0.004)
Post × Depth (per ft)	-0.001	-0.004	-0.000
	(0.006)	(0.005)	(0.004)
Boundary pairs	15	15	12
Observations	38,930	38,930	8,039
Within R^2	0.020	0.128	0.029
R^2	0.535	0.587	0.861

Notes: The same continuous-exposure specifications as Table 6, using absolute depth (-elevation in feet) instead of crest depth. Columns (1)–(3) are no hedonics, hedonics, and property fixed effects. The first two absorb tract and pair-by-year effects; the third uses property and pair-by-year effects on pre/post repeat sales without hedonics. All group-by-post-by-depth terms, including spillover terms, remain included. Levels are evaluated at zero feet NAVD88; gradients are log points per additional foot of absolute depth. The hedonic event study is Figure A.5. Standard errors clustered by boundary pair are in parentheses. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Figure A.5: Dynamic House-Price Responses: Absolute Depth



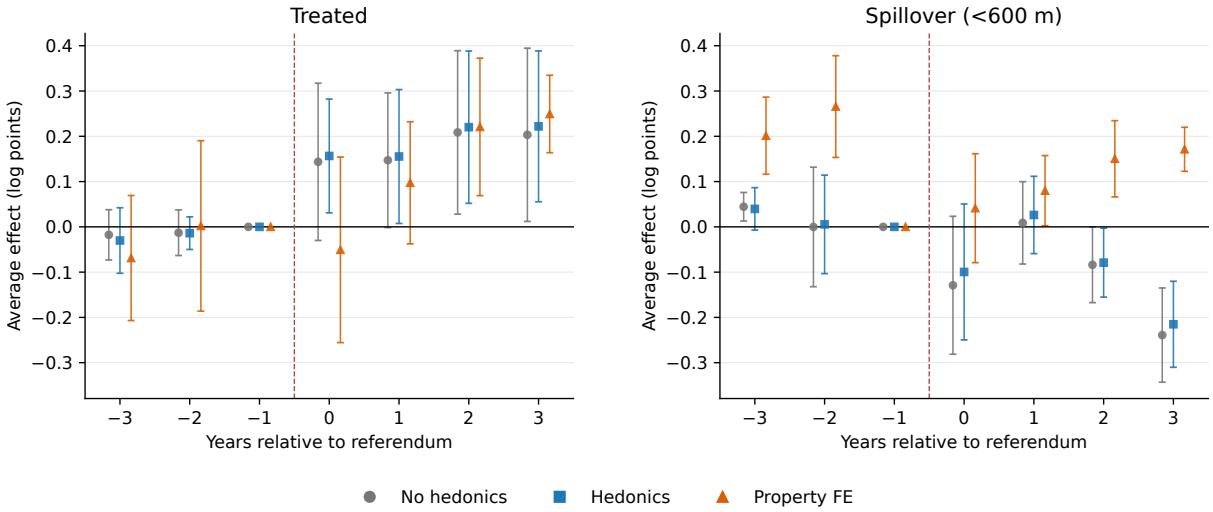
(a) Treated × Post level



(b) Treated × Post × Depth gradient, absolute depth

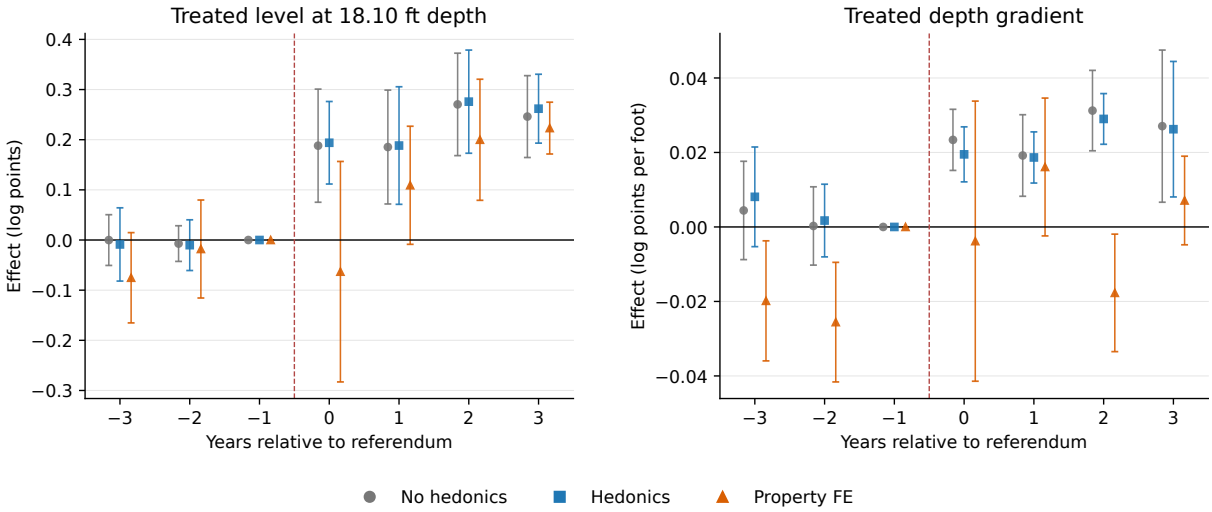
Notes: Dynamic coefficients from the hedonic absolute-depth specification (minus elevation in feet NAVD88), with tract and pair-by-year effects and main hedonic controls. Year $t-1$ is omitted. Panel (a) reports the treated level at zero feet NAVD88; panel (b) reports the treated absolute-depth gradient in log points per foot. Estimation uses event years -10 through $+10$ and displays -3 through $+3$. Bars are boundary-pair-clustered 95% confidence intervals.

Figure A.6: Average house-price dynamics across specifications



Notes: Treated (left) and neighboring spillover properties within 600 m of the treated levee (right), relative to remaining neighboring controls. Grey circles omit hedonics; blue squares include living area, lot size, stories, age, and missing indicators; orange triangles use property fixed effects on properties sold both before and after the vote. The first two specifications absorb tract and pair-by-year fixed effects; the third absorbs property and pair-by-year fixed effects. These average-effect models omit depth interactions and nearest-levee distance. Estimation spans event years -10 through $+10$, displaying -3 through $+3$, with $t-1$ omitted. Samples contain 37,683 transactions and 15 boundary pairs for the first two specifications, and 7,830 transactions and 12 pairs for property fixed effects. Bars show pair-clustered 95% confidence intervals using the regression residual degrees of freedom; horizontal offsets distinguish specifications. The property-FE comparison changes both controls and sample.

Figure A.7: House-price levels and crest-depth gradients across specifications



Notes: Dynamic full-factorial group-by-event-year-by-depth specifications. Left: the treated effect evaluated at a common crest depth of 18.10 ft, the full boundary sample's treated mean. Right: the treated depth gradient in log points per foot. Depth is positive exposure below the own-side district median crest. Specifications, samples, event windows, normalization, colors, and inference follow Figure A.6; the hedonic DDD specification additionally controls for nearest-levee distance. The left panel is a level at the stated depth, rather than the pooled average effect in the preceding figure.

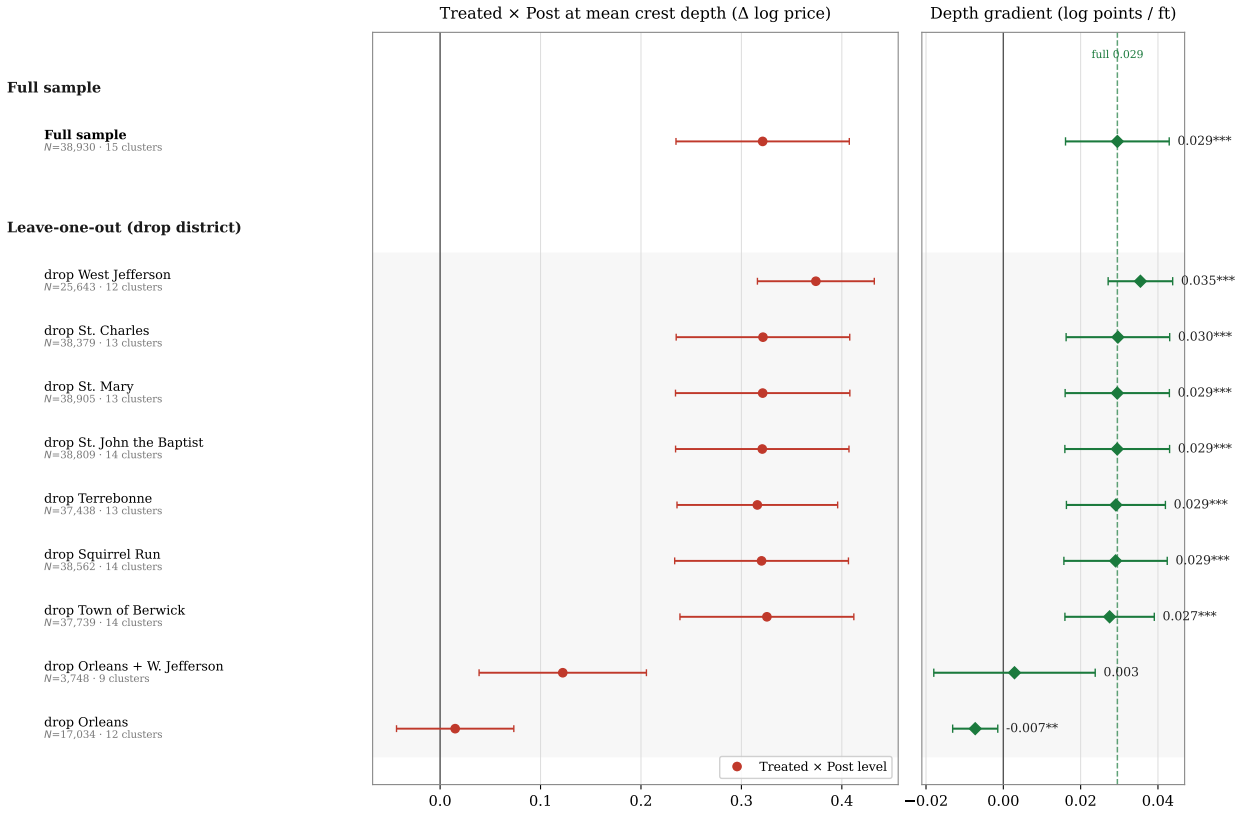
A.6 Sample Composition and Leave-One-Out Robustness

Table A.5: Sample Composition by Referendum

District	Year	Transactions	Boundary pairs	Properties	Share (%)
A. Hedonic estimation sample					
Orleans	2012	21,896	3	16,398	56.2
West Jefferson	2018	13,286	3	10,042	34.1
Terrebonne	2012	1,492	2	1,359	3.8
Town of Berwick	2020	1,191	1	930	3.1
St. Charles	2015	551	2	443	1.4
Squirrel Run	2021	368	1	284	0.9
St. John the Baptist	2017	121	1	99	0.3
St. Mary	2010	25	2	24	0.1
Total		38,930	15	29,579	100.0
B. Property-fixed-effects estimation sample					
Orleans	2012	4,115	3	1,721	51.2
West Jefferson	2018	3,358	3	1,438	41.8
Town of Berwick	2020	243	1	106	3.0
Terrebonne	2012	146	2	69	1.8
St. Charles	2015	95	1	41	1.2
Squirrel Run	2021	69	1	30	0.9
St. John the Baptist	2017	13	1	6	0.2
St. Mary	2010	0	0	0	0.0
Total		8,039	12	3,411	100.0

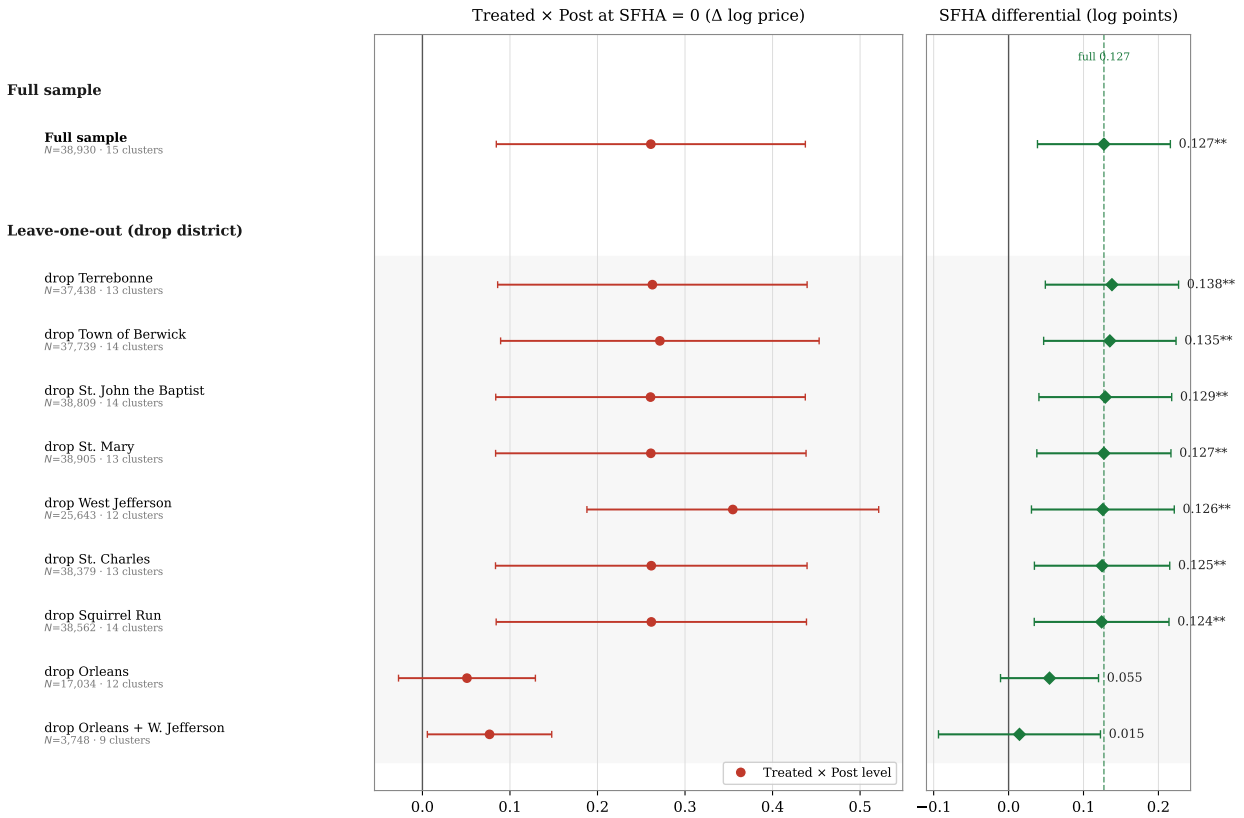
Notes: Each row describes a referendum, including transactions in the passed district and its neighboring districts, in the static continuous-crest-depth specification. Counts use actual estimation samples after singleton fixed-effect drops. Panel A uses hedonic controls, tract and boundary-pair-by-year effects, Panel B uses property and boundary-pair-by-year effects on repeat pre/post sales. Boundary pairs count represented pairs, with the overall count in the total row. Properties are distinct property identifiers within a row. Shares describe transaction-row counts.

Figure A.8: Leave-One-District-Out Robustness: Hedonic Specification



Notes: The first row is the full-sample continuous-depth price specification with hedonic levels and missing indicators, tract and boundary-pair × year fixed effects. Each remaining row excludes all referendum stacks of the named district, including their neighbor controls; the joint exclusion removes Orleans and West Jefferson. Exclusions are ordered from largest to smallest signed depth-gradient estimate. Left: Treated × Post at the full-sample treated mean crest depth, held fixed across exclusions. Right: Treated × Post × Depth per foot. Bars are normal-approximation 95% intervals (estimate ±1.96 boundary-pair clustered standard errors). Stars use the regression p-values: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure A.9: Leave-One-District-Out Robustness: SFHA-Only Exposure



Notes: SFHA (Special Flood Hazard Area) is the current FEMA 1%-annual-chance floodplain classification, obtained from the National Flood Hazard Layer (NFHL) by matching property coordinates to flood-zone polygons. The model replaces continuous-depth interactions with the current FEMA SFHA indicator, retaining the main hedonic estimation sample, controls, tract fixed effects, and boundary-pair × year fixed effects. Across main-sample transactions with observed flood-zone status, the Pearson correlation between SFHA and depth below the district median levee crest is 0.289 ($N = 39,166$), and 0.238 when Orleans is excluded ($N = 17,270$). It includes all interactions among treated/spillover/control status, Post, and SFHA. Each exclusion removes the named district's entire referendum stack. Left: Treated × Post for SFHA = 0. Right: Treated × Post × SFHA, the additional treated-versus-control post-vote change for SFHA properties relative to non-SFHA properties. Unavailable SFHA status retains the main pipeline's zero coding, with a missing-status control. Bars 95% intervals using boundary-pair clustered standard errors; stars use regression t-test p-values ($^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$). The dashed line marks the full-sample SFHA differential. The current flood map may reflect levee accreditation; SFHA is not necessarily predetermined relative to the vote.

A.7 Close Referenda Analysis

This exercise compares close passed and close failed levee-tax referenda using the entire district, not just the boundaries. A referendum qualifies when the yes-vote share is strictly between 40% and 60%, equivalently $|\text{yes share} - 50| < 10$ percentage points. The comparison is motivated by Cellini, Ferreira, and Rothstein (2010), but the small number of referenda and relatively wide vote-margin window mean that proximity to the threshold alone does not establish random assignment, but it is the closest we can get to that. This serves as an alternative design to compare with the paper's main specification - the boundary discontinuity specification.

Selected referenda and clean controls. Ten referendum cohorts satisfy the vote-margin rule of $\pm$ 10% around majority. The main comparison retains three passed cohorts: North Lafourche (2015), Algiers (2015), and West Jefferson (2018), and five failed cohorts: Iberia (2013), Lake Borgne (2015), Plaquemines (2020 and 2021), and Chenier Plain (2021). These eight cohorts span seven districts. Table A.6 lists their margins and sample support. The failed North Lafourche 2012 and West Jefferson 2015 cohorts are excluded from the control group, because they both have subsequent passes.[31]

Table A.6: Close Referenda: Selection and Estimation-Sample Support

District	Year	Result	Margin (pp)	Panel rows	Hedonic	Property FE
Included in the main clean-control comparison						
North Lafourche	2015	Passed	+7.02	5,820	5,818	1,008
Algiers	2015	Passed	+4.06	3,952	3,952	663
West Jefferson	2018	Passed	+9.61	16,927	16,927	2,870
Iberia	2013	Failed	-0.16	5,279	5,278	635
Lake Borgne	2015	Failed	-8.39	6,851	6,851	1,294
Plaquemines	2020	Failed	-6.26	1,507	1,507	301
Chenier Plain	2021	Failed	-9.83	23,126	23,126	4,222
Plaquemines	2021	Failed	-6.26	1,341	1,341	246
Total				64,803	64,800	11,239
Excluded failed cohorts						
North Lafourche	2012	Failed	-4.16	–	–	–
West Jefferson	2015	Failed	-8.43	–	–	–

Notes: Margin is yes-vote share minus 50, in percentage points. Panel rows count transaction–referendum observations. Hedonic and Property FE count the observations used in the static triple-difference regressions. Excluded cohorts are shown for transparency and enter none of the reported clean-control exhibits. Plaquemines appears twice because its two referenda are separate stacks, some transactions can therefore appear in more than one stack.

[31] Their five-year post-vote windows overlap their districts' later successful votes: North Lafourche passed in 2015 and West Jefferson in 2018. We remove these failed cohorts entirely, retaining the later passed cohorts. Lake Borgne's failed 2015 referendum remains a control despite its earlier passage in 2011.

Transactions, geography, and risk. For each retained referendum, transactions are assigned by containment in the union of its election-year voting precincts. We use clean single-parcel residential sales with non-missing log price and elevation, sale years from 2008 onward, and event years -5 through $+5$. The initial panel has no leveed-area restriction; the continuous-gradient table also reports the subset within leveed areas. The resulting panel has 64,803 transaction–referendum rows. For the raw-trend figure only, the high-risk indicator is elevation at or below the referendum-specific median among pre-vote transactions from 2008 onward; the builder computes this cutoff before imposing the five-year event window, then holds it fixed. Low-risk observations are above that median. This binary elevation split is distinct from the continuous crest-depth gradient used in the main boundary regressions.

Continuous depth and distance gradients. Table A.7 reports the hedonic comparison using continuous depth below the district median crest and distance to the nearest levee. For each measure, we report the whole-district sample and the subset within leveed areas. The sample, transformations, fully interacted specification, separate pooled average effect, and pre-period diagnostics are detailed in the table notes. These columns retain the same clean-control referendum cohorts described above.

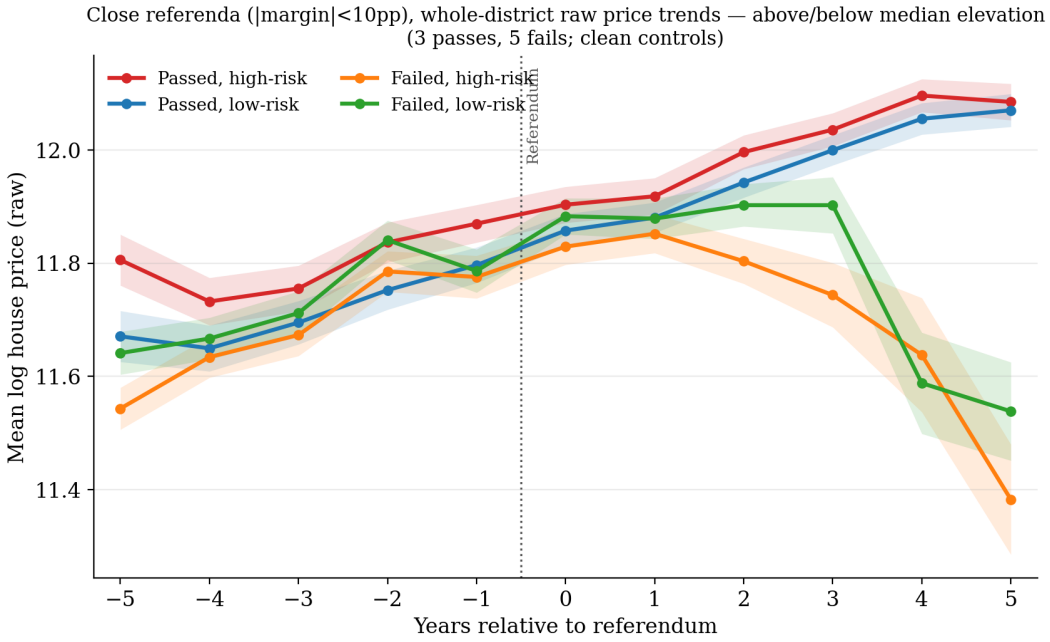
Table A.7: Close Passed versus Close Failed: Hedonic

	Depth (feet)		Distance (km)	
	(1) All areas	(2) Leveed only	(3) All areas	(4) Leveed only
Passed × Post (at mean X)	0.114*	0.074*	0.103	0.078*
	(0.049)	(0.037)	(0.055)	(0.039)
Post × X (failed)	0.006	-0.004	-0.003	-0.007*
	(0.004)	(0.008)	(0.002)	(0.003)
Passed × Post × X	-0.014**	-0.002	-0.018*	-0.013
	(0.004)	(0.008)	(0.009)	(0.009)
Gradient p-value	0.0101	0.8417	0.0819	0.2053
Average pass effect (pooled)	0.151**	0.075	0.151**	0.075
	(0.042)	(0.039)	(0.042)	(0.039)
Average effect p-value	0.0111	0.1037	0.0111	0.1037
Pre-period joint p-value	0.0249	0.0155	0.0537	0.0433
Center of X	16.448	16.320	2.034	2.048
Hedonic controls	Yes	Yes	Yes	Yes
Tract-by-referendum FE	Yes	Yes	Yes	Yes
Property-by-referendum FE	No	No	No	No
Year FE	Yes	Yes	Yes	Yes
Clusters (districts)	7	7	7	7
Observations	64,800	35,989	64,800	35,989
R^2	0.366	0.356	0.365	0.355

Sample. Close: $40 <$ yes share < 60. Passed: North Lafourche 2015, Algiers 2015, West Jefferson 2018. Failed: Iberia 2013, Lake Borgne 2015, Plaquemines 2020/2021, Chenier Plain 2021. Failed North Lafourche 2012 and West Jefferson 2015 are excluded because their post-vote windows overlap later passage, their passed cohorts remain. The control-group definition retains Lake Borgne 2015 despite passage in 2011. Sales are from 2008 onward, event years -5 through $+5$. "All areas" includes inside and outside leveed areas; "Leveed only" is properties protected by any levee in the NLD. Plaquemines stacks can repeat transactions. The leveed cut reduces Chenier Plain from 23,126 to 48 panel rows before estimation. district-clustered SEs (seven clusters; three passed) $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$

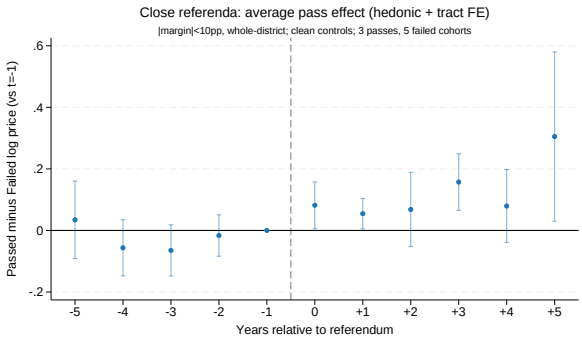
Raw trends and event studies. Figures A.10 and A.11 use the same retained clean-control cohorts. Raw trends pool transaction-level log prices within passed/failed-by-risk-by-event-year cells without cohort normalization; the bands are descriptive, unclustered intervals for the cell mean. Event studies pool risk and estimate passed-minus-failed price paths relative to event year −1, using tract-by-referendum and calendar-year fixed effects with hedonic controls in the hedonic panel, and property-by-referendum and calendar-year fixed effects without hedonic controls on repeat pre/post sales in the property-FE panel. Neither figure uses the boundary-pair sample.

Figure A.10: Close Referenda: Raw Price Trends

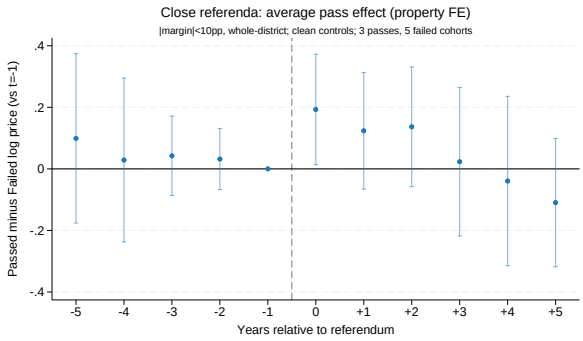


Notes: Raw pooled mean log price, three passed and five failed referendum cohorts from Table A.6. High/low risk uses the pre-vote within-referendum median elevation split. Shaded bands are mean 95% confidence intervals. Failed North Lafourche 2012 and West Jefferson 2015 are excluded, Lake Borgne 2015 remains despite passage in 2011.

Figure A.11: Close Referenda: Average Pass Effect



(a) Hedonic + tract FE



(b) Property FE (repeat pre/post)

Notes: Coefficients on passed × event-year indicators, relative to $t = -1$ (normalized to zero). Both panels follow Table A.6, with the same exclusions and retained prior-treatment control described there. Panel (a) uses hedonic controls and tract-by-referendum and calendar-year fixed effects, panel (b) uses property-by-referendum and calendar-year effects on repeat pre/post sales. Bars show normal-approximation 95% confidence intervals using district-clustered standard errors. Seven district clusters, including three passed districts, support inference.

A.8 Risk Measures and Specification Checks

The following exhibits use the current main price-analysis sample and compare alternative risk measures, identical repeat-sales observations, and the sequence of controls and fixed effects. The monthly-sales figure uses transaction dates from the exact hedonic estimation sample.

Table A.8: Risk Measures: Depth, Levee Distance, SFHA, and Levee-Protected Zone X

	(1) Depth	(2) Distance	(3) Depth + distance	(4) SFHA + X	(5) All four
Treated × Post	0.321***	0.265***	0.308***	0.234**	0.234***
	(0.044)	(0.085)	(0.032)	(0.092)	(0.060)
Treated × Post × Depth (ft)	0.030***		0.030***		0.027***
	(0.007)		(0.006)		(0.006)
Treated × Post × Distance (km)		-0.025	-0.027		-0.022
		(0.020)	(0.017)		(0.020)
Treated × Post × SFHA				0.139***	0.089*
				(0.044)	(0.049)
Treated × Post × Protected X				-0.005	0.065
				(0.091)	(0.056)
Hedonic controls	Yes	Yes	Yes	Yes	Yes
Tract FE	Yes	Yes	Yes	Yes	Yes
Pair-by-year FE	Yes	Yes	Yes	Yes	Yes
Boundary pairs	15	15	15	15	15
Transactions	38,930	38,930	38,930	38,930	38,930
R^2	0.587	0.580	0.587	0.583	0.589

Notes: All estimates use the current passed-referendum boundary price sample: the 2,354 m bandwidth, leveed-area and single-parcel restrictions, the current overlap and neighbor-treatment rules, and no additional static event-year window. Groups are treated, spillover (neighbor within 600 m of the treated levee), and control (remaining neighbors). Parentheses are boundary-pair-clustered standard errors; $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$. Spillover-group terms and their applicable risk interactions are included in every regression but omitted from the displayed coefficient rows. All five columns use exactly the same 38,930 observations and 15 boundary pairs, with tract and pair-by-year fixed effects and the main hedonic controls. Columns include (1) depth, (2) distance, (3) depth and distance, (4) separate SFHA and levee-protected Zone X indicators, and (5) all four measures. Depth is the main depth below the district median crest in feet, with main-sample centering held fixed. Distance is kilometers to the treated district's levee, centered within the treated sample and set to zero for neighbors; its main and post terms identify the within-treated distance gradient. Protected X means only FEMA's Zone X subtype with reduced flood risk due to a levee. SFHA and Protected X enter separately and are mutually exclusive in this sample, properties in neither category form the omitted map group. Each included depth or map indicator has all group, post, and exposure lower-order terms and interactions. Column (5) includes these additively, without interactions among the four risk measures. The treated level is evaluated at included continuous exposures' centers and, in columns (4)–(5), SFHA and Protected X equal to zero. Hedonic controls include age, sqft, lot size and stories.

Table A.9: Triple difference specs on repeat-sales sample

	(1) No hedonics	(2) Hedonics	(3) Property FE
Treated × Post	0.168***	0.206***	0.169***
	(0.039)	(0.046)	(0.036)
Treated × Post × Depth	0.021**	0.024***	0.020**
	(0.008)	(0.005)	(0.007)
Spillover × Post	0.109**	0.086*	0.132***
	(0.040)	(0.043)	(0.035)
Spillover × Post × Depth	-0.036***	-0.036***	-0.042***
	(0.005)	(0.005)	(0.003)
Post × Depth	-0.002	-0.005	0.000
	(0.005)	(0.005)	(0.003)
Hedonic controls	No	Yes	No
Tract FE	Yes	Yes	No
Property FE	No	No	Yes
Pair-by-year FE	Yes	Yes	Yes
Property-FE sample N	8,039	8,039	8,039
Boundary pairs	12	12	12
Transactions	8,039	8,039	8,039
R^2	0.540	0.596	0.861

Notes: All estimates use the current passed-referendum boundary price sample: the 2,354 m bandwidth, leveed-area and single-parcel restrictions, the current overlap and neighbor-treatment rules, and no additional static event-year window. Groups are treated, spillover (neighbor within 600 m of the treated levee), and control (remaining neighbors). Parentheses are boundary-pair-clustered standard errors; $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$, using the regression's finite-sample t distribution. Every column uses the same 8,039 transaction rows and 12 boundary pairs from the property-FE estimation sample. The sample is the intersection of the observations retained by all three estimators; this retains every observation in the property-FE sample. The first two columns absorb tract and pair-by-year effects; the third absorbs property and pair-by-year effects. Only column (2) adds the main hedonic controls and missingness flags. Every regression includes the full group-by-post-by-continuous-depth interaction. Depth is the main capped district-median-crest measure in feet, centered at the full main treated-sample mean, held fixed across columns. The first row is the treated level effect at that center. Matching observations isolates specification changes from the repeat-sales sample selection; it does not resolve selection into repeat selling.

Table A.10: Building the Hedonic Specification: Controls and Fixed Effects

Panel A: Estimator-specific samples

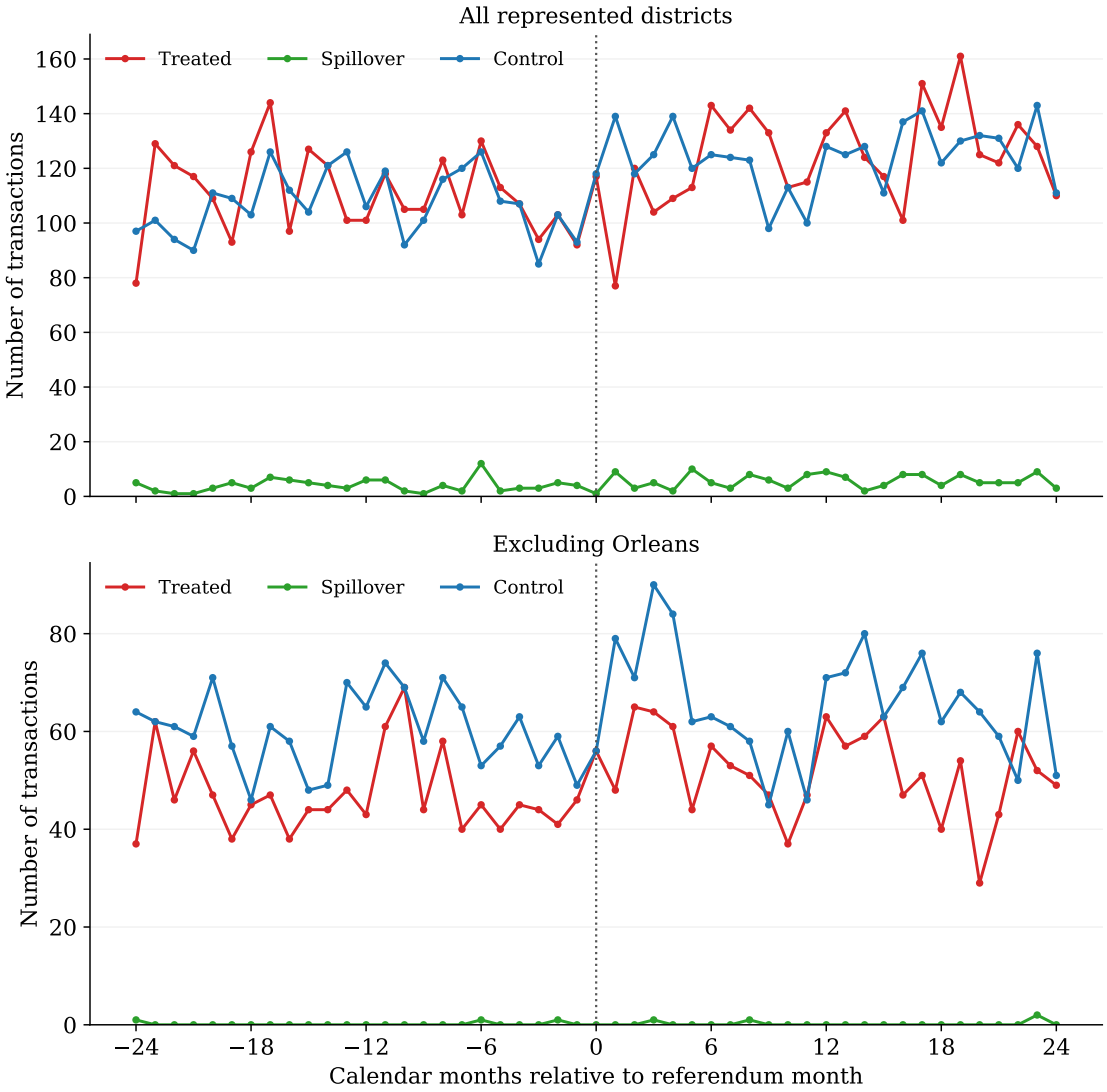
	(1) No FE	(2) Tract	(3) + Year	(4) + Pair-year	(5) + Hedonics	(6) + Parish-year
Treated × Post	0.285***	0.172**	0.185*	0.267***	0.321***	0.251***
	(0.082)	(0.074)	(0.090)	(0.037)	(0.044)	(0.059)
Treated × Post × Depth	0.026*	0.028*	0.019	0.024***	0.029***	0.020***
	(0.015)	(0.014)	(0.013)	(0.007)	(0.007)	(0.005)
Hedonic controls	No	No	No	No	Yes	Yes
Tract FE	No	Yes	Yes	Yes	Yes	Yes
Year FE	No	No	Yes	No	No	No
Pair-by-year FE	No	No	No	Yes	Yes	Yes
Parish-by-year FE	No	No	No	No	No	Yes
Boundary pairs	16	16	16	15	15	15
Transactions	38,947	38,946	38,946	38,930	38,930	38,928
Rows lost from construction sample	0	1	1	17	17	19
R^2	0.121	0.505	0.526	0.535	0.587	0.593

Panel B: Identical observations across columns

	(1) No FE	(2) Tract	(3) + Year	(4) + Pair-year	(5) + Hedonics	(6) + Parish-year
Treated × Post	0.286***	0.172**	0.185*	0.268***	0.321***	0.251***
	(0.082)	(0.074)	(0.090)	(0.037)	(0.044)	(0.059)
Treated × Post × Depth	0.026*	0.028*	0.019	0.024***	0.030***	0.020***
	(0.015)	(0.014)	(0.013)	(0.007)	(0.007)	(0.005)
Hedonic controls	No	No	No	No	Yes	Yes
Tract FE	No	Yes	Yes	Yes	Yes	Yes
Year FE	No	No	Yes	No	No	No
Pair-by-year FE	No	No	No	Yes	Yes	Yes
Parish-by-year FE	No	No	No	No	No	Yes
Boundary pairs	15	15	15	15	15	15
Transactions	38,928	38,928	38,928	38,928	38,928	38,928
Rows lost from construction sample	19	19	19	19	19	19
R^2	0.121	0.505	0.526	0.536	0.587	0.593

Notes: All estimates use the current passed-referendum boundary price sample: the 2,354 m bandwidth, leveed-area and single-parcel restrictions, the current overlap and neighbor-treatment rules, and no additional static event-year window. Groups are treated, spillover (neighbor within 600 m of the treated levee), and control (remaining neighbors). Parentheses are boundary-pair-clustered standard errors; $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$, using the regression's finite-sample t distribution. The six columns are (1) no controls or fixed effects; (2) tract fixed effects; (3) tract and year effects; (4) tract and pair-by-year effects, replacing separate year effects; (5) the same fixed effects plus hedonics; and (6) adds parish-by-year fixed effects, retaining tract effects, pair-by-year effects, and hedonics. All columns retain the full group-by-post-by-depth interaction. Panel A lets each estimator drop its own singleton observations. The construction sample has 38,947 rows; column (5) retains 38,930 and column (6) retains 38,928. The first 17-row reduction occurs before adding hedonics; parish-by-year effects remove 2 additional singleton observations. Panel B reruns the ladder on the iterated intersection of actual estimation samples, with 38,928 identical rows asserted across all six columns. Depth and its full-sample treated-mean centering are fixed throughout. Hedonic characteristics are treated as in the main specification. The table displays treated level and gradient terms; spillover terms remain in every regression.

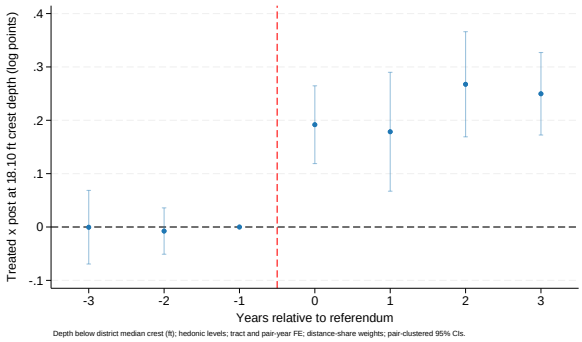
Figure A.12: Monthly Sales Around the Referendum



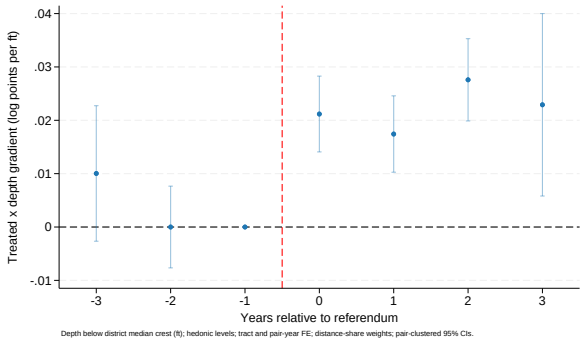
Notes: Raw transaction counts from the exact hedonic price estimation sample, within 24 calendar months before and after each referendum month. The upper panel pools all represented districts; the lower panel removes the entire Orleans stack, including its neighbors. All 15 represented pairs have complete windows after the 2008 start, data end, and neighbor-treatment censoring checks. Empty pair/group/month cells are included as zero sales. A total of 11,667 transaction rows enter this display. Month zero includes the entire referendum month and therefore mixes sales before and after the vote date. Counts retain the main overlap rule: within-event multi-pair properties are excluded, and each property is assigned to its earliest surviving referendum across events. This descriptive plot shows no isolated aggregate pre-vote spike in the immediately preceding months. It is not seasonally adjusted, a formal bunching test, or evidence that prices did not anticipate passage.

A.9 Distance-Weighted Overlapping Boundaries

Figure A.13: House-Price Triple Difference with Distance-Weighted Overlap



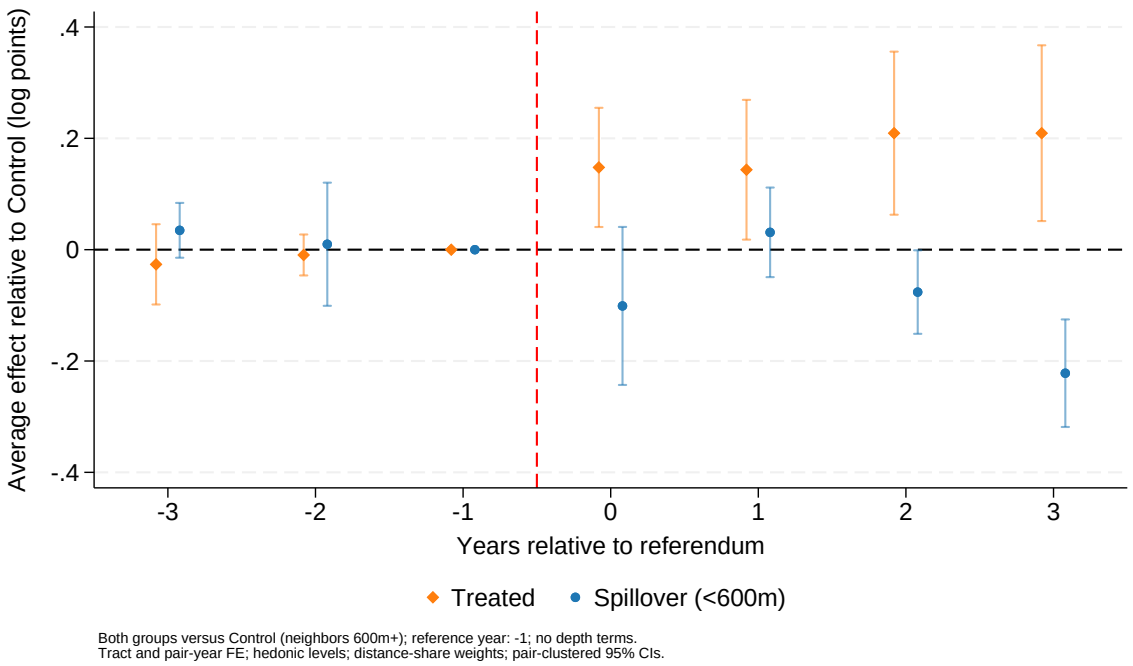
(a) Treated level at the main-sample mean depth



(b) Treated depth gradient

Notes: The continuous-depth hedonic triple difference retains properties within 2,354 projected meters of two boundaries of the same referendum. Following the slides' distance-weight rule, each property's weight in pair j is $d_{-j}/(d_j + d_{-j})$, where d_j is its minimum distance to boundary j across transactions; equal distances receive equal weights, and a property in only one pair receives weight one. Weights sum to one across pairs within a property–referendum before estimation; they are not renormalized after estimator-specific exclusions. Shared properties are removed from the West Jefferson–East Jefferson comparison on both sides if they occur in another referendum, retaining their earlier appearances. The model includes tract and pair-by-year fixed effects, the main hedonic controls, and all group-by-event-time-by-depth interactions. Depth is depth below the district median levee crest, centered at the current main-sample treated mean. Year $t-1$ is omitted; estimation spans -10 to $+10$, with -3 to $+3$ displayed. Bars show boundary-pair-clustered 95% confidence intervals.

Figure A.14: Average House-Price Event Study with Distance-Weighted Overlap



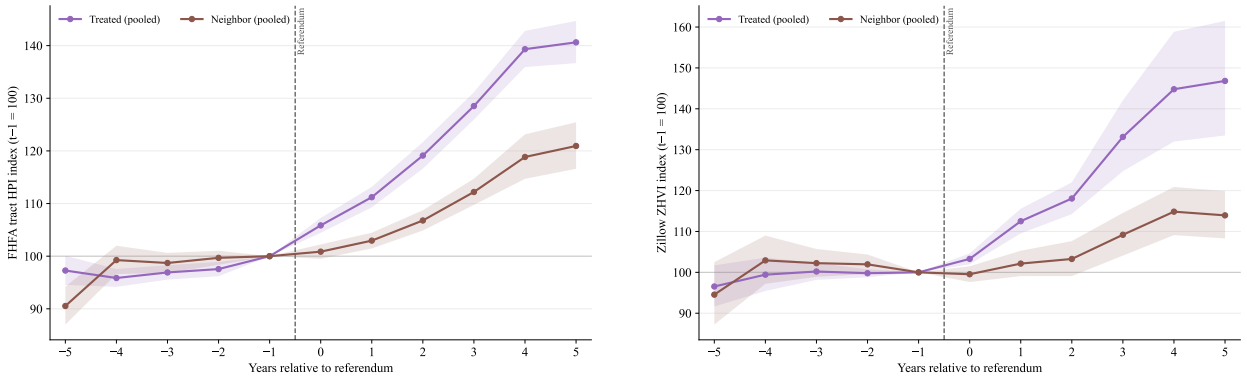
Notes: This reproduces the slides' average-effect event-study specification with the same distance weights and cross-referendum exclusion as Figure A.13. It includes tract and pair-by-year fixed effects, hedonic levels and missing-value indicators, and no depth interactions. Treated properties and neighboring spillover properties within 600 m of the treated levee are compared with the remaining neighboring controls. Year $t-1$ is omitted; estimation spans -10 to $+10$, with -3 to $+3$ displayed. Bars show boundary-pair-clustered 95% confidence intervals.

A.10 Raw Price Trends in External Repeat-Sales Indices

Figure A.15 uses the FHFA tract house-price index and the Zillow index for matched zip codes, which overlap with the boundary properties. I rebase each series to the year before the referendum and average across areas. The treated areas rise above their neighbors after the vote in both indices.

Figure A.16 splits the FHFA tracts by their mean property depth below their own district's levee crest. Panel (a) keeps the neighbors together, while panel (b) separates neighbor tracts with some properties close to the treated levee. Both treated depth groups rise above the pooled neighbors, but the shallower treated group rises slightly more. These indices capture the average pattern, but risk splitting at the tract level is coarse. The near-neighbor comparison is also diluted because the index covers the whole tract.

Figure A.15: Raw repeat-sales price trends around referendum passage: treated vs. neighbor

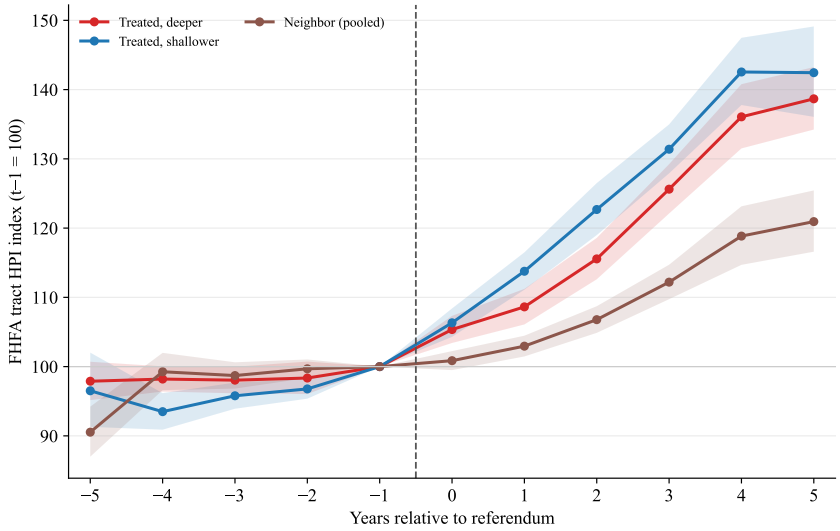


(a) FHFA census-tract HPI

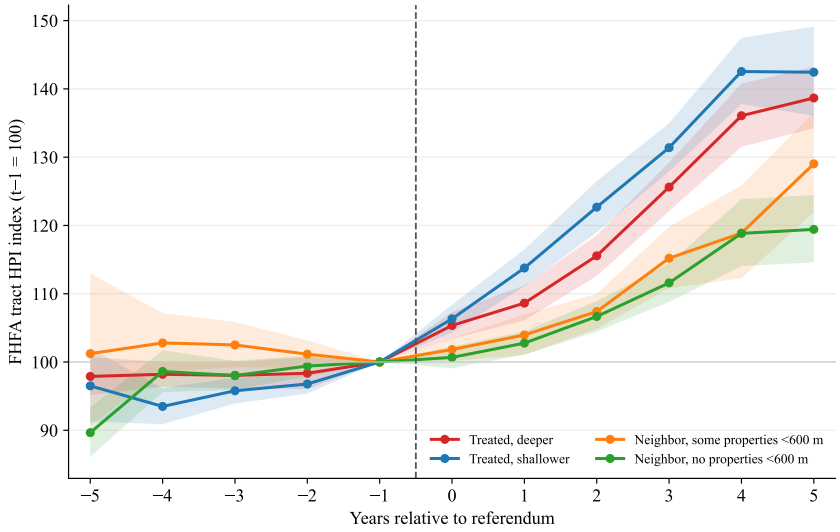
(b) Zillow ZHVI

Notes: Repeat-sales indices for areas represented by the 38,930 transactions in the current hedonic estimation sample. The sample covers 207 distinct 2020 census tracts, 131 have usable positive FHFA observations and a pre-vote baseline. Each tract or zip is rebased to $t-1=100$, lines are geometric means, giving each area–referendum observation equal weight. Years run from 2008 to 2024, with neighbors censored before their own passage; the displayed window is −5 to +5. Coverage changes across event years, and points require at least two referenda. Shading shows descriptive 95% intervals from the dispersion of log indices, not boundary-pair-clustered regression inference. The Zillow ZIP proxy uses the following crosswalk: choose the ZIP most concentrated in each tract by its share of ZIP residential addresses, then assign the ZIP to the side with more represented tracts; ties are dropped. ZIP coverage is coarser than the boundary sample.

Figure A.16: FHFA price trends by tract depth and proximity to the treated levee



(a) Treated depth groups and pooled neighbors



(b) Separating neighbor tracts with properties near the treated level

Notes: Same index construction as Figure A.15. Purple and brown identify pooled treated and neighbor areas there. Here, treated tracts split into deeper (red) and shallower (blue); pooled neighbors remain brown in panel (a). Depth is exposure below each property's own district median crest, averaged over distinct sampled properties within each tract and referendum. The depth cutoff is the pre-vote median among distinct properties on the same side of that referendum. In panel (b), orange identifies neighbor tracts with any sampled properties within 600 m of the treated district's levee; green identifies the remaining neighbors. Ten contributing tract–referendum observations contain near properties, but these account for only 0.7–29.7% of their sampled properties. No contributing neighbor tract has a majority inside the ring. This whole-tract comparison therefore cannot isolate the property-level spillover effect.

A.11 Boundary Bandwidth Sensitivity

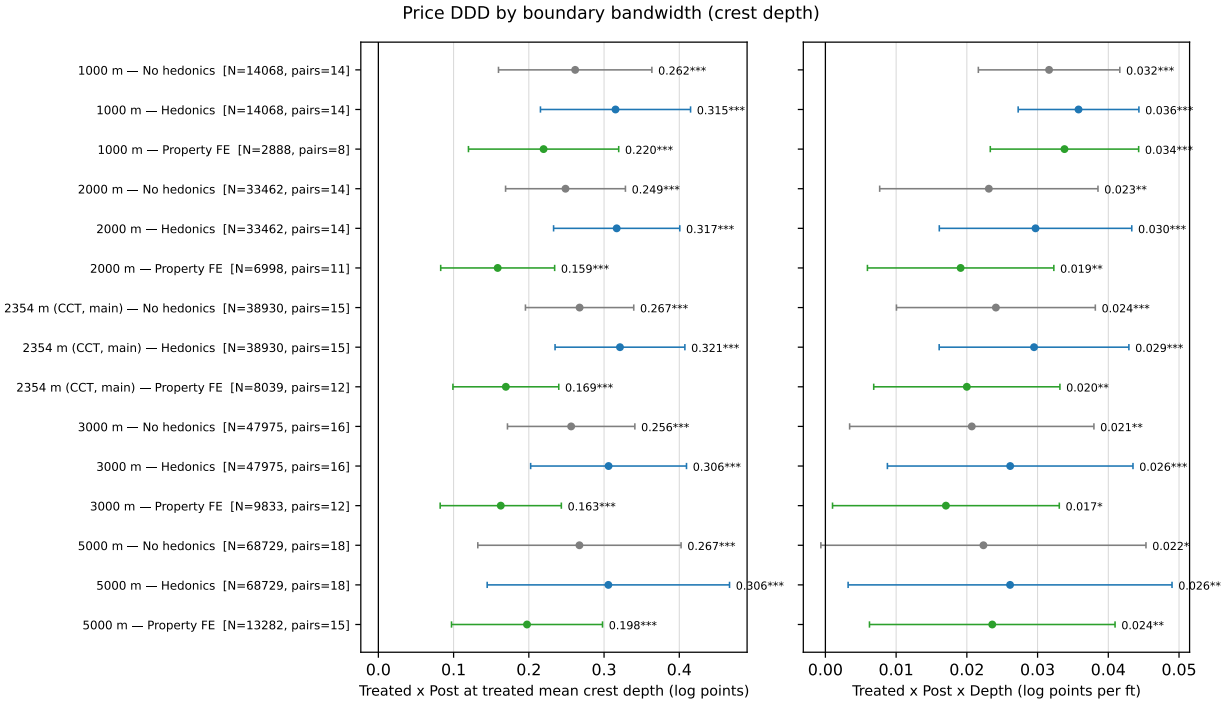
Table A.11 and Figure A.17 report the same static estimates across boundary bandwidths. Figure A.18 reports the hedonic dynamics, holding the level evaluation point fixed across bandwidths.

Table A.11: Boundary-bandwidth sensitivity: crest depth

Bandwidth	Specification	Treated level	Depth gradient	Sales	Pairs
1000 m	No hedonics	0.262*** (0.052)	0.032*** (0.005)	14,068	14
1000 m	Hedonics	0.315*** (0.051)	0.036*** (0.004)	14,068	14
1000 m	Property FE	0.220*** (0.051)	0.034*** (0.005)	2,888	8
2000 m	No hedonics	0.249*** (0.041)	0.023** (0.008)	33,462	14
2000 m	Hedonics	0.317*** (0.043)	0.030*** (0.007)	33,462	14
2000 m	Property FE	0.159*** (0.039)	0.019** (0.007)	6,998	11
2354 m (CCT, main)	No hedonics	0.267*** (0.037)	0.024*** (0.007)	38,930	15
2354 m (CCT, main)	Hedonics	0.321*** (0.044)	0.029*** (0.007)	38,930	15
2354 m (CCT, main)	Property FE	0.169*** (0.036)	0.020** (0.007)	8,039	12
3000 m	No hedonics	0.256*** (0.043)	0.021** (0.009)	47,975	16
3000 m	Hedonics	0.306*** (0.053)	0.026*** (0.009)	47,975	16
3000 m	Property FE	0.163*** (0.041)	0.017* (0.008)	9,833	12
5000 m	No hedonics	0.267*** (0.069)	0.022* (0.012)	68,729	18
5000 m	Hedonics	0.306*** (0.082)	0.026** (0.012)	68,729	18
5000 m	Property FE	0.198*** (0.051)	0.024** (0.009)	13,282	15

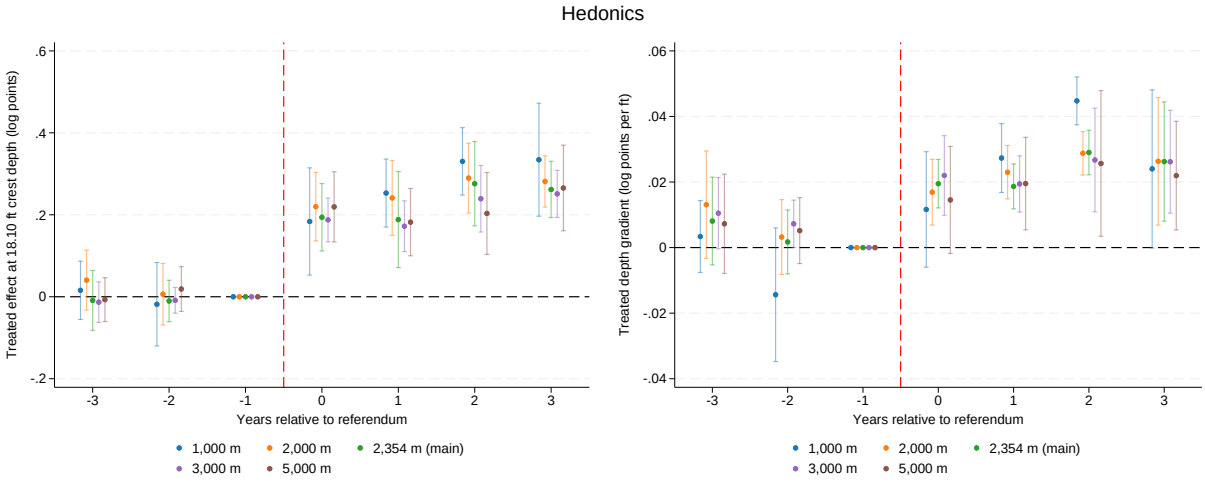
Notes: The continuous-depth price model is re-estimated at each boundary bandwidth. Entries are coefficients with boundary-pair-clustered standard errors in parentheses. Gradients are log points per foot. No-hedonics and hedonic specifications absorb tract and pair-by-year fixed effects; property FE uses pre/post repeat sales and property and pair-by-year fixed effects. The overlap rule and estimation sample are rebuilt at each bandwidth; the spillover definition remains 600 m from the treated levee. The treated level is evaluated at the treated mean crest depth of each bandwidth-specific sample; both sample composition and the evaluation point can change. Stars use finite-sample regression p-values: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure A.17: Static house-price levels and depth gradients across bandwidths



Notes: The same estimates as Table A.11, for bandwidths of 1,000, 2,000, 2,354 (main), 3,000 and 5,000 m. Left: the treated level at each bandwidth-specific sample's mean treated crest depth. Right: the treated depth gradient in log points per foot. Colors distinguish no hedonics, hedonics and property fixed effects. Samples and overlap exclusions are rebuilt at each bandwidth; the spillover cutoff remains 600 m. Whiskers are coefficient plus or minus 1.96 times the boundary-pair-clustered standard error; stars use the finite-sample regression p-values. The changing samples and level evaluation points should be considered when comparing columns.

Figure A.18: Dynamic Bandwidth Sensitivity: Hedonics



Notes: Each series uses a different boundary bandwidth: 1,000, 2,000, 2,354 (main), 3,000, or 5,000 m. Left: the treated level at the main-bandwidth treated mean crest depth (18.10 ft), held fixed across all five bandwidths. Right: the treated depth gradient per foot. Samples and overlap exclusions are rebuilt at each bandwidth; the spillover cutoff remains 600 m from the treated levee. Year $t-1$ is omitted. Bars show boundary-pair-clustered 95% confidence intervals.

A.12 Contemporaneous ballot items

Table A.12 lists the 8 passed levee-tax referenda and 42 contemporaneous ballot items recorded in their precincts. These are the levee-tax referenda represented in the main boundary sample. The inventory includes local propositions, constitutional amendments, and candidate races on the same election date. We identify precincts eligible to vote on each levee tax from recorded returns for that measure. Other items are listed if they appear in at least one eligible precinct. The levee tax appears first in each group. Brackets identify items also reported outside these precincts and give their total precinct count in the archive.

This inventory documents contemporaneous events that could affect outcomes on the treated side, complementing the school attendance-zone controls in Table **??**. For example, the St. Charles election included a proposed $42 million school bond and wastewater and disabled-persons millages; parts of the West Jefferson levee-tax electorate also voted on Gretna sewer and police taxes. Changes in school quality or other public services could contribute to house-price changes if their effects differ across the levee boundary. The table establishes electoral coincidence and precinct overlap, not passage, implementation, or exposure of the particular properties used in estimation. It therefore identifies potential confounders rather than ruling them out, and does not cover events outside the election date.

Table A.12: Passed levee-tax measures and contemporaneous ballot items

Type	Official contest title	Precincts
2010-05-01: St. Mary Levee District		
3 other items; 53 precincts eligible to vote on the levee tax		
Levee tax	PW Levee District – 5 Mills - BOC - Annually	53/53
Proposition	Mosquito Control Dist. No. 1 – 7.19 Mills - BOC - 5 Yrs.	2/53
Proposition	Recreation District No. 5 – 7-1/2 Mills - BOC - 10 Yrs.	6/53
Proposition	Water and Sewer Comm. #1 – 9.98 Mills - BOC - 10 Yrs.	7/53
2012-11-06: Orleans Levee District		
29 other items; 326 precincts eligible to vote on the levee tax		
Levee tax	Orleans Levee District – 6.07 Mills - BOC - 30 Yrs.	326/326
Amendment	CA NO. 1 (Act 873 - SB 82) – Medicaid Trust Fund for the Elderly *[366]*	326/326
Amendment	CA NO. 2 (Act 874 - SB 303) – Right of each Citizen to keep and bear arms *[366]*	326/326
Amendment	CA NO. 3 (Act 872 - SB 21) – Prefiling DL/for retirement/public employee *[366]*	326/326
Amendment	CA NO. 4 (Act 875 - SB 337) – Surv. Spouse/Deceased Veteran/Homestead Exemp. *[366]*	326/326
Amendment	CA NO. 5 (Act 868 - HB 9) – Forfeiture Retirement Benefit Pub.Off./Conv.Felon *[366]*	326/326
Amendment	CA NO. 6 (Act 869 - HB 497) – Gov.Auth. of New Iberia - prop.annexed by the City *[366]*	326/326
Amendment	CA NO. 7 (Act 870 - HB 524) – Provides filling appts/Vac. on Bds./Comm. *[366]*	326/326
Amendment	CA NO. 8 (Act 871 - HB 674) – State Bd. of Comm./Industry-Tax Exemp. contracts *[366]*	326/326

Continued on next page

Table A.12 continued

Type	Official contest title	Precincts
Amendment	CA NO. 9 (Act 876 - SB 410) – Security District/Parcel Fee w/in District *[366]*	326/326
Candidate race	Councilmember – District B	81/326
Candidate race	Councilmember – District E	57/326
Candidate race	Judge – Criminal District Court, Section B *[366]*	326/326
Candidate race	Judge, Court of Appeal – 4th Circuit, 1st District, Division H *[366]*	326/326
Candidate race	Member of School Board – District 1	50/326
Candidate race	Member of School Board – District 2	49/326
Candidate race	Member of School Board – District 3	55/326
Candidate race	Member of School Board – District 4 *[43]*	11/326
Candidate race	Member of School Board – District 6	56/326
Candidate race	Member of School Board – District 7 *[54]*	46/326
Candidate race	Presidential Electors *[366]*	326/326
Candidate race	U. S. Representative – 1st Congressional District	46/326
Candidate race	U. S. Representative – 2nd Congressional District *[320]*	280/326
Proposition	Gentilly Terrace & Gardens Sec. Dist. – Annual Fee - CC - 3 Yrs.	5/326
Proposition	Lake Vista Crime Prev. Dist. – $220 Parcel Fee - CC - 4 Yrs.	2/326
Proposition	N. Kenilworth Imprv. & Security Dist. – Annual Fee - CC - 8 Yrs.	3/326
Proposition	PW CC (HRC) – HRC Amendment - CC *[366]*	326/326
Proposition	PW CC (N.O. Reg. Bus. Park) – 20 Mills - CC - 20 Yrs. *[366]*	326/326
Proposition	PW Crescent City Bridge – 20 Yrs. - Toll Collection - Act 865 *[366]*	326/326
Proposition	PW School District – Local Option - Term Limits *[366]*	326/326
2012-12-08: Terrebonne Levee and Conservation District *1 other item; 115 precincts eligible to vote on the levee tax*		
Levee tax	PW Levee & Conserv. Dist. – 1/2% S&U Tax - BOC	115/115
Candidate race	Judge, Court of Appeal – 1st Circuit, 1st Dist., Division B *[189]*	115/115
2015-05-02: St. Charles Parish Government *3 other items; 44 precincts eligible to vote on the levee tax*		
Levee tax	PW Millage Prop. No. 1 (Levees) – 4 Mills - PC - 30 Yrs.	44/44
Proposition	PW Millage Prop. No. 2 (Wastewater) – 2.20 Mills - PC - 30 Yrs.	44/44
Proposition	PW Millage Prop. No. 3 (Disabled Persons) – .70 Mills - PC - 10 Yrs.	44/44
Proposition	PW School District No. 1 – $42M Bond - SB - 20 Yrs.	44/44
2017-04-29: St. John the Baptist Parish Government *2 other items; 39 precincts eligible to vote on the levee tax*		
Levee tax	PW Prop. – 7 Mills - PC - 30 Yrs.	39/39
Candidate race	Councilman – District 7	4/39
Candidate race	State Senator – 2nd Senatorial District	16/39
2018-04-28: West Jefferson Levee District *2 other items; 98 precincts eligible to vote on the levee tax*		
Levee tax	West Jefferson Levee District – 4.75 Mills - BOC - 10 Yrs.	98/98
Proposition	City of Gretna (Police) – 8.00 Mills - M&CC - 10 Yrs.	14/98
Proposition	City of Gretna (Sewer) – 2.4 Mills Renewal - M&CC - 10 Yrs.	12/98
2020-08-15: Town of Berwick *1 other item; 5 precincts eligible to vote on the levee tax*		
Levee tax	Town of Berwick – 6 Mills Renewal - M&TC - 20 Yrs.	5/5
Proposition	Library Proposition – 5.72 Mills Renewal - PC - 10 Yrs. *[42]*	5/5

Continued on next page

Table A.12 continued

Type	Official contest title	Precincts
2021-12-11: Squirrel Run Levee and Drainage District		
1 other item; 1 precinct eligible to vote on the levee tax		
Levee tax	Squirrel Run Levee and Drainage Dist. – 30 Mills - BOC - 10 Yrs.	1/1
Candidate race	City Marshal – City Court, City of New Iberia *[20]*	1/1

Notes: Source: precinct election returns downloaded from the Louisiana Secretary of State's Voter Portal—Election Results (`https://voterportal.sos.la.gov/Graphical`), using archived extracts. In the right column, the denominator is the number of precincts eligible to vote on the levee tax; the numerator counts those also reporting the listed item. Brackets give total archived precincts when greater than the overlap. Totals are lower bounds where parishes are missing (including statewide contests); unbracketed items may also extend beyond the archive. Early-voting and provisional aggregates are excluded; subprecincts are retained. Shared precincts need not have identical voter eligibility; unreported uncontested offices are excluded.

A.13 Wild-Cluster Bootstrap Inference

The following tables retain the main point estimates and report wild-cluster bootstrap tests using boundary pairs and referendum districts as alternative clustering units. In the full sample, 15 pairs share eight district-level referendum events; the property-fixed-effects sample has 12 pairs and seven districts. These are individual coefficient tests, without a multiple-testing adjustment.

Table A.13: Average House-Price Effects: Wild-Cluster Bootstrap

	(1) No hedonics	(2) Hedonics	(3) Property FE
Treated × Post	0.235	0.279	0.174
	(0.079)	(0.095)	(0.082)
	Pair p=0.040	Pair p=0.060	Pair p=0.281
	District p<0.001	District p=0.078	District p=0.625
Spillover × Post	-0.218	-0.249	-0.113
	(0.040)	(0.048)	(0.049)
	Pair p=0.233	Pair p=0.334	Pair p=0.109
	District p=0.312	District p=0.641	District p<0.001
Boundary pairs	15	15	12
Districts	8	8	7
Observations	38,930	38,930	8,039

Notes: Parentheses show the pair-clustered standard errors. Pair p and District p are wild-bootstrap p-values. Tests impose a zero effect and use up to 9,999 Rademacher sign-flip draws, or all patterns when fewer exist. Specifications match Table 5.

Table A.14: House-Price Depth Gradients: Wild-Cluster Bootstrap

	(1) No hedonics	(2) Hedonics	(3) Property FE
Treated × Post (at 18.10 ft crest depth)	0.267	0.321	0.169
	(0.037)	(0.044)	(0.036)
	Pair p=0.065	Pair p=0.080	Pair p=0.152
	District p=0.078	District p=0.172	District p=0.438
Treated × Post × Depth (per ft)	0.024	0.029	0.020
	(0.007)	(0.007)	(0.007)
	Pair p=0.265	Pair p=0.226	Pair p=0.375
	District p<0.001	District p=0.023	District p=0.234
Spillover × Post (at 18.10 ft crest depth)	0.106	0.106	0.132
	(0.072)	(0.045)	(0.035)
	Pair p=0.805	Pair p=0.889	Pair p=0.950
	District p=0.695	District p=0.812	District p=0.875
Spillover × Post × Depth (per ft)	-0.056	-0.059	-0.042
	(0.011)	(0.009)	(0.003)
	Pair p=0.641	Pair p=0.352	Pair p=0.249
	District p=0.648	District p=0.391	District p=0.516
Post × Depth (per ft)	-0.005	-0.007	0.000
	(0.005)	(0.005)	(0.003)
	Pair p=0.365	Pair p=0.100	Pair p=0.987
	District p=0.312	District p=0.008	District p=0.969
Boundary pairs	15	15	12
Districts	8	8	7
Observations	38,930	38,930	8,039

Notes: Parentheses show the pair-clustered standard errors. Pair p and District p are wild-bootstrap p-values. Tests impose a zero effect and use up to 9,999 Rademacher sign-flip draws, or all patterns when fewer exist. Specifications match Table 6; levels are evaluated at the treated-side mean crest depth.

Table A.15: Risk Measures Horse Race: Wild-Cluster Bootstrap

	(1) Depth	(2) Distance	(3) Depth + distance	(4) SFHA + X	(5) All four
Treated × Post	0.321	0.265	0.308	0.234	0.234
	(0.044)	(0.085)	(0.032)	(0.092)	(0.060)
	Pair p=0.080	Pair p=0.022	Pair p=0.022	Pair p=0.029	Pair p=0.030
	District p=0.172	District p=0.242	District p=0.211	District p=0.047	District p=0.109
Treated × Post × Depth (ft)	0.030		0.030		0.027
	(0.007)		(0.006)		(0.006)
	Pair p=0.226		Pair p=0.188		Pair p=0.078
	District p=0.016		District p=0.016		District p=0.086
Treated × Post × Distance (km)		-0.025	-0.027		-0.022
		(0.020)	(0.017)		(0.020)
		Pair p=0.355	Pair p=0.333		Pair p=0.303
		District p=0.656	District p=0.273		District p=0.594
Treated × Post × SFHA				0.139	0.089
				(0.044)	(0.049)
				Pair p=0.109	Pair p=0.334
				District p=0.070	District p=0.383
Treated × Post × Protected X				-0.005	0.065
				(0.091)	(0.056)
				Pair p=0.958	Pair p=0.374
				District p=0.875	District p=0.258
Boundary pairs	15	15	15	15	15
Districts	8	8	8	8	8
Transactions	38,930	38,930	38,930	38,930	38,930

Notes: Parentheses show the pair-clustered standard errors. Pair p and District p are wild-bootstrap p-values. Tests impose a zero effect and use up to 9,999 Rademacher sign-flip draws, or all patterns when fewer exist. Specifications match Table A.8. All columns include hedonics, tract and pair-year fixed effects, and spillover terms.

A.14 Quantity and insurance outcomes

Table A.16: Boundary-tract quantity and insurance summary statistics

	Mean	SD	p10	Median	p90	N
ACS occupancy (percent)	85.59	8.93	73.18	87.48	94.77	1983
USPS occupied addresses	1231.83	499.50	658.25	1185.00	1910.00	2405
ACS population	2863.71	1291.09	1352.50	2731.00	4618.00	1850
LODES resident workers	1147.20	549.44	516.00	1081.00	1867.00	2723
LODES jobs	1516.35	3976.34	112.00	537.00	3397.00	2711
ACS housing units	1277.88	497.02	719.00	1233.00	1944.00	1983
Premium per 1,000 coverage	3.64	1.50	2.04	3.23	5.81	2031
Claims / coverage (%), 2005–2024	17.20	20.37	1.33	8.36	48.81	445
Claims / coverage (%), 2009–2024	13.27	16.13	0.89	7.13	32.83	215
NFIP policy terms	644.99	412.09	206.00	557.00	1208.00	2031
Crest depth (ft)	17.06	4.81	11.44	17.49	23.39	2775
Elevation (ft NAVD88)	1.63	4.64	-4.47	1.57	7.71	2775

Notes: Available tract-year observations in the boundary panel, pooled across pre and post periods, with equal weights. The number of observations varies by source: ACS occupancy and units cover 2009–2023, USPS 2007–2024, ACS population 2010–2024, LODES 2005–2023, and NFIP policies 2009–2024. The claims samples cover 2005–2024 and 2009–2024, respectively. Counts are shown in natural units and ACS occupancy in percent; their regression transformations are defined below. ACS occupancy is one minus vacant housing units divided by total housing units, in levels. USPS occupied addresses are annual means of quarterly residential addresses net of no-stat and vacant addresses, in logs. Population and housing units are ACS five-year counts in logs. Resident workers and jobs are LODES counts transformed using the inverse hyperbolic sine. Premium here is the tract aggregate ratio of total premium to total building coverage, times 1,000. Claims are 100 times summed eligible positive payments divided by summed recorded claimant coverage, as in equation **??**; empty tract-years are missing. The two claims samples differ only in their starting year. Policy counts are term totals. The main policy-term premium sample is summarized separately in Table A.18.

Table A.17: Tract-level insurance responses to levee tax passage

	(1) Premium per 1,000 coverage	(2) Claims / coverage (%) 2005–2024	(3) Claims / coverage (%) 2009–2024	(4) Log policies
Treated × Post (at mean depth)	-0.703***	-11.198**	0.316	0.093*
	(0.214)	(4.383)	(8.436)	(0.044)
Treated × Post × Depth	-0.137***	-2.970***	-1.000	0.007
	(0.029)	(0.815)	(2.208)	(0.011)
Post × Depth	0.050**	-0.457	-1.275	0.009*
	(0.020)	(0.801)	(1.532)	(0.005)
Average treatment effect (separate DD)	-0.541*	-7.042	8.891	0.075
	(0.266)	(11.563)	(12.198)	(0.056)
Observation unit	Tract-year	Tract-year	Tract-year	Tract-year
Census tract FE	Yes	Yes	Yes	Yes
Boundary pair × year FE	Yes	Yes	Yes	Yes
Depth centering (ft)	17.52	17.52	17.52	17.52
Boundary pairs	10	10	7	10
Observations	2,031	362	147	2,031
Within R^2	0.127	0.134	0.113	0.015
R^2	0.909	0.716	0.588	0.925

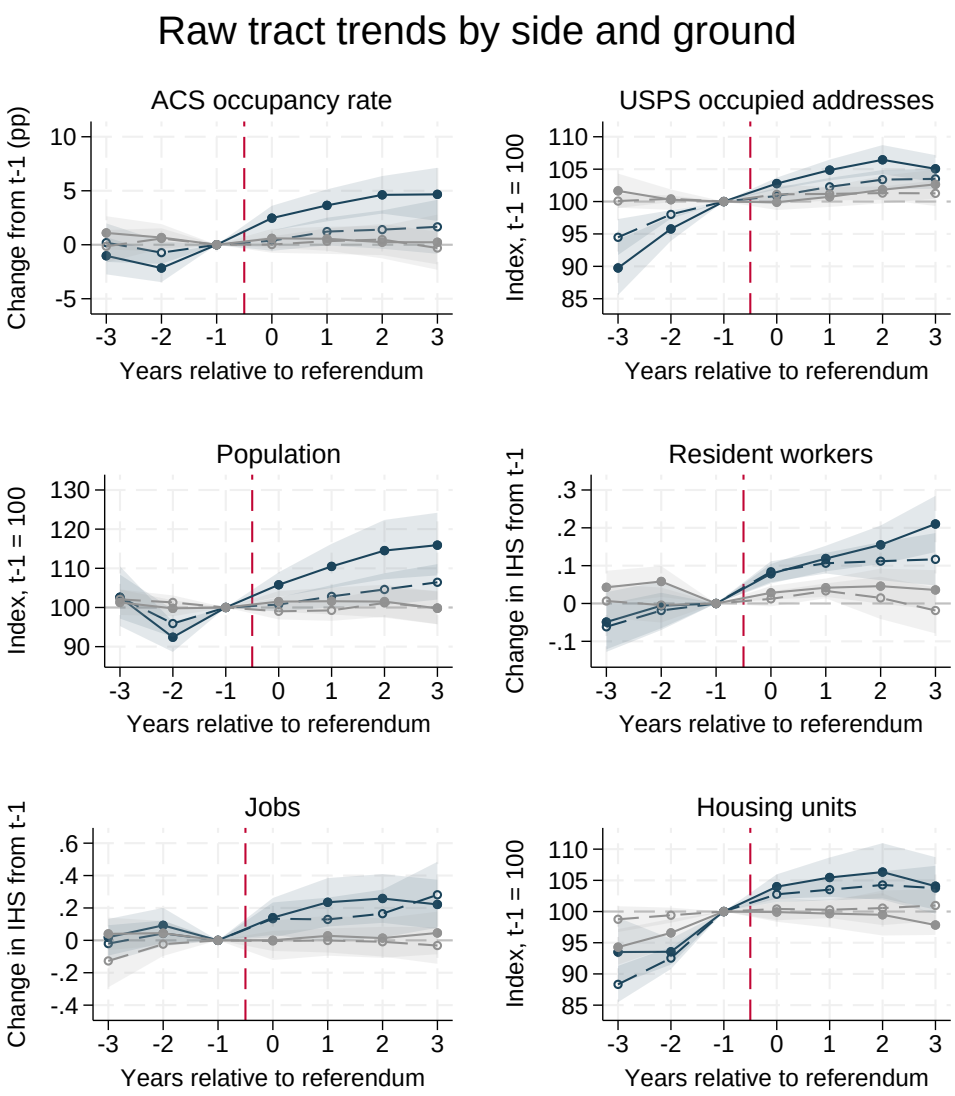
Notes: Unweighted tract-year regressions with tract and boundary-pair-by-year effects. Depth is centered at 17.52 feet, the treated tract-year mean, and uses the own-side district crest and cap. The first three rows report the full-factorial DDD. The average treatment effect is from a separate pooled DD on the same column-specific sample, with no depth interactions. Standard errors clustered by boundary pair are in parentheses; $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$. This is the all-tract counterpart to Table 9. Premium is the ratio of total premium to total building coverage, times 1,000, so it differs from the policy-level column in both observation weights and aggregation. The claims columns aggregate the same eligible positive records used in the main table to tract-year ratios of summed payments to summed claimant coverage. Each tract-year receives equal weight, whereas the main table weights individual claims equally. Within each tract-year this ratio weights individual severity by claimant coverage. The policy-count column reproduces the main table. Claims use recorded claimant coverage over 2005–2024 and 2009–2024, respectively; tract-years without eligible claims are missing. All models omit parish-by-year effects.

Table A.18: Policy-term premium summary statistics

	Mean	SD	p10	Median	p90	N
Policy premium per 1,000 coverage	4.10	4.73	1.39	2.38	8.91	1275553
Policy premium (dollars)	738.25	841.16	284.00	442.00	1483.00	1275553
Policy building coverage (dollars)	211219.68	216508.68	100000.00	250000.00	250000.00	1275553

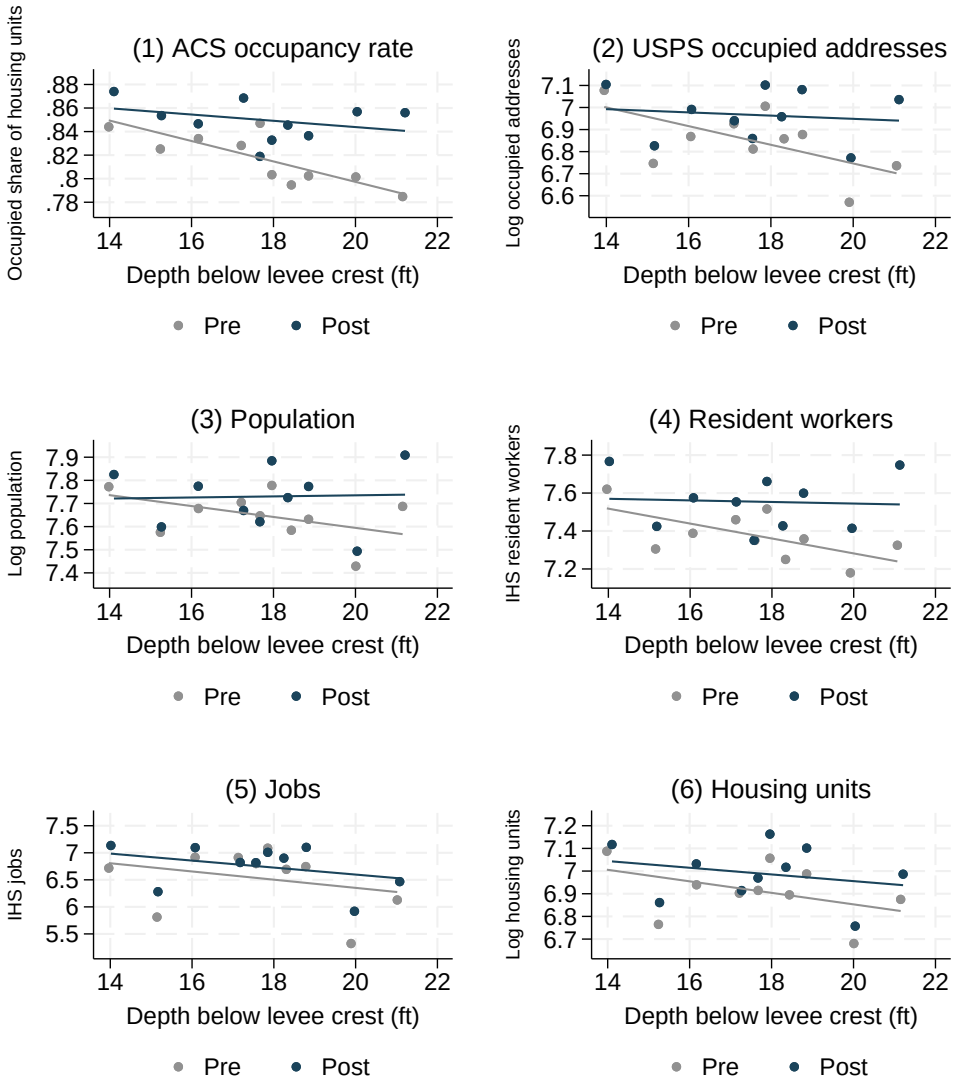
Notes: Policy-term observations in the current boundary-tract assignments and years. Premium per 1,000 is the individual term ratio; the other rows report nominal dollars. Terms must have positive premium and building coverage of at least $1,000 and cannot be cancelled at inception. No additional tail trim is applied. A term can enter different referendum samples where its tract is reused; records are unique within each referendum.

Figure A.19: Raw quantity trends by side and depth



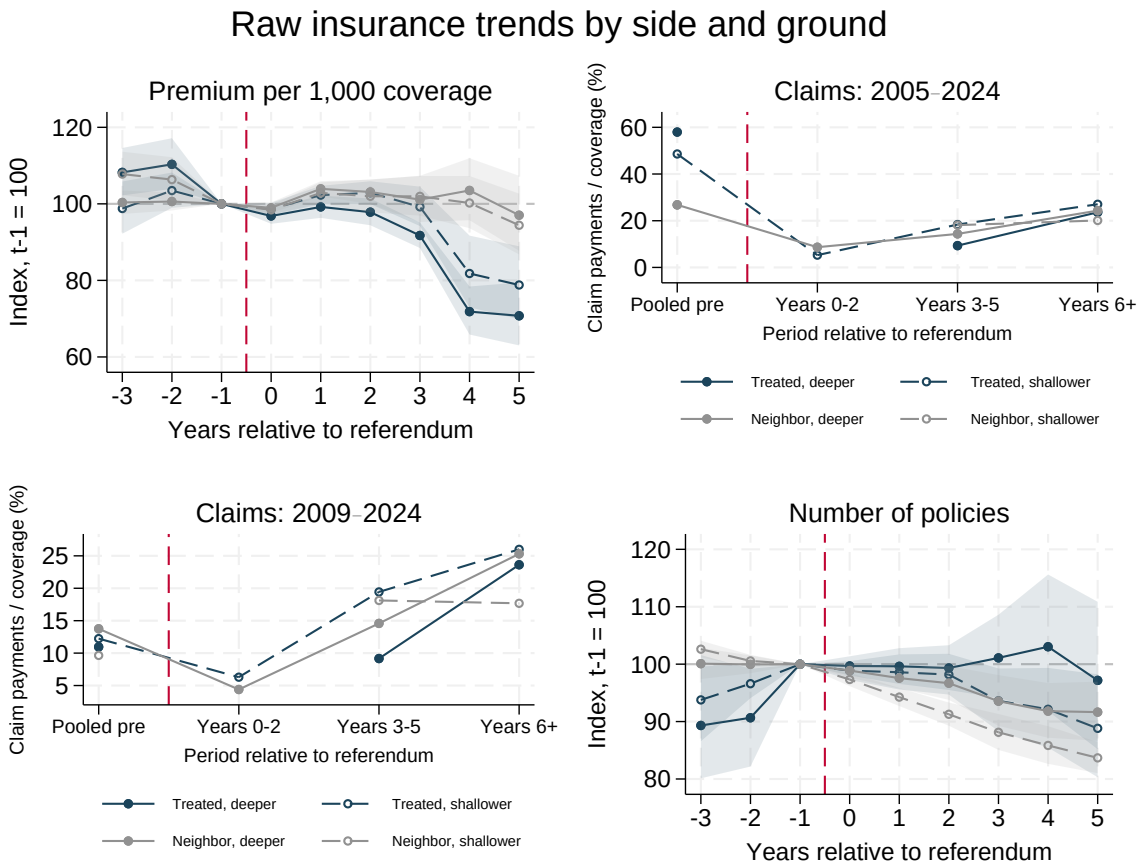
Notes: Each tract–referendum series is rebased to $t-1$. Logged counts are shown as indices ($t-1=100$); LODES workers and jobs are changes in inverse hyperbolic sine, and ACS occupancy is a percentage-point change. Blue denotes treated tracts and grey neighboring tracts; solid lines denote depth above the pair-side mean, dashed lines shallower exposure. Unweighted means and naive 95% bands, with at least two referenda per point. No controls or fixed effects are removed.

Figure A.20: Pre- and post-referendum quantity gradients



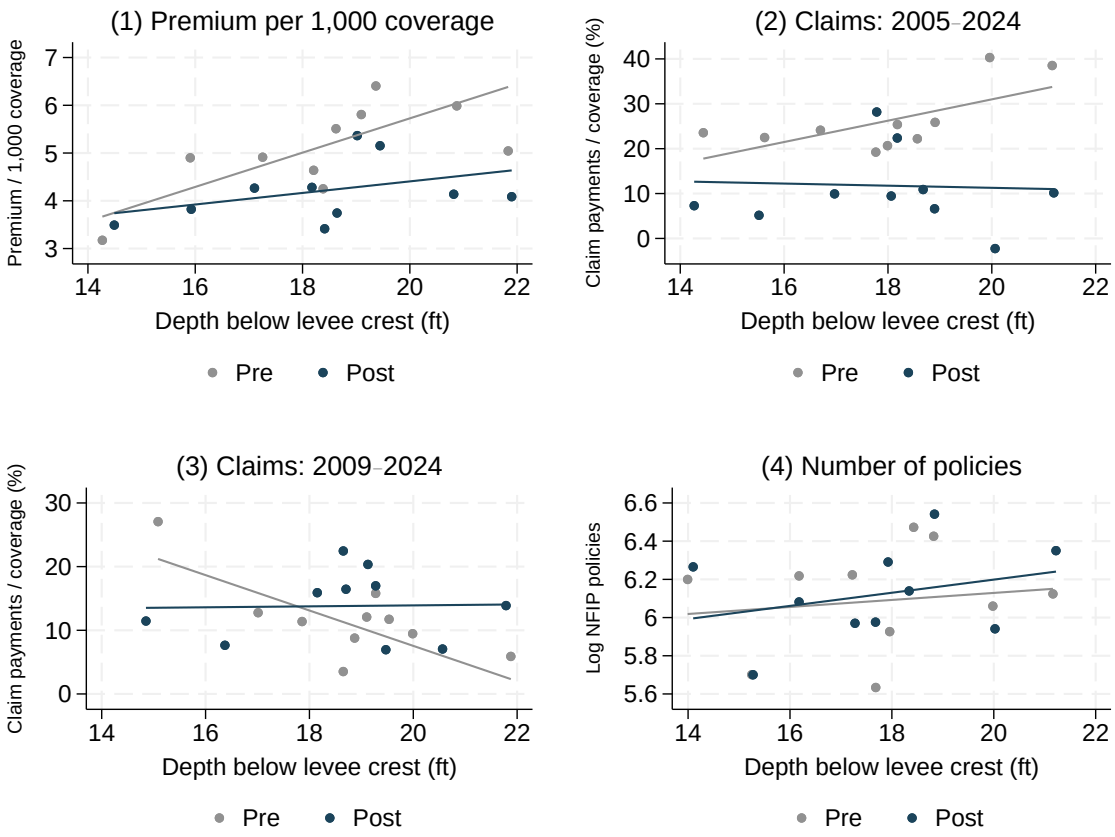
Notes: Treated-side observations, unweighted, with ten equal-count bins and separate pre/post fitted lines. Grey denotes pre-referendum and blue post-referendum. Both axes are residualized on the indicated fixed effects with their pooled means added back; the exposure measure is depth below the own-side district median levee crest; no year effects are removed. These plots are descriptive. Boundary-pair fixed effects are absorbed. ACS occupancy is one minus vacant housing units divided by total housing units, in levels. USPS occupied addresses are annual means of quarterly residential addresses net of no-stat and vacant addresses, in logs. Population and housing units are ACS five-year counts in logs. Resident workers and jobs are LODES counts transformed using the inverse hyperbolic sine.

Figure A.21: Raw insurance trends by side and depth



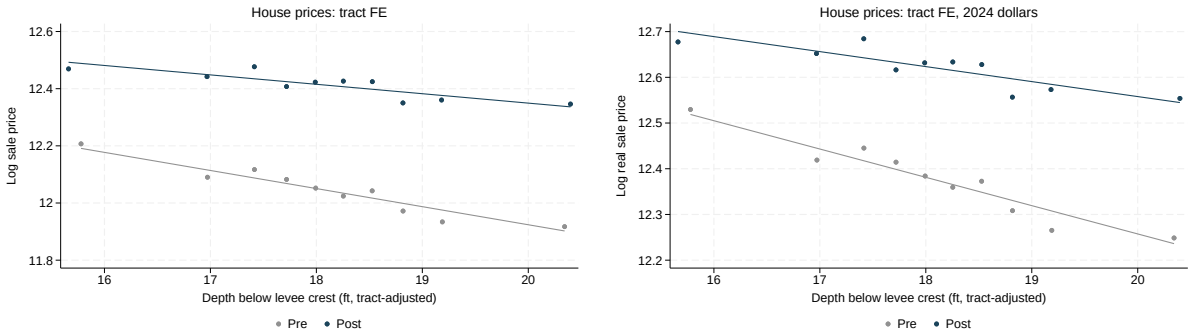
Notes: Premiums and policy counts retain tract-year aggregates rebased within each tract–referendum series to $t - 1 = 100$, displayed over years -3 through $+5$ with unweighted means and descriptive 95% bands. Claims instead show unadjusted means of individual positive-claim payments divided by recorded claimant coverage, multiplied by 100, on the same eligible records and tract–referendum–year support as Table 9. Claims are grouped into pooled pre-period, years 0–2, 3–5, and 6 onward, without requiring a positive claim in year -1 or rebasing. Each claim receives equal weight. Claims panels show descriptive means without confidence bands; clustered inference is reported in Figure 7. Points require at least two referenda and two boundary pairs. Unsupported points are omitted, lines break at missing periods, and no-claim years remain missing. Blue denotes treated tracts and grey neighboring tracts; solid lines indicate deeper exposure and dashed lines shallower exposure. Claims depth groups use the pair-side mean crest depth over the structural tract-panel rows in years -3 through $+3$, fixed before selecting claim records. These are descriptive comparisons without controls or fixed effects; common storms and changing claimant composition can affect the paths. Payments and coverage include building plus contents, with ICC excluded. The claims calendar windows are 2005–2024 and 2009–2024, respectively.

Figure A.22: Pre- and post-referendum insurance gradients



Notes: Treated-side observations, unweighted, with ten equal-count bins and separate pre/post fitted lines. Grey denotes pre-referendum and blue post-referendum. Both axes are residualized on the indicated fixed effects with their pooled means added back; the exposure measure is depth below the own-side district median levee crest; no year effects are removed. These plots are descriptive. Boundary-pair fixed effects are absorbed. The main premium outcome is an individual policy premium divided by its building coverage, multiplied by 1,000. It uses positive-premium terms with at least $1,000 of building coverage, drops flat cancellations, and applies no additional tail trim or building controls. The remaining columns are tract-year observations; redacted claims cannot be reliably linked to policy terms to construct individual zeros. Claims are 100 times summed building-plus-contents payments divided by summed recorded building-plus-contents coverage over the same eligible positive claims in a tract-year (cause codes 1, 2, 3, and B; ICC excluded). Claims without positive, observed coverage or with payments exceeding total coverage are excluded from both sums. Tract-years without eligible claims are missing. The claims columns use 2005–2024 and 2009–2024, respectively, with the latter matching the calendar-year coverage of the premium data. Both retain all eligible claims in their window; no storm is excluded separately. Policies and claims use the same one-to-one tract assignment: current codes match directly, while retired 2010 codes map to the current tract with the largest overlap. Unmapped records are excluded and counted in the geography audit. No geographic or regression weights are used. This outcome measures severity among claimants, not losses across all insured properties. Policy take-up is the log count of policy terms by effective-date year, not the fraction of households insured. No SFHA-share outcome is included.

Figure A.23: Price gradients with tract fixed effects



Notes: Nominal-price comparison to the CPI-adjusted introductory price panel in Figure 5. The left panel uses nominal prices; the right deflates them to 2024 dollars with annual CPI-U (BLS CUUR0000SA0). Both log sale price and crest depth are residualized on census-tract fixed effects with pooled means added back. No year effects, hedonics, or manual centering are used. Ten equal-count bins; grey pre and blue post. An analogous tract-FE occupancy binscatter is not identified because depth is constant within every treated occupancy tract.

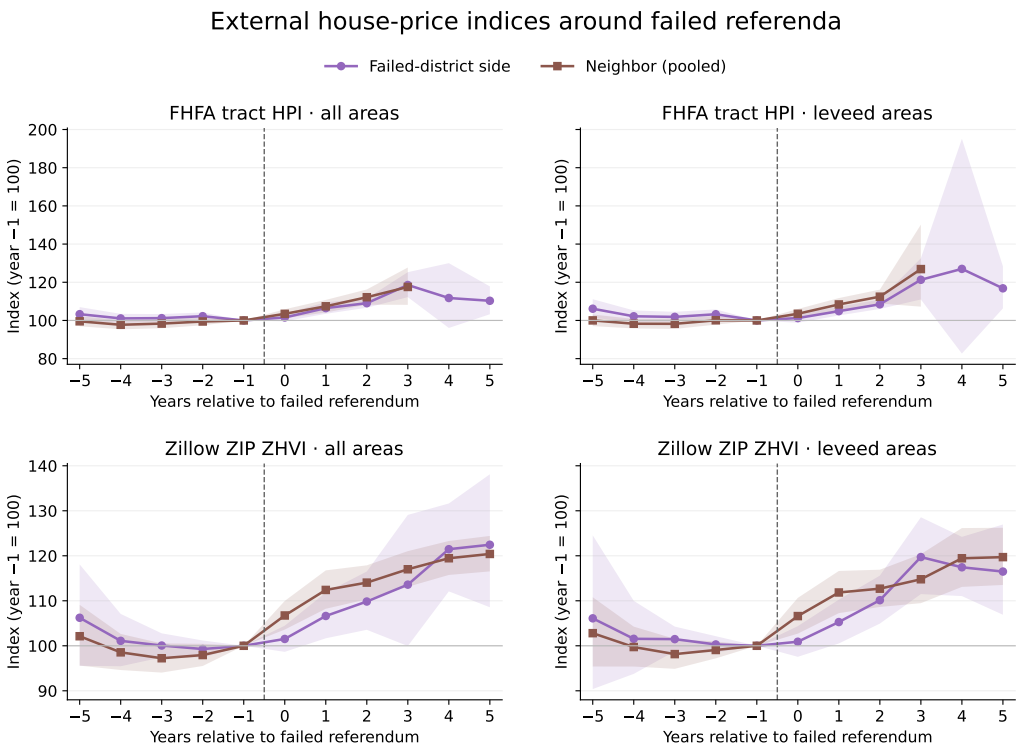
A.15 Failed Referenda

Table A.19: House-price responses following failed referenda

Panel A: Average treatment effects		
	(1) All areas	(2) Leveed areas
Treated × Post	-0.076*	-0.091**
	(0.038)	(0.031)
Spillover × Post	-0.206*	-0.470***
	(0.109)	(0.111)
Panel B: Crest-depth gradient		
Treated × Post (at mean depth)	-0.057	-0.046
	(0.044)	(0.052)
Treated × Post × Depth (per ft)	-0.009*	-0.008
	(0.004)	(0.007)
Spillover × Post (at mean depth)	-0.511*	-0.560***
	(0.235)	(0.100)
Spillover × Post × Depth (per ft)	0.075	0.099
	(0.077)	(0.101)
Post × Depth (per ft)	0.013***	0.018***
	(0.002)	(0.002)
Treated mean crest depth (ft)	10.52	13.41
Census tract FE	Yes	Yes
Boundary pair × year FE	Yes	Yes
Parish × year FE	No	No
Leveed-area restriction	No	Yes
Boundary pairs	13	9
Observations	13,074	9,845
Within R^2 (gradient)	0.116	0.085
R^2 (gradient)	0.460	0.397

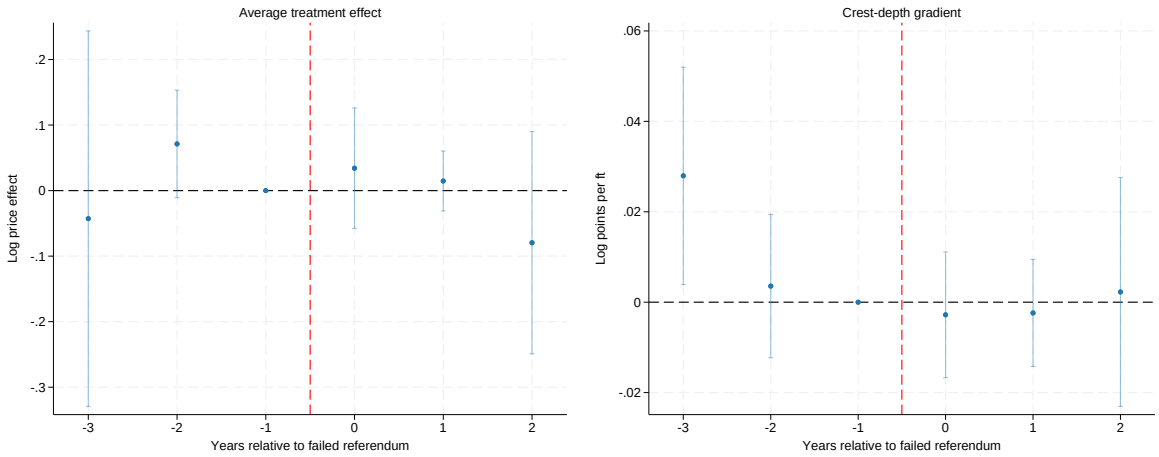
Notes: Both columns use the main paper's hedonic controls, census-tract and boundary-pair-by-year fixed effects, and boundary-pair-clustered standard errors. Column (1) includes all qualifying properties; column (2) restricts to leveed areas. Panel A estimates pooled average effects without depth interactions. Panel B uses the full treatment-by-post-by-continuous-depth factorial, with levels evaluated at each column's treated mean depth (reported below). Depth is the positive difference between the own-side district median crest and property elevation; as in the existing failed-vote pipeline, a side without a crest anchor receives zero exposure (613 unrestricted-sample sales before singleton removal). Treated denotes properties inside the district holding the failed vote; spillover denotes neighboring properties within 600 m of its levee; other neighbors are controls. Both columns apply the main bandwidth, single-parcel and property-overlap rules. Each district's first failed vote is retained; prior-treated own or control districts are excluded and later passage is censored using the paper's 2010-onward inventory. Earlier levies are baseline history. Average and gradient models use identical observations within each column. The leveed sample has only 17 spillover sales before singleton removal. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Figure A.24: External house-price indices around failed referenda



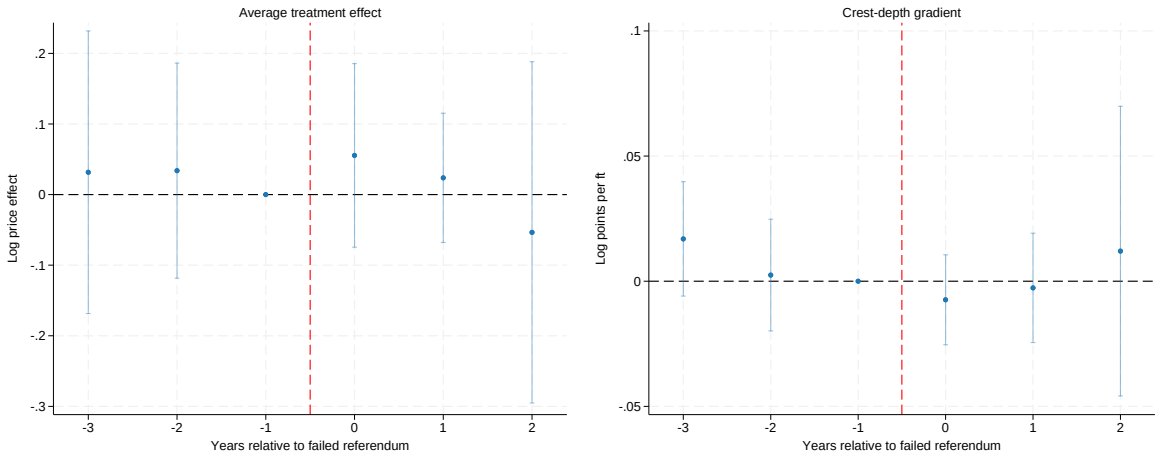
Notes: FHFA tract HPI (top) and Zillow ZIP-level ZHVI (bottom), on geographic support represented in the hedonic samples of Appendix Table A.19. Left: all qualifying properties; right: leveed areas. Each geographic unit–referendum index is rebased to 100 in year −1; lines are geometric means with equal unit–referendum weights. Neighbors are pooled, including the spillover ring. FHFA tracts represented on both sides of a referendum are excluded. Zillow uses the same coarse HUD ZIP–tract proxy as the passed-referendum plots, with majority represented-tract side assignment and ties excluded; monthly ZHVI is averaged within calendar year. Shading shows descriptive 95% bands based on standard errors of mean log changes, not clustered regression inference or a formal test of equality of the paths. Points require at least two contributing referenda. Index years begin in 2008 and stop before later own or neighbor passage under the paper's 2010-onward treatment definition. Earlier levies are baseline history. Coverage falls sharply after +2 when West Jefferson and North Lafourche are censored; changes thereafter also reflect changing composition.

Figure A.25: Failed-referendum dynamics: all qualifying properties



Notes: Hedonic regressions on all qualifying properties, corresponding to Table A.19, column (1). Left: pooled treated-by-event-time effects without depth interactions. Right: treated-by-event-time-by-crest-depth gradients from the full factorial model. Both include census-tract and boundary-pair-by-year fixed effects. The sample, spatial groups, depth construction, and passage-censoring rules follow the table. Estimation uses event years -10 through $+10$; the displayed window is -3 through $+2$, retaining the earlier failed-vote pipeline's cutoff because later periods have sparse support. Only $t-1$ is normalized to zero; unidentified coefficients are not plotted as zero. Bars are pair-clustered 95% confidence intervals.

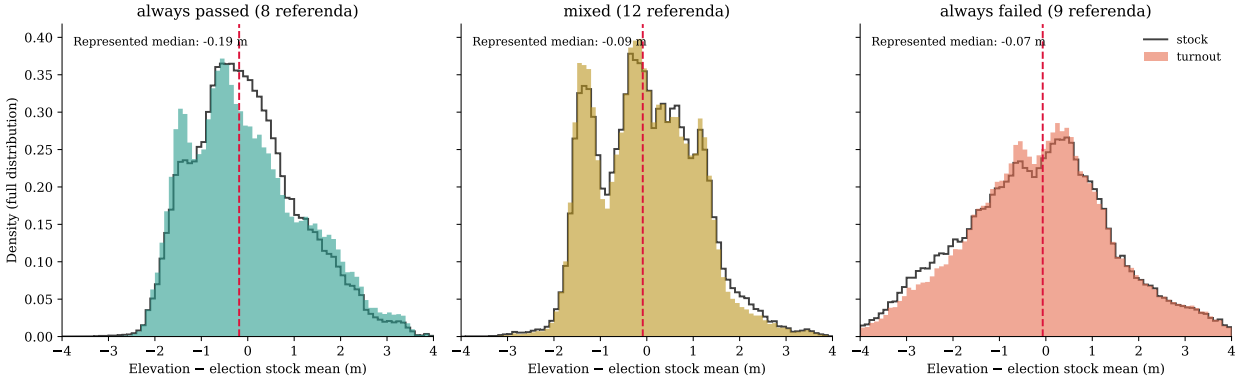
Figure A.26: Failed-referendum dynamics: leveed areas



Notes: Hedonic regressions restricted to leveed areas, corresponding to Table A.19, column (2). Left: pooled treated-by-event-time effects without depth interactions. Right: continuous crest-depth gradients from the full factorial model. Controls, fixed effects, estimation and display windows, normalization, and confidence intervals follow Figure A.25.

A.16 Precinct voting patterns and characteristics

Figure A.27: Residential stock and the turnout-represented electorate



Notes: Parcel elevations are centered on the mean stock elevation of their own district-election. The outline gives each residential parcel equal weight; the shaded distribution allocates each precinct's ballot-day turnout equally among its parcels with observed elevation. Both curves retain elections with positive represented turnout. Dashed lines are the represented medians computed from the full distributions, including tails beyond the display range. Densities are normalized by full mass; observations outside $[-4, 4]$ meters are not reassigned to displayed bins. Pooled stock curves weight elections by parcel counts, and electorate curves by represented turnout; neither is an equal-election average. Official passage histories define the groups over 2010–2025, including the 2021 constitutional amendment; Chenier Plain is retained.

Table A.20: Precinct elevation relative to the district stock mean and levee-tax approval

	Risk only		+ Characteristics	
	(1) No wt.	(2) Turnout wt.	(3) No wt.	(4) Turnout wt.
Risk				
Precinct median - district mean (per m)	-0.58 (0.46)	-0.18 (0.77)	-0.68 (0.54)	-0.61 (0.40)
Salience				
Distance to levee			1.52 (1.94)	0.63 (1.80)
Levee in precinct			2.40* (1.19)	0.22 (0.65)
Experience				
NFIP claims history			1.61 (1.36)	0.39 (0.92)
Other characteristics				
Median household income			0.57 (1.32)	-1.24 (1.20)
Voter density			1.73** (0.79)	0.87 (0.69)
BA-or-higher share			1.23 (1.06)	1.97 (1.18)
Renter share			2.20** (0.91)	1.08 (0.64)
NRI social vulnerability			-0.96* (0.54)	-0.58 (0.49)
Political				
Democratic voter share			0.55 (1.93)	-0.91 (2.70)
Constant	52.12*** (0.17)	51.95*** (0.12)	51.46*** (0.38)	50.91*** (1.13)
Referendum (district × date) FE	Yes	Yes	Yes	Yes
Other characteristics	–	–	Yes	Yes
Turnout weighted	–	Yes	–	Yes
Clustering		Precinct + ballot item		
Stable precincts	1,092	1,092	1,092	1,092
Ballot items	24	24	24	24
Observations	1,724	1,724	1,724	1,724
Within R-squared	0.007	0.001	0.054	0.073
Overall R-squared	0.669	0.726	0.684	0.746

Notes: The risk regressor is median residential-parcel elevation in the precinct minus mean residential-parcel elevation in its district-election. The district mean gives equal weight to every observed residential parcel in the matched election precincts and is constructed before regression exclusions. It is the stock benchmark underlying the district-level figure and electorate histograms; the precinct median here is a stock median, not an observed voter's location. All four columns use the same 1,724 precinct–proposition observations as Table 10, with the same controls, scaling, referendum fixed effects, turnout weights and two-way precinct/ballot-item clustering. The elevation coefficient is per meter; the levee-presence indicator is binary and other characteristics are per standard deviation. Subtracting the district-election mean leaves the precinct-median slope unchanged under referendum fixed effects. These are conditional associations. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table A.21: Precinct voting sample: characteristics in original units

	N	Mean	SD	p10	Median	p90
Approval (percent)	1,726	52.08	19.49	26.32	53.51	76.27
Item votes	1,726	142.49	114.13	26.00	108.00	310.00
Median minus mean elevation (m)	1,726	-0.00	0.14	-0.11	-0.01	0.12
Distance to levee (km)	1,726	8.41	13.02	0.41	1.87	30.58
Levee in precinct (0/1)	1,726	0.29	0.45	0.00	0.00	1.00
Log NFIP claims history	1,726	1.03	1.09	0.07	0.55	2.64
Log median household income	1,726	10.76	0.42	10.21	10.76	11.27
Log voter density	1,726	5.48	1.84	2.68	6.10	7.32
BA-or-higher share	1,726	0.21	0.16	0.07	0.16	0.44
Renter share	1,726	0.33	0.19	0.12	0.29	0.61
NRI social vulnerability	1,726	57.77	24.08	21.95	60.34	86.28
Democratic voter share	1,726	0.56	0.21	0.27	0.55	0.85

Notes: Unweighted complete-case levee-tax precinct–item observations with at least ten votes, before requiring positive recorded turnout for the main regression. Logs remain in their transformed units, but no standardization has been applied. Item votes are yes plus no; the regression's turnout weight counts all recorded ballot-day voters. Thus the summary sample can exceed the main table's sample.

Table A.22: Levee-tax approval when the tax is the sole proposition or shares the ballot

	Sole item		Shared ballot	
	(1) No wt.	(2) Turnout wt.	(3) No wt.	(4) Turnout wt.
Risk				
Median minus mean elevation (per m)	1.49 (5.02)	1.42 (5.21)	-3.06** (1.25)	-1.93 (1.39)
Salience				
Distance to levee	6.00 (4.58)	12.43** (4.91)	0.02 (1.17)	-0.84 (0.82)
Levee in precinct	3.52 (1.99)	2.11 (1.69)	1.81 (1.46)	-0.10 (0.88)
Experience				
NFIP claims history	4.13*** (1.20)	2.96 (1.80)	1.33 (1.42)	0.34 (0.87)
Other characteristics				
Median household income	2.81 (2.20)	0.31 (1.73)	-0.32 (1.64)	-1.77 (1.54)
Voter density	2.39 (1.58)	1.84 (1.24)	1.60 (0.94)	0.94 (0.92)
BA-or-higher share	-0.44 (2.28)	1.33 (2.43)	1.39 (1.40)	2.00 (1.67)
Renter share	2.69 (1.67)	1.80* (0.92)	2.01* (1.12)	0.70 (0.75)
NRI social vulnerability	-0.08 (0.80)	-1.14 (0.68)	-1.30* (0.63)	-0.15 (0.53)
Political				
Democratic voter share	0.76 (1.63)	0.86 (1.44)	0.38 (2.96)	-1.62 (3.63)
Constant	57.93*** (1.41)	53.44*** (1.81)	49.01*** (0.59)	51.51*** (1.68)
Referendum (district × date) FE	Yes	Yes	Yes	Yes
Turnout weighted	–	Yes	–	Yes
Clustering		Precinct + ballot item		
Stable precincts	368	368	885	885
Ballot items	11	11	15	15
Observations	596	596	1,128	1,128
Within R-squared	0.082	0.140	0.055	0.096
Overall R-squared	0.746	0.811	0.619	0.722

Notes: The fully controlled referendum-fixed-effect specification of Table 10, split by the presence of another proposition in the retained precinct ballot records. "Sole" refers to propositions, not candidate races; items with fewer than ten votes are excluded before classifying ballot type. Weighted and unweighted columns use the same eligible rows within each subset. Scaling and two-way clustering follow the main table. The weighted shared-ballot regression requires Stata's adjustment of a non-positive-semidefinite two-way covariance matrix. Differences across subsamples are descriptive.

Table A.23: How levee-tax approval slopes differ from other propositions

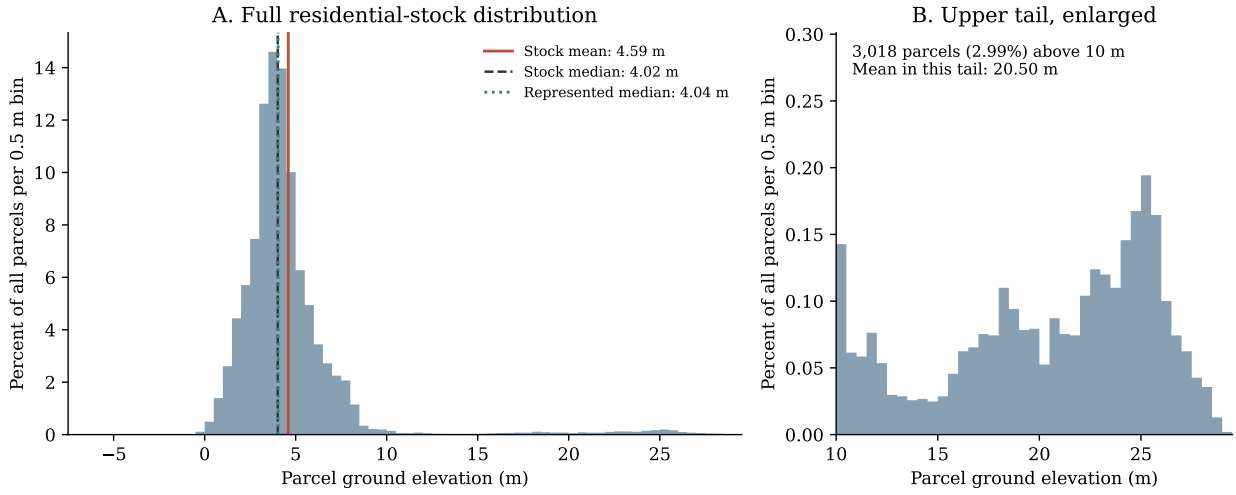
	(1) Date + district FE	(2) District × item FE
Risk		
Median minus mean elevation (per m)	-3.05* (1.75)	-1.28 (0.83)
Salience		
Distance to levee	-2.06 (1.74)	-1.57 (1.14)
Levee in precinct	1.48 (1.27)	1.01 (1.33)
Experience		
NFIP claims history	0.95 (1.33)	0.83 (1.35)
Other characteristics		
Median household income	2.05* (1.16)	0.76 (1.50)
Voter density	0.03 (1.06)	-0.71 (0.92)
BA-or-higher share	-0.57 (1.56)	1.04 (1.43)
Renter share	2.27** (0.94)	1.95* (1.04)
NRI social vulnerability	-1.14** (0.55)	-1.18** (0.48)
Political		
Democratic voter share	2.68 (2.44)	2.77 (3.01)
Characteristic levels	Yes	Yes
Levee-tax item indicator	Yes	absorbed
Clustering	Precinct + ballot item	
Stable precincts	885	885
Ballot items	89	88
Observations	8,579	8,574
Within R-squared	0.052	0.068
Overall R-squared	0.463	0.743

Notes: Only precinct-dates with a levee-tax proposition and at least one other retained proposition enter. Both columns regress item approval on characteristic levels, a levee-tax indicator, and its interactions with all characteristics. Reported coefficients are the interaction terms: the levee-tax approval slope minus the other-proposition slope, not the total levee-tax slope. Column 1 absorbs ballot-date and district fixed effects; Column 2 absorbs district-by-item fixed effects, which also absorb the levee-tax indicator. Both columns are unweighted, with the same scaling as Table 10 and standard errors clustered by stable precinct and ballot item. Fixed-effect singleton removal can change the estimation sample across columns. Stata adjusts a non-positive-semidefinite two-way covariance matrix in both specifications; inference should therefore be treated cautiously. These within-ballot comparisons are descriptive and do not identify causal preference differences. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

A.17 Chenier Plain: the full residential elevation distribution

Figure A.28 explains Chenier Plain's position in Figure 9. The stock mean is 4.59 m, compared with a stock median of 4.02 m and a turnout-represented median of 4.04 m. The 2.99% of parcels above 10 m have mean elevation 20.50 m and the remaining parcels average 4.10 m. Including the upper group therefore raises the mean by 0.49 m relative to the lower group's mean. This accounting exercise explains much of the negative median–mean gap without attributing it to turnout composition. The 10 m threshold describes the tail, no parcels are removed from either figure or the regressions.

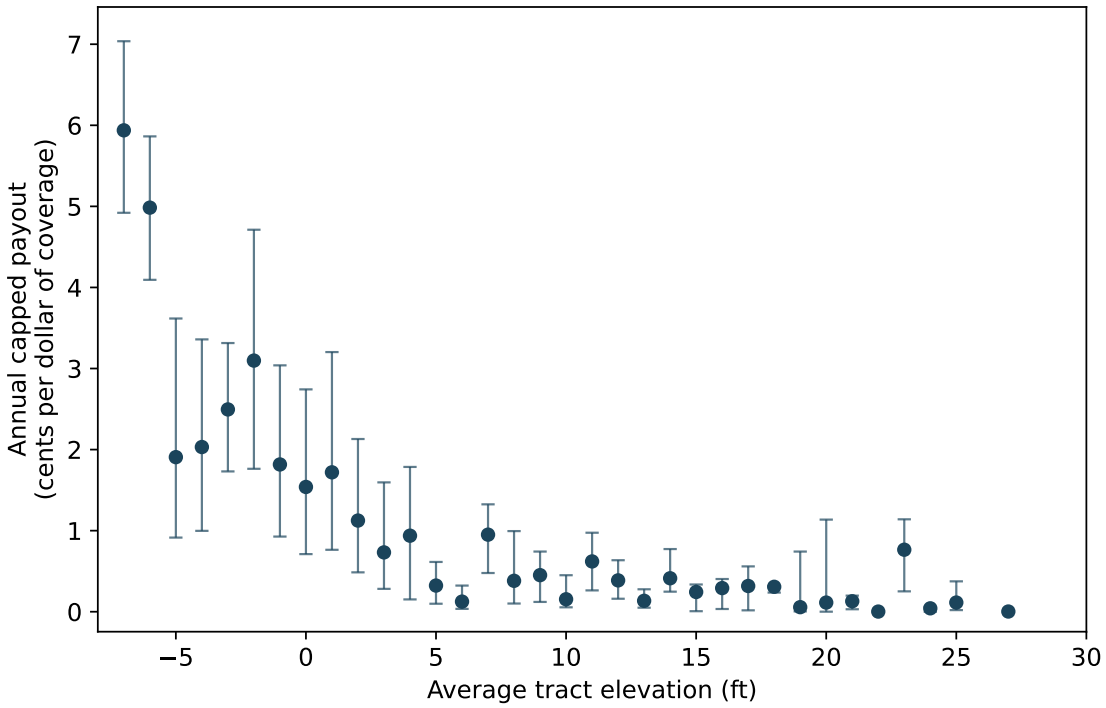
Figure A.28: Chenier Plain residential parcel elevations, including the upper tail



Notes: All 100,893 distinct residential parcels with coordinates assigned to the matched November 13, 2021 precinct geography; all have observed elevation. This is the observed assessor residential stock used in the main figure, not a census of all property types or dwelling units. Elevation is ground elevation sampled from the available USGS 3DEP rasters, not first-floor height or a historical 2021 measurement. Panel A displays the full observed range, including negative elevations, without trimming or winsorization. Panel B enlarges the part above 10 m; both panels express bin counts as percentages of the full stock and use 0.5 m bins. The represented median allocates each precinct's recorded turnout equally over its observed residential parcels; it is not an observed individual voter's elevation.

A.18 Elevation and annual insured flood losses

Figure A.29: Average tract elevation and annual insured flood losses



Notes: Each equal-size dot is 100 times total NFIP building-plus-contents payouts divided by building-plus-contents coverage-years in a one-foot bin of average tract elevation. Payouts and coverage use nominal dollars without inflation adjustment. Before pooling, payouts are capped at coverage-years within each tract-year, including setting payouts to zero when recorded exposure is zero. This affects 16 tract-years and reduces nominal payouts by $41.0 million (1.44%). Elevation is the unadjusted mean parcel elevation in the district housing stock within each tract; tracts are assigned to their modal district, and tract means above 30 feet are excluded. The sample pools referendum districts and A-zone contracts. Claims have loss dates from 2005 strictly before the district's first passed referendum, or first referendum for districts that never passed, and damage cause codes 1, 2, or 3. Coverage-years include policies without claims, apportion contract days to calendar years, and prorate the vote year by the fraction elapsed before the vote. Exposure in 2005–08 is proxied by recorded 2009 exposure. The policy file begins with 2009 policy starts and omits contracts begun in 2008 that remained active in 2009, so the proxy may understate full-year exposure. Bins with fewer than 500 policy-years are omitted (26, 28, and 30 feet). Bars are pointwise 95% percentile intervals from 2,000 tract-bootstrap draws within each bin, keeping each tract's years together; the 22- and 27-foot bins each have one tract and no interval; they do not account for common storm shocks across tracts or uncertainty in the early-period exposure proxy. The plot is descriptive and does not establish exact linearity, a sharp zero-loss threshold, or the causal effect of protection revenue.